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MuonEq: Balancing Before Orthogonalization with Lightweight Equilibration

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Abstract

Orthogonalized-update optimizers such as Muon improve training of matrix-valued parameters, but existing extensions typically either rescale updates after orthogonalization or use heavier whitening-based preconditioners before it. We introduce **MuonEq**, a lightweight family of pre-orthogonalization equilibration schemes for Muon with three forms: two-sided row/column normalization (RC), row normalization (R), and column normalization (C). By rebalancing the momentum matrix before finite-step Newton–Schulz orthogonalization, **MuonEq** improves the geometry seen by orthogonalization. We show that finite-step orthogonalization is governed by the input spectrum, especially stable rank and condition number, and that row/column normalization acts as a zeroth-order surrogate for whitening. For hidden matrix weights, R is the default variant. Theoretically, **MuonEq** (R) retains the standard $\widetilde{\mathcal{O}}(T^{-1/4})$ Muon-type nonconvex stationarity guarantee with decoupled weight decay and a horizon-free diminishing learning-rate schedule, and extends it to finite-step NS5 up to an explicit inexactness constant. In LLaMA2 pretraining on C4, **MuonEq** (R) consistently outperforms Muon on 130M, 350M, and 1B models, with faster convergence and lower validation perplexity. The code is available at the MuonEq codebase.

1 Introduction

As the computational and data costs of large language model pretraining grow, optimization increasingly determines sample efficiency, throughput, and scalability. Beyond coordinate-wise adaptive methods such as AdamW [1], Adan [2], and MARS [3], recent work has shifted toward a geometric view of matrix-valued updates. Muon [4], for instance, uses a Newton–Schulz approximation to the polar factor of the momentum matrix to produce spectrally aligned updates [5, 6]. It has shown strong empirical scalability in LLM pretraining, often improving the trade-off among data efficiency, compute, and wall-clock time at fixed target loss [7, 8, 9]. Recent theory further characterizes Muon as spectral-norm steepest descent and establishes nonconvex convergence guarantees for both exact and approximate orthogonalization [10, 11, 12, 13, 14, 15, 16].

Follow-up work around Muon mainly proceeds along two directions. *(i) Post-orthogonalization rescaling* augments an already constructed orthogonal update with additional magnitude correction. Muon+ [17], NorMuon [18], and AdaMuon [19] respectively add row/column normalization, neuron-wise normalization, or element-wise variance adaptation after orthogonalization. These variants show that orthogonal updates remain compatible with later adaptive scaling, but they do not

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MuonEq：在正交化之前进行平衡，采用轻量级均衡

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摘要

正交化更新优化器如Muon可以提高矩阵值参数的训练效果，但现有的扩展通常在正交化后重新缩放更新，或者在使用更重的白化预处理器之前进行。我们引入**MuonEq**，一个轻量级的预正交化均衡方案家族，包含三种形式：双边行/列归一化（RC）、行归一化（R）和列归一化（C）。通过在有限步Newton-Schulz正交化之前重新平衡动量矩阵，**MuonEq** 改进了正交化所看到的几何结构。我们证明有限步正交化受输入谱的影响，特别是稳定秩和条件数，并且行/列归一化充当白化的零阶替代品。对于隐藏的矩阵权重，R是默认变体。理论上，**MuonEq** (R)保留了标准的 $\widetilde{\mathcal{O}}(T^{-1/4})$ Muon型非凸平稳性保证，具有解耦的权重衰减和没有视野的递减学习率计划，并将其扩展到有限步NS5，直到一个显式的非精确性常数。在C4上的LLaMA2预训练中，**MuonEq** (R)在130M、350M和1B模型上始终优于Muon，收敛速度更快，验证困惑度更低。代码可在MuonEq代码库中获取。

1 简介

随着大型语言模型预训练的计算和数据处理成本不断增长，优化越来越决定了样本效率、吞吐量和可扩展性。除了 AdamW [1], Adan [2], 和 MARS [3], 等坐标自适应方法之外，最近的研究工作已转向矩阵值更新的几何视角。例如，Muon [4], 使用动量矩阵极因子的 Newton–Schulz 近似来生成频谱对齐的更新 [5, 6]。它在大型语言模型预训练中已展现出强大的经验可扩展性，通常在固定目标损失 [7, 8, 9]下，能够改善数据效率、计算量和墙钟时间之间的权衡。最近的理论进一步将 Muon 视为频谱范数最速下降法，并为精确和近似正交化 [10, 11, 12, 13, 14, 15, 16]建立了非凸收敛保证。

围绕Muon的后续工作主要沿着两个方向进行。*(i)* 正交化后重缩放 通过额外的幅度校正来增强已构建的正交更新。Muon+ [17]、NorMuon [18], 和AdaMuon [19]分别在校正后添加行/列归一化、神经元归一化或元素方差自适应。这些变体表明正交更新与后续自适应缩放仍然兼容，但它们不

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预印本。

change the matrix that finite-step Newton–Schulz actually sees. *(ii) Pre-orthogonalization geometric correction* instead modifies the coordinate system before spectral optimization. SOAP [20] connects Shampoo [21] with Adam/Adafactor [22, 23] through a preconditioned basis, while FISMO [24] and Mousse [25] further perform spectral optimization in whitened coordinates. These methods motivate studying geometry correction before orthogonalization as a promising lever, but existing approaches typically incur substantially higher state and computational overhead.

These works motivate a more precise question:

Is there a lightweight design between post-orthogonalization rescaling and full whitening-based preconditioning that directly improves the input geometry for orthogonalization?

Our starting point is that, in Muon, the relevant object is not only the exact polar factor but also the finite-step Newton–Schulz approximation used in practice. Section 3.1 shows that both the convergence speed of Newton–Schulz iterations and the gap between finite-step iterations and exact orthogonalization are governed by input spectral geometry, especially stable rank and condition number. This points to a simple design principle: improve the geometry *before* orthogonalization, but do so with the smallest possible intervention. A natural formalization is the optimal diagonal preconditioning problem:

$$\min_{\mathbf{D}_1,\mathbf{D}_2\succ 0} \kappa\big(\mathbf{D}_1^{1/2}\mathbf{A}\mathbf{D}_2^{-1/2}\big).$$

Qu et al. [26] derive an optimal diagonal preconditioner, but applying it online typically requires an auxiliary iterative solve. Classical row/column equilibration is less exact yet much lighter-weight [27]; moreover, recent operator-geometry results identify row and column normalization as steepest-descent directions under the $p \to \infty$ and $1 \to q$ geometries, respectively [28]. This makes diagonal equilibration a natural lightweight way to improve the matrix presented to Muon-style orthogonalization.

Motivated by this observation, we propose **MuonEq**, a lightweight family of pre-orthogonalization equilibration schemes for Muon, instantiated as two-sided row/column normalization (RC), row normalization (R), and column normalization (C). All variants rescale the momentum matrix before Newton–Schulz, using row/column squared-norm reductions from the current momentum. Section 3 provides the theory: Theorem 3.1 quantifies the spectral sensitivity of finite-step orthogonalization; Proposition 3.2 shows that row/column normalization is a zeroth-order whitening surrogate; and Proposition 3.4, Theorem 3.5, and Corollary 3.6 formalize the RC/R trade-off. In particular, the default R variant satisfies the standard $\widetilde{\mathcal{O}}(T^{-1/4})$ nonconvex stationarity guarantee while explicitly accounting for decoupled weight decay, horizon-free learning-rate schedules, and finite-step NS5. For the hidden Transformer weight matrices targeted by **MuonEq**, recent architecture-aware analyses [28] identify row-sided geometry as the natural one-sided choice, while column-sided geometry is more aligned with embeddings; hence we use R by default and keep RC/C for analysis and ablation. Compared with Muon+ [17], which rescales updates after orthogonalization, **MuonEq** changes the input to orthogonalization. Unlike whitening-based methods such as FISMO and Mousse [24, 25], **MuonEq** avoids Kronecker factors, matrix-valued second-order statistics, eigendecomposition, and momentum transport; its diagonal rescaling is computed on the fly from row/column squared norms, with no persistent optimizer state.

Our contributions are threefold. *(i)* First, we propose **MuonEq**, a lightweight family of pre-orthogonalization equilibration schemes for Muon with three forms: RC, R, and C; these variants rebalance the current momentum matrix before Newton–Schulz using row/column squared norms computed on the fly, with R as the default for hidden matrix weights. *(ii)* Second, we analyze how pre-orthogonalization geometry affects finite-step Newton–Schulz: Theorem 3.1 links its behavior to stable rank and condition number, Proposition 3.2 identifies row/column normalization as a zeroth-order whitening surrogate, and Proposition 3.4 together with Theorem 3.5 and Corollary 3.6 formalize the trade-off between stronger two-sided spectral correction and cleaner one-sided geometry. In particular, **MuonEq** (R) satisfies the standard $\widetilde{\mathcal{O}}(T^{-1/4})$ nonconvex stationarity guarantee while accounting for decoupled weight decay, horizon-free learning-rate schedules, and finite-step NS5. *(iii)* Third, empirically, the default R variant consistently outperforms Muon across model sizes and token budgets, while ablations over RC, R, and C show that R is the default form for the hidden-weight setting targeted by **MuonEq**.

改变有限步Newton-Schulz实际看到的矩阵。*(ii)* 正交化前几何校正 则修改了频谱优化前的坐标系。SOAP [20] 通过预条件基将Shampoo [21] 与Adam/Adafactor [22, 23] 连接起来，而FISMO [24] 和Mousse [25]进一步在白化坐标中进行频谱优化。这些方法促使人们研究正交化前的几何校正作为一个有前景的杠杆，但现有方法通常会导致显著更高的状态和计算开销。

这些工作促使我们提出了一个更精确的问题：

是否存在一种轻量级设计，它介于正交化后的重缩放和基于完全白化的预条件化之间，并能直接改进用于正交化的输入几何？

我们的出发点是，在Muon中，相关对象不仅是精确的极因子，还包括实践中使用的有限步Newton-Schulz近似。3.1节表明，Newton-Schulz迭代的收敛速度以及有限步迭代与精确正交化之间的差距，都受输入谱几何的支配，特别是稳定秩和条件数。这指向了一个简单的设计原则：在正交化之前改进几何，但要以最小的干预程度来实现。一个自然的公理化表述是最优对角预条件化问题：

$$\min_{\mathbf{D}_1,\mathbf{D}_2\succ 0} \kappa\big(\mathbf{D}_1^{1/2}\mathbf{A}\mathbf{D}_2^{-1/2}\big).$$

Qu等人 [26] 推导出了一个最优对角预条件器，但在线应用它通常需要辅助迭代求解。经典的行/列均衡不够精确，但轻量级 [27]；此外，最近的算子几何结果识别出行和列归一化分别是 $p \to \infty$和 $1 \to q$ 几何下的最速下降方向 [28]。这使得对角均衡成为改进呈现给Muon风格正交化的矩阵的一种自然轻量级方法。

基于这一观察，我们提出了 **MuonEq**，这是一系列轻量级的预正交化均衡方案，用于Muon，具体实例化为双边行/列归一化（RC）、行归一化（R）和列归一化（C）。所有变体在 Newton–Schulz 之前对动量矩阵进行重新缩放，使用当前动量的行/列平方范数减少。第 3 节提供了理论：定理 3.1 定量了有限步正交化的谱敏感性；命题 3.2 表明行/列归一化是一种零阶白化替代方案；命题 3.4、定理 3.5 和推论 3.6 形式化了 RC/R 的权衡。特别是，默认的 R 变体满足标准的 $\widetilde{\mathcal{O}}(T^{-1/4})$ 非凸平稳性保证，同时明确考虑了解耦权重衰减、无视野学习率调度和有限步 NS5。对于 **MuonEq** 目标隐藏 Transformer 权重矩阵，最近的架构感知分析 [28]确定行侧几何是自然的一侧选择，而列侧几何更符合嵌入；因此我们默认使用 R，并保留 RC/C 用于分析和消融。与 Muon+ [17]（在正交化后重新缩放更新）相比，**MuonEq** 改变了正交化的输入。与基于白化的方法（如 FISMO 和 Mousse [24, 25], **MuonEq** ）不同，MuonEq 避免了 Kronecker 因子、矩阵值二阶统计、特征分解和动量传输；其对角重新缩放是在线计算，基于行/列平方范数，且没有持久的优化器状态。

我们的贡献有三方面。*(i)* 首先，我们提出了**MuonEq**，一种轻量级的预正交化均衡方案家族，用于Muon，包含三种形式：RC、R和C；这些变体在Newton-Schulz之前重新平衡当前动量矩阵，使用实时计算的行/列平方范数，默认情况下R用于隐藏矩阵权重。*(ii)* 其次，我们分析了预正交化几何如何影响有限步Newton-Schulz：定理3.1将其行为与稳定秩和条件数联系起来，命题3.2将行/列归一化识别为零阶白化替代方案，命题3.4连同定理3.5和推论3.6形式化了更强的双边谱校正与更干净的单边几何之间的权衡。特别是，**MuonEq** (R) 在考虑解耦权重衰减、无视野学习率调度和有限步NS5的情况下，满足标准的 $\widetilde{\mathcal{O}}(T^{-1/4})$非凸平稳性保证。*(iii)*第三，从经验上看，默认的R变体在所有模型大小和token预算中始终优于Muon，而RC、R和C的消融实验表明，R是**MuonEq**针对隐藏权重设置的目标形式的默认形式。

2 Method

Notation and setup. Scalars are denoted by non-bold letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. On $\mathbb{R}^d$, we use the Euclidean inner product $\langle \mathbf{x}, \mathbf{y} \rangle_2 := \mathbf{x}^\top \mathbf{y}$ and norm $\|\mathbf{x}\|_2$. For matrices, $\langle \mathbf{A}, \mathbf{B} \rangle_F := \mathrm{tr}(\mathbf{A}^\top \mathbf{B})$ and $\|\mathbf{A}\|_F$ denote the Frobenius inner product and norm, and $\|\mathbf{A}\|_*$ denotes the nuclear norm. We write $[m] := \{1, 2, \ldots, m\}$ and let $\mathbb{N}$ be the set of non-negative integers. The model parameter is a matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, and, unless otherwise stated, we assume $m \geq n$. For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with rank $r \leq n$, we write its compact singular value decomposition as $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$, where $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ and $\sigma_1 \geq \cdots \geq \sigma_r > 0$, and define the polar factor, which we also call the orthogonalized form, as $\mathrm{Orth}(\mathbf{A}) := \mathbf{U}\mathbf{V}^\top$. Here, orthogonalization and polar step mean applying or approximating this map, whereas whitening denotes full Gram inverse-square-root preconditioning. The row and column rescalings introduced below are diagonal equilibration operations. In finite-data form, the training objective can be written as $f(\mathbf{X}) := \frac{1}{N} \sum_{i \in [N]} f_i(\mathbf{X})$, where $f_i(\mathbf{X})$ is the loss associated with the i-th sample $\mathbf{z}_i$. Equivalently, we use the stochastic formulation

$$\min_{\mathbf{X} \in \mathbb{R}^{m \times n}} f(\mathbf{X}), \qquad f(\mathbf{X}) = \mathbb{E}_{\xi \sim \mathcal{D}}[f(\mathbf{X}; \xi)], \tag{1}$$

where f may be nonconvex, ξ is independent of $\mathbf{X}$, and $\mathbb{E}_\xi[\cdot]$ denotes expectation with respect to ξ. For any matrix $\mathbf{A}$ with singular values $\{\sigma_i(\mathbf{A})\}_{i=1}^r$, we further use $\mathrm{sr}(\mathbf{A}) := \frac{\|\mathbf{A}\|_F^2}{\|\mathbf{A}\|_2^2}, \kappa(\mathbf{A}) := \frac{\sigma_1(\mathbf{A})}{\sigma_r(\mathbf{A})}, \kappa_i(\mathbf{A}) := \frac{\sigma_1(\mathbf{A})}{\sigma_i(\mathbf{A})}, p_i := \frac{\sigma_i(\mathbf{A})^2}{\|\mathbf{A}\|_F^2}, H(p) := -\sum_{i=1}^r p_i \log p_i$.

Muon as finite-step orthogonalized momentum. For matrix parameters, Muon constructs the update by orthogonalizing the momentum matrix. Let $\mathbf{M}_t$ denote the momentum at iteration t. The abstract Muon update is

$$\mathbf{O}_t = \mathrm{Orth}(\widetilde{\mathbf{M}}_t), \qquad \mathbf{X}_{t+1} = \mathbf{X}_t - a\eta_t \mathbf{O}_t, \tag{2}$$

where $\widetilde{\mathbf{M}}_t$ is the standard momentum or its Nesterov variant [4, 7, 10]; following LLM Muon practice [7], we set $a = 0.2\sqrt{\max\{m, n\}}$. In practice, $\mathrm{Orth}(\widetilde{\mathbf{M}}_t)$ is implemented by a fixed number of Newton–Schulz iterations rather than exact polar decomposition [4].

Pre-orthogonalization equilibration family. MuonEq inserts diagonal preconditioning before orthogonalization:

$$\hat{\mathbf{M}}_t = \mathbf{DiagPre}(\widetilde{\mathbf{M}}_t, s, \varepsilon), \qquad \mathbf{O}_t = \mathrm{Orth}(\hat{\mathbf{M}}_t), \qquad \mathbf{X}_{t+1} = \mathbf{X}_t - a\eta_t \mathbf{O}_t, \tag{3}$$

where the mode $s \in \{\mathrm{RC}, \mathrm{R}, \mathrm{C}\}$ selects one of three equilibration forms. Let

$$\mathbf{D}_{r,t} = \mathrm{diag}\big(\mathrm{rowsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon\big), \qquad \mathbf{D}_{c,t} = \mathrm{diag}\big(\mathrm{colsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon\big).$$

We consider the following three maps:

$$\mathbf{DiagPre}(\widetilde{\mathbf{M}}_t, s, \varepsilon) = \begin{cases} \mathbf{D}_{r,t}^{-1/2} \widetilde{\mathbf{M}}_t \mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{RC}, \\ \mathbf{D}_{r,t}^{-1/2} \widetilde{\mathbf{M}}_t, & s = \mathrm{R}, \\ \widetilde{\mathbf{M}}_t \mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{C}. \end{cases} \tag{4}$$

Unlike methods that rely on matrix-valued or adaptive preconditioners [21, 22, 23, 24], **MuonEq** changes only the transient input fed to Newton–Schulz. Relative to Muon [4], the orthogonalization routine and persistent optimizer state are unchanged; relative to Muon+, NorMuon, and AdaMuon [17, 18, 19], **MuonEq** rebalances the input rather than rescales the output.

Why the default variant is R. This choice is block-dependent rather than a universal preference for row normalization. The trade-off is guided by

$$\|\mathrm{NS}_K(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F \leq \underbrace{\|\mathrm{NS}_K(S(\mathbf{M})) - \mathrm{Orth}(S(\mathbf{M}))\|_F}_{\text{finite-step approximation error}} + \underbrace{\|\mathrm{Orth}(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F}_{\text{preconditioning bias}}, \tag{5}$$

where S is one of RC, R, or C. The first term is the finite-step Newton–Schulz error and depends on the spectrum of $S(\mathbf{M})$; Theorem 3.1 therefore explains why the more aggressive two-sided RC map often gives the largest spectral correction. The second term is the bias induced by preprocessing.

2 方法

符号和设置。标量用非粗体字母表示，向量用粗体小写字母表示，矩阵用粗体大写字母表示。在 $\mathbb{R}^d$上，我们使用欧几里得内积 $\langle \mathrm{x}, \mathrm{y} \rangle_2 := \mathrm{x}^\top \mathrm{y}$和范数 $\|\mathrm{x}\|_2$。对于矩阵， $\langle \mathrm{A}, \mathrm{B} \rangle_F := \mathrm{tr}(\mathrm{A}^\top \mathrm{B})$和 $\|\mathrm{A}\|_F$表示Frobenius内积和范数， $\|\mathrm{A}\|_*$表示核范数。我们写 $[m] := \{1, 2, \ldots, m\}$并让 $\mathbb{N}$是非负整数的集合。模型参数是一个矩阵$\mathrm{X} \in \mathbb{R}^{m \times n}$，除非另有说明，我们假设 $m \geq n$。对于任何秩为 $r \leq n$的矩阵$\mathrm{A} \in \mathbb{R}^{m \times n}$，我们将其紧凑奇异值分解写为$\mathrm{A} = \mathrm{U}\Sigma\mathrm{V}^\top$，其中 $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$和 $\sigma_1 \geq \cdots \geq \sigma_r > 0$，并定义极因子，我们称之为正交化形式，为 $\mathrm{Orth}(\mathrm{A}) := \mathrm{UV}^\top$。在这里，正交化和极化步骤意味着应用或近似这个映射，而白化表示完整的Gram逆平方根预条件化。下面引入的行和列重缩放是对角均衡操作。在有限数据形式中，训练目标可以写为$f(\mathrm{X}) := \frac{1}{N} \sum_{i \in [N]} f_i(\mathrm{X})$，其中 $f_i(\mathrm{X})$是与第 i-个样本z_i相关的损失。等效地，我们使用随机公式

$$\min_{\mathbf{X} \in \mathbb{R}^{m \times n}} f(\mathbf{X}), \qquad f(\mathbf{X}) = \mathbb{E}_{\xi \sim \mathcal{D}}[f(\mathbf{X}; \xi)], \tag{1}$$

其中 f 可能是非凸的， ξ 与 X 无关， $\mathbb{E}_\xi[\cdot]$ 表示关于 ξ 的期望。对于任何具有奇异值 $\{\sigma_i(\mathrm{A})\}_{i=1}^r$ 的矩阵 A，我们进一步使用 $\mathrm{sr}(\mathrm{A}) := \frac{\|\mathrm{A}\|_F^2}{\|\mathrm{A}\|_2^2}$ A， $\kappa(\mathrm{A}) := \sigma_1(\mathrm{A})\, \sigma_r(\mathrm{A})$ ~~$\kappa_i(\mathrm{A}) := \sigma_1(\mathrm{A})\, \sigma_i(\mathrm{A})$~~ $p_i := \frac{\sigma_i(\mathrm{A})^2}{\|\mathrm{A}\|_F^2}$ A ~~, $H(p) := -\sum_{i=1}^r p_i \log p_i$~~.

Muon 作为有限步正交化的动量。 对于矩阵参数，Muon 通过正交化动量矩阵来构建更新。令 $\mathbf{M}_t$ 表示迭代 t 时的动量。抽象的 Muon 更新是

$$\mathbf{O}_t = \mathrm{Orth}(\widetilde{\mathbf{M}}_t), \qquad \mathbf{X}_{t+1} = \mathbf{X}_t - a\eta_t \mathbf{O}_t, \tag{2}$$

其中 $\widetilde{\mathbf{M}}_t$ 是标准动量或其 Nesterov 变体 [4, 7, 10]； 遵循 LLM Muon 实践 [7]，我们设置 $a = 0.2\sqrt{\max\{m, n\}}$。在实践中， $\mathrm{Orth}(\widetilde{\mathbf{M}}_t)$ 通过固定数量的 Newton–Schulz 迭代实现，而不是精确的极分解 [4]。

预正交化均衡族。MuonEq 在正交化之前插入对角预处理：

$$\hat{\mathbf{M}}_t = \mathbf{DiagPre}(\widetilde{\mathbf{M}}_t, s, \varepsilon), \qquad \mathbf{O}_t = \mathrm{Orth}(\hat{\mathbf{M}}_t), \qquad \mathbf{X}_{t+1} = \mathbf{X}_t - a\eta_t \mathbf{O}_t, \tag{3}$$

其中模式 $s \in \{\mathrm{RC}, \mathrm{R}, \mathrm{C}\}$ 选择三种均衡形式中的一种。令

$$\mathbf{D}_{r,t} = \mathrm{diag}\big(\mathrm{rowsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon\big), \qquad \mathbf{D}_{c,t} = \mathrm{diag}\big(\mathrm{colsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon\big).$$

我们考虑以下三种映射：</style>

$$\mathbf{DiagPre}(\widetilde{\mathbf{M}}_t, s, \varepsilon) = \begin{cases} \mathbf{D}_{r,t}^{-1/2} \widetilde{\mathbf{M}}_t \mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{RC}, \\ \mathbf{D}_{r,t}^{-1/2} \widetilde{\mathbf{M}}_t, & s = \mathrm{R}, \\ \widetilde{\mathbf{M}}_t \mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{C}. \end{cases} \tag{4}$$

与依赖矩阵值或自适应预处理的正交化方法不同 [21, 22, 23, 24], **MuonEq**仅改变输入到 Newton–Schulz的瞬态输入。相对于Muon [4],正交化例程和持久优化器状态保持不变；相对于Muon+，NorMuon和AdaMuon [17,18, 19], **MuonEq** 重新平衡输入，而不是缩放输出。

为什么默认变体是R。 这种选择取决于块，而不是对行归一化的普遍偏好。权衡考虑的是

$$\|\mathrm{NS}_K(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F \leq \underbrace{\|\mathrm{NS}_K(S(\mathbf{M})) - \mathrm{Orth}(S(\mathbf{M}))\|_F}_{\text{finite-step approximation error}} + \underbrace{\|\mathrm{Orth}(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F}_{\text{preconditioning bias}}, \tag{5}$$

S 是RC、R或C中的一种。第一个项是有限步Newton-Schulz误差，取决于 S（$\mathbf{M}$）的谱；因此定理3.1解释了为什么更激进的双边RC映射通常能给出最大的谱校正。第二个项是预处理引入的偏差。

Algorithm 1 MuonEq (mode $s \in \{\mathrm{RC}, \mathrm{R}, \mathrm{C}\}$; default $s = \mathrm{R}$)

Input: Initial parameters $\mathbf{X}_1 \in \mathbb{R}^{m\times n}$, learning rates $\eta_t > 0$, momentum coefficients $\{\beta_t\}_{t\geq 1} \subset [0, 1)$, weight decay $\lambda_t \geq 0$, mode $s \in \{\mathrm{RC}, \mathrm{R}, \mathrm{C}\}$, stability term $\varepsilon \geq 0$, Nesterov flag, scaling constant $a = 0.2\sqrt{\max(m, n)}$.
Initialize $\mathbf{M}_0 = \mathbf{0}$
for $t = 1$ **to** T **do**
  $\mathbf{G}_t = \nabla f(\mathbf{X}_t; \xi_t)$
  $\mathbf{M}_t = \beta_t \mathbf{M}_{t-1} + (1 - \beta_t)\mathbf{G}_t$
  if Nesterov **then**
    $\widetilde{\mathbf{M}}_t = \beta_{t+1}\mathbf{M}_t + (1 - \beta_{t+1})\mathbf{G}_t$
  else
    $\widetilde{\mathbf{M}}_t = \mathbf{M}_t$
  end if
  $\hat{\mathbf{M}}_t = \mathbf{DiagPre}(\widetilde{\mathbf{M}}_t, s, \varepsilon)$
  $\mathbf{O}_t = \mathrm{NS5}(\hat{\mathbf{M}}_t)$
  $\mathbf{X}_{t+1} = (1 - \lambda_t\eta_t)\mathbf{X}_t - a\eta_t\mathbf{O}_t$
end for

Algorithm 2 DiagPre

Input: Momentum $\widetilde{\mathbf{M}}_t$, mode $s \in \{\mathrm{RC}, \mathrm{R}, \mathrm{C}\}$, stability term $\varepsilon \geq 0$
if $s \in \{\mathrm{RC}, \mathrm{R}\}$ **then**
  $\mathbf{D}_{r,t} = \mathrm{diag}(\mathrm{rowsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon)$
end if
if $s \in \{\mathrm{RC}, \mathrm{C}\}$ **then**
  $\mathbf{D}_{c,t} = \mathrm{diag}(\mathrm{colsum}(\widetilde{\mathbf{M}}_t \odot \widetilde{\mathbf{M}}_t) + \varepsilon)$
end if

$$\hat{\mathbf{M}}_t = \begin{cases} \mathbf{D}_{r,t}^{-1/2}\widetilde{\mathbf{M}}_t\mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{RC}; \\ \mathbf{D}_{r,t}^{-1/2}\widetilde{\mathbf{M}}_t, & s = \mathrm{R}; \\ \widetilde{\mathbf{M}}_t\mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{C}. \end{cases}$$

return $\hat{\mathbf{M}}_t$

Remark 2.1. *Unless otherwise stated, we use $s = \mathrm{R}$; RC and C are retained for analysis and ablation. For varying β_t, the Nesterov line follows the EMA-buffer convention; for $\beta_t \equiv \beta$, it reduces to the Nesterov form [3, 10].*

Proposition 3.2 shows that row/column normalization is a zeroth-order surrogate for whitening, removing marginal scale mismatch before orthogonalization. Among the one-sided variants, recent operator-geometry analyses identify row-sided geometry as the natural choice for the hidden Transformer weight matrices targeted by **MuonEq**, whereas column-sided geometry is more closely associated with embedding matrices [28]. This makes R the most appropriate default in our setting. Consistent with this picture, Proposition 3.4 characterizes the pure RC regime as a stronger two-sided spectral corrector, whereas Theorem 3.5 gives the cleanest nonconvex guarantee for the row-normalized variant.

3 Analysis

The analysis follows the logic of **MuonEq**: Section 3.1 establishes the spectrum sensitivity of finite-step Newton–Schulz; Section 3.2 interprets row/column normalization as a zeroth-order surrogate for whitening; and Section 3.3 motivates the R rather than RC, with row normalization admitting a clean stochastic nonconvex guarantee. Together, these results motivate the **MuonEq** family and clarify the trade-off between stronger spectral correction and preprocessing bias. Within the hidden-weight setting studied here, they support using R as our default working variant.

3.1 Why spectral geometry matters for finite-step Newton–Schulz orthogonalization

For clarity, the theorem below is stated for the wide case $p \leq q$; the tall case follows by transposition.

Theorem 3.1. *Let $\mathbf{G} \in \mathbb{R}^{p\times q}$ have rank $r \geq 1$ and compact SVD $\mathbf{G} = \mathbf{U}\Sigma\mathbf{V}^\top$, where $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and $p \leq q$. Consider the Newton–Schulz iteration*

$$\mathbf{X}_0 = \frac{\mathbf{G}}{\|\mathbf{G}\|_F}, \quad \mathbf{X}_{k+1} = \left(a\mathbf{I}_p + b\mathbf{X}_k\mathbf{X}_k^\top + c(\mathbf{X}_k\mathbf{X}_k^\top)^2\right)\mathbf{X}_k,$$

where $\phi(s) := as + bs^3 + cs^5$ and $q(t) := a + bt + ct^2$ satisfy $a > 1, 0 < q(t) \leq a, t \in [0, 1]$. Such coefficient choices include those used in [4, 15]. Then, for each $k \geq 0$,

$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)}, \ldots, s_r^{(k)}\big)\mathbf{V}^\top, \qquad s_i^{(0)} = \frac{\sigma_i}{\|\mathbf{G}\|_F}, \qquad s_i^{(k+1)} = \phi\big(s_i^{(k)}\big).$$

Moreover,

$$\frac{1}{\sqrt{r}}\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\|_F \geq \frac{1}{\sqrt{r}}\left(\sum_{i=1}^{r}\left(1 - \frac{a^k}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}}\right)_+^2\right)^{1/2}.$$

算法1 MuonEq (模式 $s \in \{\mathrm{RC,R,C}\}$; 默认 $s = \mathrm{R}$)

输入: 初始参数$\mathbf{X}_1 \in \mathbb{R}^{m\times n}$, 学习率 $\eta_t > 0$, 动量系数 $\{\beta_t\}_{t\geq 1} \subset [0, 1)$, 权重衰减 $\lambda_t \geq 0$, 模式 $s \in \{\mathrm{RC,R,C}\}$, 稳定性项 $\varepsilon \geq 0$, Nesterov标志, 缩放常数 $a = 0.2\sqrt{\max(m, n)}$
初始化$\mathbf{M}_0 = \mathbf{0}$
for $t = 1$ **to** T **do**
$\mathbf{G}_t = \nabla f(\mathbf{X}_t; \xi_t)$
$\mathbf{M}_t = \beta_t\mathbf{M}_{t-1} + (1 - \beta_t)\mathbf{G}_t$
  如果Nesterov **则**
    $\widetilde{\mathbf{M}}_t = \beta_{t+1}\mathbf{M}_t + (1 - \beta_{t+1})\mathbf{G}_t$
  else
    $\widetilde{\mathbf{M}}_t = \mathbf{M}_t$
  end if
  $\hat{\mathbf{M}}_t = \mathbf{DiagPre}(\mathbf{M}_t, s, \varepsilon)$
  $\mathbf{O}_t = \mathrm{NS5}(\hat{\mathbf{M}}_t)$
  $\mathbf{X}_{t+1} = (1 - \lambda_t\eta_t)\mathbf{X}_t - a\eta_t\mathbf{O}_t$
结束for

算法2 DiagPre

输入: 动量 $\widetilde{\mathbf{M}}_t$, 模式 $s \in \{\mathrm{RC,R,C}\}$, 稳定性项 $\varepsilon \geq 0$
如果 $s \in \{\mathrm{RC,R}\}$ **则**
  $\mathbf{D}_{r,t} = \mathrm{diag}(\text{行和}(\mathbf{M}_t \odot \mathbf{M}_t) + \varepsilon)$
否则
如果 $s \in \{\mathrm{RC,C}\}$ **则**
  $\mathbf{D}_{c,t}$ =对角矩阵(列和($\widetilde{\mathbf{M}}_t \odot$ **模**$_t$) $+ \varepsilon$)
否则

$$\hat{\mathbf{M}}_t = \begin{cases} \mathbf{D}_{r,t}^{-1/2}\widetilde{\mathbf{M}}_t\mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{RC}; \\ \mathbf{D}_{r,t}^{-1/2}\widetilde{\mathbf{M}}_t, & s = \mathrm{R}; \\ \widetilde{\mathbf{M}}_t\mathbf{D}_{c,t}^{-1/2}, & s = \mathrm{C}. \end{cases}$$

返回 $\hat{\mathbf{M}}_t$

备注2.1。除非另有说明，我们使用$s = \mathrm{R}$; RC 和 C 保留用于分析和消融。对于变化的 β_t，*Nesterov*线遵循*EMA*缓冲约定；对于 $\beta_t \equiv \beta$，它减少为*Nesterov*形式 *[3, 10]*。

命题3.2表明行/列归一化是对白化的零阶替代，在正交化之前移除边缘尺度不匹配。在单边变体中，最近的算子几何分析确定了行侧几何是针对由MuonEq（**MuonEq**）目标隐藏Transformer权重矩阵的自然选择，而列侧几何更紧密地与嵌入矩阵 [28]相关联。这使得R成为我们设置中最合适的默认值。与此图景一致，命题3.4将纯RC状态描述为更强的双边谱校正器，而定理3.5为行归一化变体给出了最干净的非凸保证。

3 分析

分析遵循 **MuonEq** 的逻辑：第3.1节建立了有限步Newton–Schulz的谱敏感性；第3.2节将行/列归一化解释为白化的零阶替代；第3.3节说明了为什么选择R而不是RC，因为行归一化允许干净的随机非凸保证。这些结果共同支持 **MuonEq** 系列，并阐明了更强的谱校正与预处理偏差之间的权衡。在本研究中研究的隐藏权重设置中，它们支持将R作为我们的默认工作变体。

3.1 为什么谱几何对于有限步Newton–Schulz正交化很重要

为清晰起见，下面的定理针对宽情况 $p \leq q$陈述；高情况通过转置得到。**定理3.1**。 设 $\mathbf{G} \in \mathbb{R}^{p\times q}$ 的秩为 $r \geq 1$，且具有紧凑*SVD*$\mathbf{G} = \mathbf{U}\Sigma\mathbf{V}^\top$，其中 $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$，且 $\sigma_1 \geq \cdots \geq \sigma_r > 0$ 和 $p \leq q$。考虑Newton–Schulz迭代

$$\mathbf{X}_0 = \frac{\mathbf{G}}{\|\mathbf{G}\|_F}, \quad \mathbf{X}_{k+1} = \left(a\mathbf{I}_p + b\mathbf{X}_k\mathbf{X}_k^\top + c(\mathbf{X}_k\mathbf{X}_k^\top)^2\right)\mathbf{X}_k,$$

在 $\phi(s) := as + bs^3 + cs^5$ 和 $q(t) := a + bt + ct^2$ *s* 满足 $a > 1$, $0 < q(t) \leq a$, $t \in [0, 1]$. 的情况下，此类系数选择包括 *[4, 15]* 中使用的那些。然后，对于每个 $k \geq 0$,

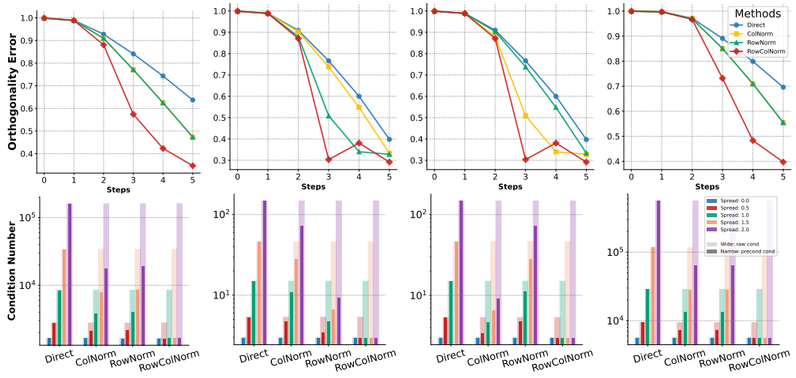
$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)}, \ldots, s_r^{(k)}\big)\mathbf{V}^\top, \qquad s_i^{(0)} = \frac{\sigma_i}{\|\mathbf{G}\|_F}, \qquad s_i^{(k+1)} = \phi\big(s_i^{(k)}\big).$$

此外，

$$\frac{1}{\sqrt{r}}\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\|_F \geq \frac{1}{\sqrt{r}}\left(\sum_{i=1}^{r}\left(1 - \frac{a^k}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}}\right)_+^2\right)^{1/2}.$$

Equivalently, the i-th singular direction leaves the linear regime at $\tau_i = \log_a\left(\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}\right)$. In particular, $\tau_1 = \log_a\sqrt{\mathrm{sr}(\mathbf{G})}, \tau_r = \log_a\left(\kappa(\mathbf{G})\sqrt{\mathrm{sr}(\mathbf{G})}\right), \tau_r - \tau_1 = \log_a\kappa(\mathbf{G})$. See Appendix B for details.

Remark 3.1. *Theorem 3.1 shows that the stable rank controls the onset of the linear to nonlinear transition, while the condition number controls its width across singular directions.*



(a) 1024×1024 (b) 1024×4096 (c) 4096×1024 (d) 4096×4096

Figure 1: Random Gaussian matrices with controlled shapes and spectral spreads. Top: finite-step relative Frobenius error to the exact polar factor across Newton–Schulz steps. Bottom: raw and post-normalization condition numbers. Two-sided row/column normalization yields the smallest error and the most consistent spectral compression.

To isolate this effect, Figure 1 uses random Gaussian matrices with controlled shapes and spectral spreads. The top row shows the finite-step error in Theorem 3.1, and the bottom row shows the corresponding condition numbers before and after normalization.

To verify the same mechanism on real training trajectories, Figure 2 uses pre-NS momentum matrices from a 2.6B-token Muon run on LLaMA2-130M [29] trained on C4 [30]. The top row shows the module-wise median of the error in Theorem 3.1, and the bottom row shows the corresponding mean. Appendix J provides a more detailed module-wise view of the same phenomenon: Figure 6 reports singular-value entropy and stable-rank diagnostics for the pre-NS momentum matrices, showing that equilibration reshapes the spectrum seen by finite-step Newton–Schulz.

3.2 Row/column normalization as a zeroth-order surrogate for whitening

The next result shows that normalization is not merely heuristic rescaling. It is the leading-order term of exact whitening/orthogonalization when the normalized Gram matrix is close to identity. Consequently, normalization removes marginal scale mismatch first and leaves only the correlation correction to the polar step.

Proposition 3.2. *The following first-order expansions hold.*

(i) Column/right whitening.

Let $\mathbf{M} \in \mathbb{R}^{p\times q}$ have full column rank, and set $\mathbf{D}_c := \mathrm{diag}(\|\mathbf{M}_{:,1}\|_2, \dots, \|\mathbf{M}_{:,q}\|_2) \succ 0$, $\mathbf{N}_c := \mathbf{M}\mathbf{D}_c^{-1}$, and $\mathbf{C}_c := \mathbf{N}_c^\top\mathbf{N}_c - \mathbf{I}_q$. If $\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2$ is sufficiently small, then $\mathrm{Orth}_r(\mathbf{M}) := \mathbf{M}(\mathbf{M}^\top\mathbf{M})^{-1/2} = \mathbf{N}_c - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}\big(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2\big)$, where $\mathbf{L}_c$ is the unique solution of $\mathbf{D}_c\mathbf{L}_c + \mathbf{L}_c\mathbf{D}_c = \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c$. In particular,

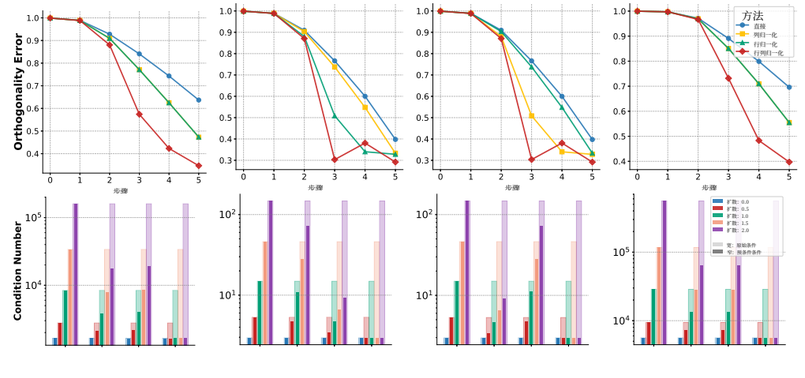
$$\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c + \mathcal{O}\big(\|\mathbf{C}_c\|_2\big).$$

(ii) Row/left whitening.

Let $\mathbf{M} \in \mathbb{R}^{p\times q}$ have full row rank, and set $\mathbf{D}_r := \mathrm{diag}(\|\mathbf{M}_{1,:}\|_2, \dots, \|\mathbf{M}_{p,:}\|_2) \succ 0$, $\mathbf{N}_r := \mathbf{D}_r^{-1}\mathbf{M}$, and $\mathbf{C}_r := \mathbf{N}_r\mathbf{N}_r^\top - \mathbf{I}_p$. If $\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2$ is sufficiently small, then $\mathrm{Orth}_\ell(\mathbf{M}) := (\mathbf{M}\mathbf{M}^\top)^{-1/2}\mathbf{M} =$

等价地，i-个奇异方向在 $\tau_i = \log_a\left(\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}\right)$ 处离开线性区域。特别是，$\tau_1 = \log_a\sqrt{\mathrm{sr}(\mathbf{G})}$，$\tau_r = \log_a\left(\kappa(\mathbf{G})\sqrt{\mathrm{sr}(\mathbf{G})}\right)$，$\tau_r - \tau_1 = \log_a\kappa(\mathbf{G})$。详情请参见附录*B*。

备注3.1。 定理*3.1*表明，稳定秩控制了非线性转换的起始，而条件数控制了其在奇异方向上的宽度。



(a) 1024×1024 (b) 1024×4096 (c) 4096×1024 (d) 4096×4096

图1：具有可控形状和谱扩散的随机高斯矩阵。顶部：Newton-Schulz步的有限步相对Frobenius误差与精确极因子的关系。底部：原始和后归一化条件数。双边行/列归一化产生最小的误差和最一致的谱压缩。

为了隔离这种效应，图1使用了具有可控形状和谱扩散的随机高斯矩阵。顶行显示了定理3.1中的有限步误差，底行显示了归一化前后的相应条件数。

为了在真实的训练轨迹上验证相同机制，图2使用了来自在LLaMA2-130M [29] 上训练的2.6B-token Muon运行的预NS动量矩阵。顶行显示了定理3.1中模块误差的中位数，底行显示了相应的均值。附录J提供了更详细的模块视图，展示了相同现象：图6报告了预NS动量矩阵的奇异值熵和稳定秩诊断，表明均衡重塑了有限步Newton-Schulz看到的谱。

3.2 行/列归一化为白化的零阶替代

下一个结果表明，归一化不仅仅是启发式重缩放。当归一化后的格拉姆矩阵接近单位矩阵时，它是精确白化/正交化的主导项。因此，归一化首先消除边缘尺度不匹配，并将相关性校正留给极化步骤。

命题3.2。 以下是一阶展开成立。

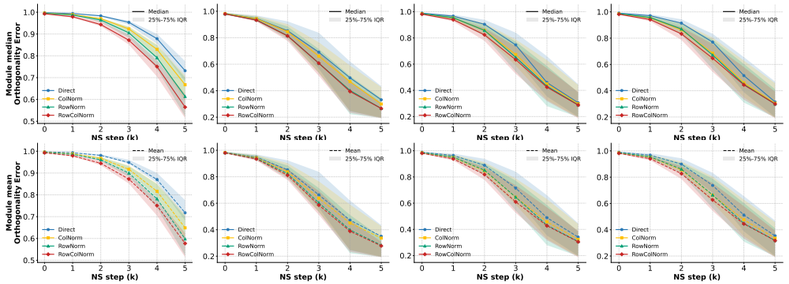
(i) 列/右白化。

设$M \in \mathbb{R}^{p\times q}$具有满列秩，并设置$D_c := diag(\|M_{:,1}\|_2, \dots, \|M_{:,q}\|_2) \succ 0$, $N_c := MD_c^{-1}$, 和 $C_c := N_c^\top N_c - I_q$。如果 $\|D_cC_cD_c\|_2$ 足够小，则 $Orth_r(M) := M(M^\top M)^{-1/2} = N_c - N_cL_cD_c^{-1} + \mathcal{O}\big(\|D_cC_cD_c\|_2^2\big)$，其中$L_c$ 是$D_cL_c + L_cD_c = D_cC_cD_c$ 的唯一解。特别地，

$$\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c + \mathcal{O}\big(\|\mathbf{C}_c\|_2\big).$$

(ii) 行/左白化。

设 $M \in \mathbb{R}^{p\times q}$ 满行秩，并定义 $D_r :=$ 对角矩阵$(\|M_{1,:}\|_2, \dots, \|M_{p,:}\|_2) \succ 0$，$N_r := D_r^{-1}M$，以及 $C_r := N_rN_r^\top - I_p$。若 $\|D_rC_rD_r\|_2$ 足够小，则 $Orth_\ell(M) := (MM^\top)^{-1/2}M =$



(a) Step 1% (b) Step 10% (c) Step 50% (d) Step 100%

Figure 2: Finite-step orthogonalization error across Newton–Schulz steps at 1%, 10%, 50%, and 100% of training. In each column, the top panel shows the module-wise median and the bottom panel shows the module-wise mean; shaded bands denote the 25%–75% range. Two-sided row/column normalization decays fastest.

$\mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}\big(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2\big)$, *where* $\mathbf{L}_r$ *is the unique solution of* $\mathbf{D}_r\mathbf{L}_r + \mathbf{L}_r\mathbf{D}_r = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$. *In particular,*

$$\mathrm{Orth}_\ell(\mathbf{M}) = \mathbf{N}_r + \mathcal{O}\big(\|\mathbf{C}_r\|_2\big).$$

See Appendix D for details.

Corollary 3.3. *Under the assumptions of the row/left part of Proposition 3.2, if* $\|\mathbf{C}_r\|_2$ *is sufficiently small, then*

$$\mathrm{Orth}_\ell(\mathbf{N}_r) = (\mathbf{I}_p + \mathbf{C}_r)^{-1/2}\mathbf{N}_r = \mathbf{N}_r - \frac{1}{2}\mathbf{C}_r\mathbf{N}_r + \mathcal{O}\big(\|\mathbf{C}_r\|_2^2\big),$$

and hence

$$\|\,\mathrm{Orth}_\ell(\mathbf{N}_r) - \mathbf{N}_r\|_2 \le \frac{1}{2}\|\mathbf{N}_r\|_2\|\mathbf{C}_r\|_2 + \mathcal{O}\big(\|\mathbf{C}_r\|_2^2\big).$$

In particular, since $\mathrm{diag}(\mathbf{N}_r\mathbf{N}_r^\top) = \mathbf{I}_p$, *row normalization removes the marginal scale mismatch, so the leading whitening correction acts only on the residual off-diagonal Gram error.*

Remark 3.2. *By contrast, the corresponding bound for* $\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M}$ *contains the additional zeroth-order term*

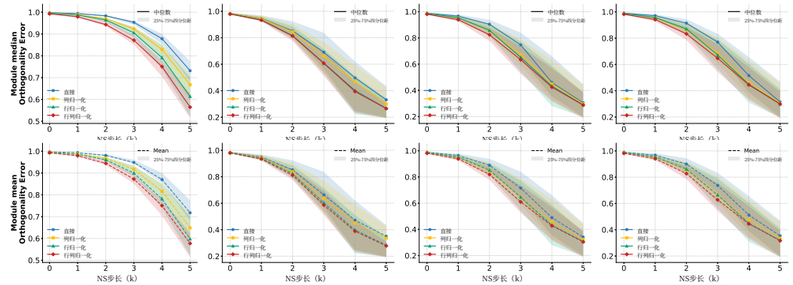
$$\|\mathbf{D}_r - \mathbf{I}_p\|_2\|\mathbf{N}_r\|_2.$$

Thus row normalization separates marginal rescaling from the genuine whitening step. The column-normalized/right analogue and the corresponding two-sided normalization statements are completely symmetric; see Appendix E.

Together, Proposition 3.2, Corollary 3.3, and Remark 3.2 explain the division of labor in **MuonEq**: normalization removes marginal-scale mismatch at zeroth order, and the subsequent Newton–Schulz orthogonalization handles the remaining correlation structure. This is exactly why **DiagPre** Eq. (4) is a meaningful lightweight surrogate for a much more expensive whitening step.

3.3 Convergence analysis of the RC and R variants

To analyze the **MuonEq** family, we focus on the two variants that are most informative theoretically: RC and the default row-normalized R. RC gives stronger two-sided spectral correction but leads to a more involved alignment term, while R has a cleaner one-sided geometry. Accordingly, Proposition 3.4 studies RC. For R, Theorem 3.5 gives the main guarantee for the exact-polar update, and Corollary 3.6 extends it to the finite-step NS5 implementation. Compared with recent Muon analyses [10, 11, 13, 15, 16, 31], our contribution is not to improve the asymptotic rate, but to establish a guarantee closer to the optimizer setting used in practice: the same $\widetilde{\mathcal{O}}(T^{-1/4})$ stationarity rate is proved while explicitly covering decoupled weight decay, a learning-rate schedule independent of the training horizon, and finite-step NS5 through the inexactness constant $\varepsilon_{\mathrm{ns}}$.



(a) 步骤1% (b) 步骤10% (c) 步骤50% (d) 步骤100%

图2：在训练的1%、10%、50%和100%处，Newton–Schulz步骤的有限步正交化误差。每一列中，顶部面板显示模块中位数，底部面板显示模块均值；阴影带表示25%–75%范围。双向行/列归一化衰减最快。

$\mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}\big(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2\big)$，其中 $\mathbf{L}_r$ 是 $\mathbf{D}_r\mathbf{L}_r + \mathbf{L}_r\mathbf{D}_r = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$ 的唯一解。特别地，

$$\mathrm{Orth}_\ell(\mathbf{M}) = \mathbf{N}_r + \mathcal{O}\big(\|\mathbf{C}_r\|_2\big).$$

详见附录*D*。

推论3.3。 在命题*3.2*的行/左部分的假设下，如果 $\|\mathbf{C}_r\|_2$ 足够小，则

$$\mathrm{Orth}_\ell(\mathbf{N}_r) = (\mathbf{I}_p + \mathbf{C}_r)^{-1/2}\mathbf{N}_r = \mathbf{N}_r - \frac{1}{2}\mathbf{C}_r\mathbf{N}_r + \mathcal{O}\big(\|\mathbf{C}_r\|_2^2\big),$$

并且因此

$$\|\,\mathrm{Orth}_\ell(\mathbf{N}_r) - \mathbf{N}_r\|_2 \le \frac{1}{2}\|\mathbf{N}_r\|_2\|\mathbf{C}_r\|_2 + \mathcal{O}\big(\|\mathbf{C}_r\|_2^2\big).$$

特别是，由于 $\mathrm{diag}(\mathbf{N}_r\mathbf{N}_r^\top) = \mathbf{I}_p$，行归一化消除了边缘尺度不匹配，因此主要的白化校正仅作用于残差对角线外*Gram*误差。

备注 3.2。 相比之下，对于 $\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M}$ 的相应界限包含额外的零阶项

$$\|\mathbf{D}_r - \mathbf{I}_p\|_2\|\mathbf{N}_r\|_2.$$

因此行归一化将边缘重缩放与真正的白化步骤分开了。列归一化/右侧类似物以及相应的双边归一化语句是完全对称的；参见附录 *E*。

Together, 命题3.2、推论3.3和注解3.2解释了**MuonEq**中的分工：归一化在零阶上消除了边际尺度不匹配，而随后的Newton–Schulz正交化处理了剩余的相关结构。这正是为什么 **DiagPre** Eq. (4)是白化步骤的一个有意义的轻量级替代品。

3.3 RC和R变种的收敛分析

为分析 **MuonEq** 系列，我们关注理论上最具有信息量的两种变体：RC 和默认的行归一化 R。RC 提供更强的双边谱校正，但会导致更复杂的对齐项，而 R 具有更简洁的单边几何结构。因此，命题3.4 研究 RC。对于 R，定理3.5 为精确极性更新提供了主要保证，而推论 3.6 将其扩展到有限步的 NS5 实现。与最近的 Muon 分析 [10, 11, 13, 15, 16, 31]，相比，我们的贡献并非改进渐近率，而是建立更接近实践中使用的优化器设置保证：证明了相同的 $\widetilde{\mathcal{O}}(T^{-1/4})$ 稳定性率，同时明确覆盖了解耦权重衰减、与训练范围无关的学习率调度，以及通过不精确性常数 $\varepsilon_{\mathrm{ns}}$ 的有限步 NS5。

Assumption 3.1. *The objective is bounded below: there exists $f^\star > -\infty$ such that $f(\mathbf{X}) \ge f^\star$ for all $\mathbf{X} \in \mathbb{R}^{m\times n}$.*

Assumption 3.2. *The function f is L-smooth, that is, $\|\nabla f(\mathbf{Y}) - \nabla f(\mathbf{X})\|_F \le L\|\mathbf{Y}-\mathbf{X}\|_F$ for all $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m\times n}$.*

Assumption 3.3. *The stochastic gradient is unbiased with bounded variance. Specifically, there exists $\sigma > 0$ such that, for all $\mathbf{X} \in \mathbb{R}^{m\times n}$, $\mathbb{E}[\nabla f(\mathbf{X};\xi)] = \nabla f(\mathbf{X}), \mathbb{E}\|\nabla f(\mathbf{X};\xi) - \nabla f(\mathbf{X})\|_F^2 \le \sigma^2$.*

These assumptions are standard in analyses of adaptive and Muon-type methods [2, 6, 32, 33, 34, 35, 36].

Proposition 3.4. *Let $\eta_t = t^{-3/4}$, $\beta_t = 1-t^{-1/2}$, $h_t := 1-\beta_t = t^{-1/2}$, and $a := 0.2\sqrt{\max\{m,n\}}$. Assume Assumptions 3.1, 3.2, and 3.3. Moreover, assume there exists $G_\infty > 0$ such that $\|\mathbf{G}_t\|_\infty \le G_\infty$ almost surely. For Algorithm 1, consider the exact-polar RC variant in* (4)*, with Nesterov momentum disabled and $\lambda_t = 0$. Suppose further that $\varepsilon \ge \frac{4}{5}G_\infty^2 \max\{m,n\}$, and define $\rho_r := \sqrt{1+\frac{nG_\infty^2}{\varepsilon}}, \rho_c := \sqrt{1+\frac{mG_\infty^2}{\varepsilon}}, \chi_\varepsilon := \frac{(\rho_r+1)(\rho_c+1)}{4\rho_r\rho_c} - \frac{3\rho_r\rho_c-\rho_r-\rho_c-1}{4}$. Then $\rho_r, \rho_c \le 3/2$ and hence $\chi_\varepsilon \ge 1/144$. Consequently,*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_1(1+\ln T) + C_2}{a\chi_\varepsilon \cdot T^{1/4}},$$

where $C_1 = \frac{a}{L}\big(2\sqrt{2}L^2a^2n + \sigma^2\big) + aL\big(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\big)^2, C_2 = \frac{a\sigma^2}{L} + \frac{3La^2n}{2}$. See Appendix G for details.

Theorem 3.5. *Let $\eta_t = t^{-3/4}$, $\beta_t = 1-t^{-1/2}$, $h_t := 1-\beta_t = t^{-1/2}$, $\varepsilon = 0$ and $a := 0.2\sqrt{\max\{m,n\}}$. Assume Assumptions 3.1, 3.2, and 3.3. For Algorithm 1, consider the exact-polar R variant in* (4)*, with Nesterov momentum disabled and $0 \le \rho < \frac{a}{\sqrt{m}}, 0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a\sqrt{n}}\, t^{-1/4}$. Then, for every $T \ge 1$,*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ep}}(1+\ln T) + C_{2,\rho}^{\mathrm{ep}}}{\big(a/\sqrt{m} - \rho\big)\cdot T^{1/4}},$$

where $C_{1,\rho}^{\mathrm{ep}} = \frac{a}{L}\Big(2\sqrt{2}L^2(a\sqrt{n}+\rho)^2 + \sigma^2\Big) + aL\Big(\frac{1}{\sqrt{m}} + \sqrt{n}\Big)^2, C_{2,\rho}^{\mathrm{ep}} = \frac{a\sigma^2}{L} + \frac{3L}{2}(a\sqrt{n}+\rho)^2$.

Corollary 3.6. *Under the assumptions of Theorem 3.5, replace the exact-polar step by the finite-step NS5 step $\mathbf{O}_t := \mathrm{NS5}(\hat{\mathbf{M}}_t)$, where $\mathrm{NS5}$ is the five-step pre-scaled Newton–Schulz map in Lemma H.2. For the fixed horizon T, we define*

$$\varepsilon_{\mathrm{ns}} := \varepsilon_{\mathrm{ns},T} = \max_{1\le t\le T}\big\|\mathrm{NS5}(\hat{\mathbf{M}}_t) - \mathrm{Orth}(\hat{\mathbf{M}}_t)\big\|_2.$$

By Lemma H.2, $\varepsilon_{\mathrm{ns}} < 1$. Replace the decoupled weight decay condition by $0 \le \rho < \frac{a(1-\varepsilon_{\mathrm{ns}})}{\sqrt{m}}, 0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}}\, t^{-1/4}$. Then, for every $T \ge 1$,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}}{\big(a(1-\varepsilon_{\mathrm{ns}})/\sqrt{m} - \rho\big)\cdot T^{1/4}},$$

where $C_{1,\rho}^{\mathrm{ns}} = \frac{a}{L}\Big(2\sqrt{2}L^2\big(a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}+\rho\big)^2 + \sigma^2\Big) + aL\Big(\frac{1-\varepsilon_{\mathrm{ns}}}{\sqrt{m}} + (1+\varepsilon_{\mathrm{ns}})\sqrt{n}\Big)^2$, $C_{2,\rho}^{\mathrm{ns}} = \frac{a\sigma^2}{L} + \frac{3L}{2}\big(a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}+\rho\big)^2$. See Appendix I for details.

Remark 3.3. *When $\varepsilon = 0$, $\mathbf{D}_{r,t}^{-1/2}$ is interpreted as $(\mathbf{D}_{r,t}^{1/2})^\dagger$, leaving zero rows unchanged. Theorem 3.5 is the main exact-polar guarantee for the default R variant. Corollary 3.6 transfers the same $T^{-1/4}$ dependence to the NS5 implementation, up to the explicit $(1\pm\varepsilon_{\mathrm{ns}})$ constants. In Algorithm 1, enabling Nesterov momentum simply amounts to replacing $\mathbf{M}_t$ with $\widetilde{\mathbf{M}}t$ in both $\mathbf{D}r,t$ and $\hat{\mathbf{M}}_t$; this should not change the stated $T^{-1/4}$ stationarity result, consistent with [6, 10].*

Remark 3.4. *Equation* (5) *separates finite-step orthogonalization error from preprocessing bias. RC is the stronger two-sided spectral corrector, but Proposition 3.4 requires the auxiliary condition $\chi_\varepsilon > 0$. By contrast, the row-normalized map satisfies*

$$\langle \mathbf{M}, \mathrm{Orth}(\mathbf{P}(\mathbf{M})\mathbf{M})\rangle \ge \frac{1}{\sqrt{m}}\|\mathbf{M}\|_F, \qquad \mathbf{P}(\mathbf{M}) := \big(\mathrm{diag}(\mathrm{rowsum}(\mathbf{M}\odot\mathbf{M}))^{1/2}\big)^\dagger,$$

假设3.1。 目标是下界有界：存在 $f^\star > -\infty$ 使得 $f(\mathbf{X}) \ge f^\star$ 对所有 $\mathbf{X} \in \mathbb{R}^{m\times n}$ 成立。

假设3.2。 函数 f 是 L-平滑的，即， $\|\nabla f(\mathbf{Y}) - \nabla f(\mathbf{X})\|_F \le L\|\mathbf{Y}-\mathbf{X}\|_F$对所有$\mathbf{X},\mathbf{Y} \in \mathbb{R}^{m\times n}$成立。

假设3.3。 随机梯度是无偏的且方差有界。具体来说，存在$\sigma > 0$使得，对所有 $\mathbf{X} \in \mathbb{R}^{m\times n}$, $\mathbb{E}[\nabla f(\mathbf{X};\xi)] = \nabla f(\mathbf{X})$ $\mathbb{E}\|\nabla f(\mathbf{X};\xi) - \nabla f(\mathbf{X})\|_F^2 \le \sigma^2$.

这些假设在自适应和Muon型方法的分析中是标准的 [2, 6, 32, 33, 34, 35,36]。

命题3.4。 设 $\eta_t = t^{-3/4}$, $\beta_t = 1-t^{-1/2}$, $h_t := 1-\beta_t = t^{-1/2}$, 且 $a := 0.2\sqrt{\max\{m,n\}}$。假设假设*3.1*、*3.2*和*3.3*成立。此外，假设存在 $G_\infty > 0$ 使得 $\|\mathbf{G}_t\|_\infty \le G_\infty$ 几乎必然成立。对于算法*1*，考虑 (4) 中的精确极化*RC*变体，禁用*Nesterov*动量且 $\lambda_t = 0$。进一步假设 $\varepsilon \ge \frac{4}{5}G_\infty^2\max\{m,n\}$，并定义 $\rho_r := \sqrt{1+\frac{nG_\infty^2}{\varepsilon}}, \rho_c := \sqrt{1+\frac{mG_\infty^2}{\varepsilon}}, \chi_\varepsilon := \frac{(\rho_r+1)(\rho_c+1)}{4\rho_r\rho_c} - \frac{3\rho_r\rho_c-\rho_r-\rho_c-1}{4}$ 然后 $\rho_r, \rho_c \le 3/2$ 并且因此 $\chi_\varepsilon \ge 1/144$。因此，

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_1(1+\ln T) + C_2}{a\chi_\varepsilon \cdot T^{1/4}},$$

where $C_1 = \frac{a}{L}\big(2\sqrt{2}L^2a^2n + \sigma^2\big) + aL\big(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\big)^2, C_2 = \frac{a\sigma^2}{L} + \frac{3La^2n}{2}$. 参见附录*G*了解详情。

定理3.5。 令 $\eta_t = t^{-3/4}$, $\beta_t = 1-t^{-1/2}$, $h_t := 1-\beta_t = t^{-1/2}$, $\varepsilon = 0$ 和 $a := 0.2\sqrt{\max\{m,n\}}$。假设假设*3.1*、*3.2*和*3.3*。对于算法*1*，考虑在(4)中的精确极化*R*变体，禁用*Nesterov*动量以及 $0 \le \rho < \frac{a}{\sqrt{m}}, 0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a\sqrt{n}}\, t^{-1/4}$。然后，对于每一个 $T \ge 1$,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ep}}(1+\ln T) + C_{2,\rho}^{\mathrm{ep}}}{\big(a/\sqrt{m} - \rho\big)\cdot T^{1/4}},$$

模式$C_{1,\rho}^{\mathrm{ep}} = \frac{a}{L}\Big(2\sqrt{2}L^2(a\sqrt{n}+\rho)^2 + \sigma^2\Big) + aL\Big(\frac{1}{\sqrt{m}} + \sqrt{n}\Big)^2, C_{2,\rho}^{\mathrm{ep}} = \frac{a\sigma^2}{L} + \frac{3L}{2}(a\sqrt{n}+\rho)^2$.
whe

推论3.6。 在定理*3.5*的假设下，将精确极化步骤替换为有限步^*NS5*步骤$\mathbf{O}_t := \mathrm{NS5}(\mathbf{M}_t)$，其中 $\mathrm{NS5}$ 是引理*H.2*中的五步预缩放牛顿-舒尔茨映射。对于固定的视界 T，我们定义

$$\varepsilon_{\mathrm{ns}} := \varepsilon_{\mathrm{ns},T} = \max_{1\le t\le T}\big\|\mathrm{NS5}(\hat{\mathbf{M}}_t) - \mathrm{Orth}(\hat{\mathbf{M}}_t)\big\|_2.$$

根据引理*H.2*, $\varepsilon_{\mathrm{ns}} < 1$。将解耦权重衰减条件替换为 $0 \le \rho < \frac{a(1-\varepsilon_{\mathrm{ns}})}{\sqrt{m}}, 0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}}\, t^{-1/4}$. 然后，对于每个 $T \ge 1$,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}}{\big(a(1-\varepsilon_{\mathrm{ns}})/\sqrt{m} - \rho\big)\cdot T^{1/4}},$$

在 $C_{1,\rho}^{\mathrm{ns}} = \frac{a}{L}\Big(2\sqrt{2}L^2\big(a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}+\rho\big)^2 + \sigma^2\Big) + aL\Big(\frac{1-\varepsilon_{\mathrm{ns}}}{\sqrt{m}} + (1+\varepsilon_{\mathrm{ns}})\sqrt{n}\Big)^2$, $C_{2,\rho}^{\mathrm{ns}} = \frac{a\sigma^2}{L} + \frac{3L}{2}\big(a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}+\rho\big)^2$。详情请参见附录*I*。

备注3.3。 当 $\varepsilon = 0$, $\mathbf{D}_{r,t}^{-1/2}$ 被解释为 $(\mathbf{D}_{r,t}^{1/2})^\dagger$，并保留零行不变。定理*3.5*是默认*R*变体的主要精确极化保证。推论*3.6*将相同的 $T^{-1/4}$ 依赖关系转移到*NS5*实现中，直到显式的 $(1\pm\varepsilon_{\mathrm{ns}})$ 常数。在算法*1*中， ^启用*Nesterov*动量只需将 $\mathbf{M}_t$ 替换为 $\mathbf{M}t$ 在 $\mathbf{D}r,t$ 和 $\mathbf{M}_t$ 中；这不应改变所声明的 $T^{-1/4}$ 稳定性结果，这与 *[6, 10]* 一致。

备注3.4。 方程<code id='3'>(5)</code>将有限步正交化误差与预处理偏差分离。*RC*是更强的双边谱校正器，但命题*3.4*需要辅助条件$\chi_\varepsilon > 0$。相比之下，行归一化映射满足

$$\langle \mathbf{M}, \mathrm{Orth}(\mathbf{P}(\mathbf{M})\mathbf{M})\rangle \ge \frac{1}{\sqrt{m}}\|\mathbf{M}\|_F, \qquad \mathbf{P}(\mathbf{M}) := \big(\mathrm{diag}(\mathrm{rowsum}(\mathbf{M}\odot\mathbf{M}))^{1/2}\big)^\dagger,$$

so Theorem 3.5 needs no χ_ε-type condition, and Corollary 3.6 only degrades this coefficient by the explicit factor $1-\varepsilon_{\mathrm{ns}}$. Hence the default R variant obeys

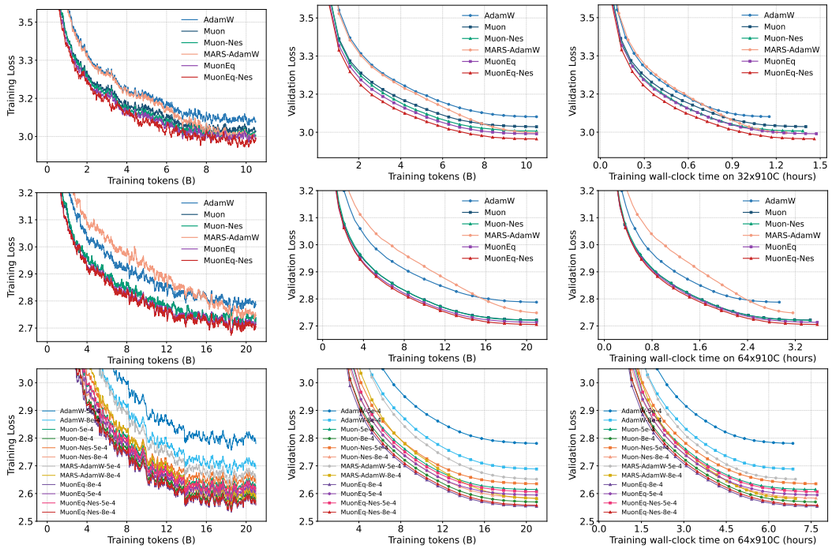
$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F = \mathcal{O}\left(\frac{1+\ln T}{T^{1/4}}\right),$$

matching the standard Muon-type baseline up to logarithmic [6, 10, 15, 16] and constant factors, while C is retained as the embedding-aligned companion and studied empirically [28]. Improvements beyond this baseline typically require additional mechanisms such as variance reduction [3, 10, 37, 38, 39, 40, 41, 42].

4 Experiments

4.1 Main Results

In this section, we evaluate the performance of the proposed **MuonEq** optimizers on LLM pretraining. We use the same 8-node Ascend 910C cluster for all LLaMA experiments; 130M uses 32 NPUs, whereas 350M and 1B use 64 NPUs. Detailed experimental settings are provided in Appendix K. We will refer to the **MuonEq** with Nesterov acceleration enabled as **MuonEq**-Nes.



(a) Training Loss (b) Validation Loss (c) Wall-clock Time

Figure 3: Comparison of LLaMA2 130M, 350M, and 1B from three perspectives. In each column, the top panel corresponds to the 130M model, the middle panel corresponds to the 350M model, and the bottom panel corresponds to the 1B model. From left to right, the panels show training loss versus tokens, validation loss versus tokens, and validation loss versus wall-clock time.

► **LLaMA2 on C4 Dataset.** We compare **MuonEq/MuonEq**-Nes with AdamW, MARS-AdamW, Muon, and Muon-Nes on LLaMA2-130M, 350M and 1B trained on C4. In the main comparison, 130M is trained on 10.5B tokens, while 350M and 1B are trained on 21.0B tokens. Learning rates are selected through shorter pilot runs using search grids $\{5e{-}4, 1e{-}3, 2e{-}3, 3e{-}3, 5e{-}3, 8e{-}3, 1e{-}2\}$ for 130M, $\{5e{-}4, 1e{-}3, 1.5e{-}3, 2e{-}3, 3e{-}3, 5e{-}3\}$ for 350M, and $\{5e{-}4, 8e{-}4\}$ for 1B. Appendix K. Figure 3 shows training loss versus tokens, validation loss versus tokens, and validation loss versus wall-clock time for all three model sizes, while Table 1 summarizes the best validation

因此定理3.5不需要 χ_ε-类型条件，而推论3.6仅通过显式因子 $1-\varepsilon_{\mathrm{ns}}$降低了该系数。因此默认的R变体遵循

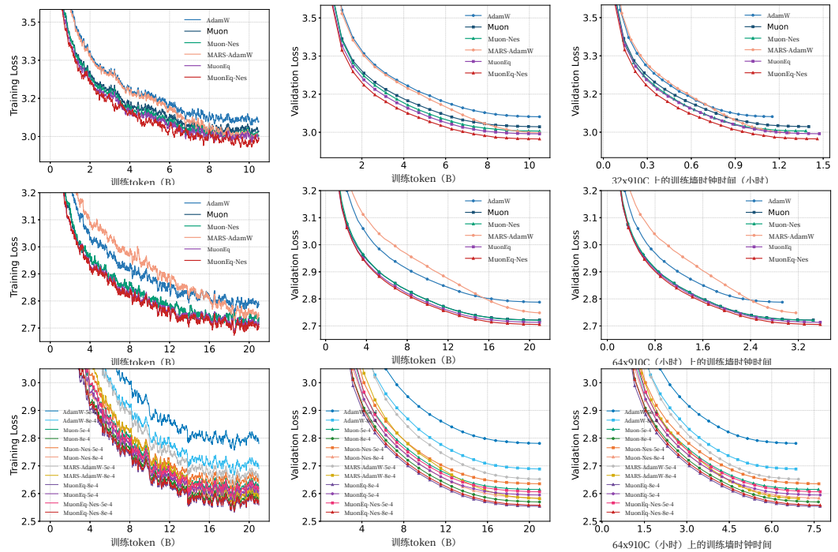
$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F = \mathcal{O}\left(\frac{1+\ln T}{T^{1/4}}\right),$$

与标准的 Muon 类型基线匹配，直到对数 [6, 10, 15, 16] 和常数因子，同时保留 C 作为嵌入对齐的同伴并进行经验研究 [28]。超越此基线的改进通常需要额外的机制，例如方差减少 [3, 10, 37,38, 39, 40, 41, 42]。

4 实验

4.1 主要结果

在本节中，我们评估了所提出的**MuonEq**优化器在LLM预训练上的性能。我们为所有LLaMA实验使用相同的8节点Ascend 910C集群；130M使用32个NPU，而350M和1B使用64个NPU。详细的实验设置在附录K中提供。我们将启用Nesterov加速的**MuonEq** 称为**MuonEq**-Nes。



(a) 训练损失 (b) 验证损失 (c) 墙上时间

图3：从三个角度对LLaMA2 130M、350M和1B的比较。每一列中，顶部面板对应130M模型，中间面板对应350M模型，底部面板对应1B模型。从左到右，面板分别显示训练损失与token的关系、验证损失与token的关系以及验证损失与墙上时间的关系。

► **LLaMA2 在 C4 数据集上。** 我们在 LLaMA2-130M、350M 和 1B 上比较 **MuonEq/MuonEq**-Nes 与 AdamW、MARS-AdamW、Muon 和 Muon-Nes。在主要比较中，130M 是在 10.5B 个 token 上训练的，而 350M 和 1B 是在 21.0B 个 token 上训练的。学习率是通过使用搜索网格 $\{5e{-}4,\ 1e{-}3,2e{-}3,\ 3e{-}3,\ 5e{-}3,\ 8e{-}3,\ 1e{-}2\}$for 130M，$\{5e{-}4,\ 1e{-}3,1.5e{-}3,\ 2e{-}3,\ 3e{-}3,\ 5e{-}3\}$ for 350M，和 $\{5e{-}4,\ 8e{-}4\}$ for 1B。附录 K。图 3 显示了所有三种模型大小的训练损失与 token 、验证损失与 token 以及验证损失与墙上时间，而表 1 总结了最佳验证

perplexity, peak memory usage (GB), and training time (s/step). Overall, **MuonEq/MuonEq**-Nes consistently outperforms Muon/Muon-Nes.

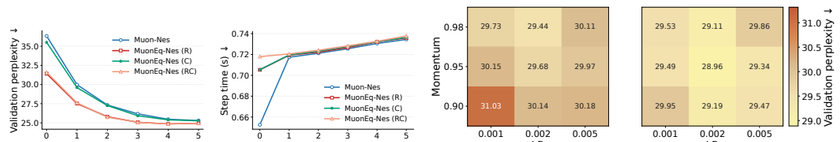
Table 1: Best validation perplexity, peak memory usage, and wall-clock training time on LLaMA2/C4.

Optimizer	130M			350M			1B		
	Perplexity	Memory	Time	Perplexity	Memory	Time	Perplexity	Memory	Time
AdamW [1]	21.35	12.94	0.21	16.25	29.43	0.53	14.71	39.56	2.47
MARS-AdamW [3]	20.03	12.99	0.23	15.62	29.47	0.57	13.22	39.57	2.50
Muon [4, 7]	20.52	12.52	0.25	15.21	28.18	0.61	13.06	34.81	2.76
Muon-Nes [4, 7, 10]	20.18	12.52	0.25	15.19	28.18	0.61	13.24	34.81	2.76
MuonEq	**19.96**↓0.56	12.52	0.26	**15.07**↓0.14	28.18	0.63	**12.86**↓0.20	34.81	2.78
MuonEq-Nes	**19.56**↓0.62	12.52	0.26	**14.96**↓0.23	28.18	0.63	**12.91**↓0.33	34.81	2.78
Training Tokens	10.5B			21.0B			21.0B		

4.2 Ablation Study

We ablate the three static **MuonEq**-Nes variants, RC, R, and C, to isolate the effect of pre-orthogonalization geometry. All ablations run on 4 RTX Pro6000 (96GB) GPUs. Detailed experimental settings are provided in Appendix K.2.

Tables 2 and 3 show that, with results averaged over three random seeds, R is consistently the strongest static variant across both benchmarks, achieving higher test accuracy than Muon-Nes on CIFAR-10 with ResNet-18 and lower validation perplexity on FineWeb-10.5B tokens with GPT2-small.



(a) Validation Perplexity (b) Step Time (c) Muon-Nes | K=5 (d) MuonEq-Nes (R) | K=5

Figure 4: (a,b) Validation perplexity and step time vs. Newton–Schulz iterations K on GPT2-small/FineWeb (10.5B tokens); (c,d) validation perplexity heatmaps over learning rate and momentum at $K = 5$ (2.6B tokens).

Table 2: Ablation on CIFAR-10 with ResNet-18.

Method	Component		Test Acc. (%)
	Row	Col	
Baseline Experiments			
SGD			$92.82_{\pm 0.04}$
AdamW			$93.36_{\pm 0.10}$
Muon			$94.39_{\pm 0.04}$
Muon-Nes			$94.44_{\pm 0.05}$
Ablation Experiments			
MuonEq-Nes (RC)	✓	✓	$94.29_{\pm 0.10}$
MuonEq-Nes (C)		✓	$94.45_{\pm 0.02}$
MuonEq-Nes (R)	✓		**94.57**$_{\pm 0.07}$ ↑0.13

Table 3: Ablation on FineWeb with GPT2-small.

Method	Component		Validation Perplexity
	Row	Col	
Baseline Experiments			
AdamW			$26.48_{\pm 0.12}$
Muon-Nes			$25.23_{\pm 0.08}$
AdaMuon			$24.99_{\pm 0.13}$
Muon+			$25.12_{\pm 0.08}$
Mousse			$25.16_{\pm 0.10}$
Ablation Experiments			
MuonEq-Nes (RC)	✓	✓	$24.94_{\pm 0.04}$
MuonEq-Nes (C)		✓	$25.23_{\pm 0.06}$
MuonEq-Nes (R)	✓		**24.88**$_{\pm 0.02}$ ↓0.35

Figure 4a and 4b show that the default R variant of **MuonEq**-Nes is markedly less sensitive to the NS iteration count K than Muon-Nes: its validation perplexity remains stable over a wider range of K, while step time stays comparable throughout the sweep. Complementary robustness evidence is given in Figure 4c and 4d, where **MuonEq**-Nes (R) exhibits a broader low-perplexity region over learning-rate and momentum sweeps than Muon-Nes, especially at $K = 5$. This pattern is consistent with Eq. (5) and Section 3. RC applies stronger two-sided spectral correction and can reduce finite-step NS error more aggressively, but at the cost of larger preconditioning bias. For the hidden-weight settings we tested, R provides a favorable trade-off between spectral correction

困惑度、峰值内存使用量（GB）和训练时间（每步）。总体而言，**MuonEq**/**MuonEq**-Nes始终优于Muon/Muon-Nes。

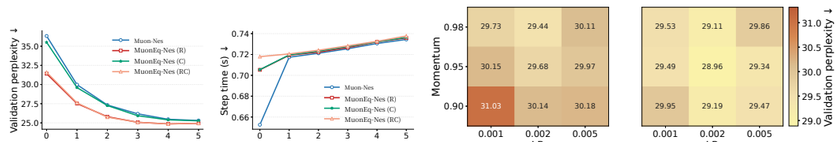
表1：在LLaMA2/C4上最佳验证困惑度、峰值内存使用和墙钟训练时间。

优化器	130M			350M			1B		
	困惑度	内存	Time	困惑度	内存	Time	困惑度	内存	Time
AdamW [1]	21.35	12.94	0.21	16.25	29.43	0.53	14.71	39.56	2.47
MARS-AdamW [3]	20.03	12.99	0.23	15.62	29.47	0.57	13.22	39.57	2.50
Muon [4, 7]	20.52	12.52	0.25	15.21	28.18	0.61	13.06	34.81	2.76
Muon-Nes [4, 7, 10]	20.18	12.52	0.25	15.19	28.18	0.61	13.24	34.81	2.76
MuonEq	**19.96**↓0.56	12.52	0.26	**15.07**↓0.14	28.18	0.63	**12.86**↓0.20	34.81	2.78
MuonEq-Nes	**19.56**↓0.62	12.52	0.26	**14.96**↓0.23	28.18	0.63	**12.91**↓0.33	34.81	2.78
训练令牌	10.5B			21.0B			21.0B		

4.2 消融研究

我们消融了三种静态 **MuonEq**-Nes 变体，RC、R 和 C，以隔离预正交化几何形状的影响。所有消融实验都在 4 块 RTX Pro6000（96GB）GPU 上运行。详细的实验设置在附录 K.2 中提供。

表2和表3显示，在三个随机种子上平均的结果中，R始终是两个基准测试中最强的静态变体，在CIFAR-10上使用ResNet-18时测试精度高于Muon-Nes，在FineWeb-10.5B token上使用GPT2-small时验证困惑度更低。



(a) 验证困惑度 (b) 步骤时间 (c) Muon-Nes | K=5 (d) MuonEq-Nes (R) | K=5

图4： (a,b) 在GPT2-small/FineWeb (10.5B token) 上Newton–Schulz迭代 K 时的验证困惑度和步骤时间；(c,d) 在 $K = 5$ (2.6B token) 处学习率和动量的验证困惑度热图。

表2：在CIFAR-10上对ResNet-18进行消融实验。

方法	组件		测试准确率 (%)
	Row	Col	
基线实验			
SGD			$92.82_{\pm 0.04}$
AdamW			$93.36_{\pm 0.10}$
Muon			$94.39_{\pm 0.04}$
Muon-Nes			$94.44_{\pm 0.05}$
消融实验			
MuonEq-Nes (RC)	✓	✓	$94.29_{\pm 0.10}$
MuonEq-Nes (C)		✓	$94.45_{\pm 0.02}$
MuonEq-Nes (R)	✓		**94.57**$_{\pm 0.07}$ ↑0.13

表3：在FineWeb上对GPT2-small进行消融实验。

方法	组件		验证困惑度
	Row	Col	
基线实验			
AdamW			$26.48_{\pm 0.12}$
Muon-Nes			$25.23_{\pm 0.08}$
AdaMuon			$24.99_{\pm 0.13}$
Muon+			$25.12_{\pm 0.08}$
Mousse			$25.16_{\pm 0.10}$
消融实验			
MuonEq-Nes (RC)	✓	✓	$24.94_{\pm 0.04}$
MuonEq-Nes (C)		✓	$25.23_{\pm 0.06}$
MuonEq-Nes (R)	✓		**24.88**$_{\pm 0.02}$ ↓0.35

图4a和4b显示，**MuonEq**-Nes的默认R变体对NS迭代次数 K 的敏感度明显低于Muon-Nes：其验证困惑度在更宽的范围内保持稳定，而步长时间在整个扫描过程中保持相当。补充的鲁棒性证据在图4c和4d中给出，其中MuonEq-Nes (R)在学习率和动量扫描中表现出比Muon-Nes更广泛的低困惑度区域，尤其是在 $K = 5$处。这种模式与公式(5)和第3节一致。RC应用更强的双边谱校正，可以更激进地减少有限步NS误差，但代价是更大的预条件偏差。对于我们测试的隐藏权重设置，R在谱校正和预处理偏差之间提供了有利权衡，这与一致。与和 K 的比较需要谨慎，因为行/列项因布局和任务而异；参见附录K.3。

and preprocessing bias, consistent with [28]. Comparisons with [43] and [44] require caution, as row/column terms vary by layout and task; see Appendix K.3.

5 Conclusion

We improve the matrix geometry encountered by finite-step orthogonalization in Muon-style optimization without resorting to full whitening. **MuonEq** reshapes the input to Newton–Schulz using lightweight pre-orthogonalization equilibration in three forms, RC, R, and C; R is used by default for hidden weights. Theoretically, these schemes serve as zeroth-order whitening surrogates: RC provides stronger two-sided correction, whereas R yields cleaner one-sided geometry and supports the main $\widetilde{\mathcal{O}}(T^{-1/4})$ guarantee. Empirically, R consistently outperforms Muon in LLaMA2 pretraining on C4 across all tested scales and budgets.

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和预处理偏差，这与 [28]一致。与 [43] 和 [44] 的比较需要谨慎，因为行/列项因布局和任务而异；参见附录K.3。

5 结论

我们改进了 Muon 风格优化中有限步正交化遇到的矩阵几何，而无需进行完全白化。**MuonEq** 使用轻量级预正交化均衡将输入重塑为 Newton–Schulz，形式包括 RC、R 和 C；默认情况下，R 用于隐藏权重。理论上，这些方案作为零阶白化替代方案：RC 提供更强的双边校正，而 R 产生更干净的单边几何，并支持主要的 $\widetilde{\mathcal{O}}(T^{-1/4})$ 保证。实践中，R 在所有测试规模和预算下，始终在 C4 上的 LLaMA2 预训练中优于 Muon。

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Appendix

A Related Work

MuonEq is viewed as a lightweight pre-balanced Muon family rather than as a fully whitened optimizer. It inserts diagonal equilibration before Newton–Schulz orthogonalization and includes three forms: two-sided row/column normalization (RC), row normalization (R), and column normalization (C), with R as the default choice for the hidden matrix weights considered in this paper. All three compute row/column squared norms from the current momentum and add no persistent optimizer state beyond Muon.

Muon and orthogonalized-update optimizers. Muon [4] introduced the core design of orthogonalizing matrix-valued momentum with a small fixed number of Newton–Schulz iterations. Subsequent work has expanded this idea into a broader family of orthogonalized-update optimizers, including post-orthogonalization normalization and rescaling methods such as Muon+, NorMuon, and AdaMuon [17, 18, 19]; adaptive stepsize variants such as AdaGO and NAMO [45, 46]; distributed and communication-aware extensions such as Dion and MuLoCo [47, 48]; reduced-state implementations [49]; and fine-tuning-oriented variants [50]. These methods mainly alter the orthogonalized update itself or the surrounding optimization machinery. By contrast, **MuonEq** modifies the matrix fed into orthogonalization through lightweight pre-orthogonalization equilibration.

Theory of Muon-style methods. Recent theory interprets Muon as steepest descent under spectral-norm or nuclear-norm geometry [5, 6], establishes stochastic and nonconvex convergence guarantees [13, 10, 31, 51], and analyzes practical ingredients such as weight decay, critical batch size, and inexact orthogonalization [11, 15, 16]. This literature is directly relevant to **MuonEq** because we retain the same finite-step orthogonalization primitive as Muon and change only the geometry of its input. Our analysis further highlights the role of input spectrum in finite-step Newton–Schulz and clarifies how diagonal equilibration trades stronger spectral correction against preprocessing bias.

Structured preconditioning, factorized statistics, and whitening-inspired methods. A separate line of work studies matrix-aware preconditioning, including Shampoo [21], K-FAC [52, 53], SOAP [20], Sketchy [54], and ASGO [55]. Adafactor [22] is especially relevant from a systems perspective because it shows how row- and column-factorized second-moment statistics reduce optimizer state to $\mathcal{O}(m+n)$ for matrix-valued parameters. Recent Muon-adjacent whitening or preconditioned-polar methods, including FISMO and Mousse [24, 25], pursue a similar high-level goal: improve the geometry before imposing spectral constraints. The key difference is that these methods rely on richer matrix- or Kronecker-structured preconditioners, whereas **MuonEq** restricts the intervention to diagonal row/column equilibration. In this sense, **MuonEq** is closer to low-cost equilibration than to full whitening.

Orthogonalization algorithms. There is also a complementary numerical literature on faster and more accurate polar/orthogonalization routines, including improved Newton–Schulz variants and matrix-sign methods such as CANS and Polar Express [56, 57]. This literature matters because Muon-style optimizers operate in the regime of approximate orthogonalization, where only a small number of iterations is affordable; **MuonEq** is designed explicitly for this setting.

B Proofs of Theorem 3.1

Proof. Let $\alpha > 0$. For $k = 0$, the desired form follows directly from

$$\mathbf{X}_0 = \alpha^{-1}\mathbf{G} = \mathbf{U}\,\mathrm{diag}\Big(\frac{\sigma_1}{\alpha}, \ldots, \frac{\sigma_r}{\alpha}\Big)\mathbf{V}^\top.$$

Assume that, for some $k \geq 0$,

$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)}, \ldots, s_r^{(k)}\big)\mathbf{V}^\top.$$

Then

$$\mathbf{X}_k\mathbf{X}_k^\top = \mathbf{U}\,\mathrm{diag}\big((s_1^{(k)})^2, \ldots, (s_r^{(k)})^2\big)\mathbf{U}^\top.$$

Inserting this expression into the iteration and using $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_r$ gives

$$\mathbf{X}_{k+1} = \mathbf{U}\,\mathrm{diag}\big(\phi(s_1^{(k)}), \ldots, \phi(s_r^{(k)})\big)\mathbf{V}^\top.$$

附录

相关工作

MuonEq 被视为一种轻量级的预平衡 Muon 系列，而不是一个完全白化的优化器。它在 Newton–Schulz 正交化之前插入对角均衡，并包含三种形式：双边行/列归一化（RC）、行归一化（R）和列归一化（C），其中 R 是本文考虑的隐藏矩阵权重默认选择。这三种方法都从当前动量计算行/列平方范数，并且除了 Muon 之外，不会添加任何持久的优化器状态。

Muon 和正交化更新优化器。 Muon [4] 引入了正交化矩阵值动量的核心设计，使用少量固定的 Newton–Schulz 迭代。后续工作将这一思想扩展到更广泛的正交化更新优化器系列，包括正交化后归一化和重缩放方法，如 Muon+、NorMuon 和 AdaMuon [17, 18, 19]；自适应步长变体，如 AdaGO 和 NAMO [45, 46]；分布式和感知通信的扩展，如 Dion 和 MuLoCo [47, 48]；减少状态实现 [49]；以及面向微调的变体 [50]。这些方法主要改变正交化更新本身或周围的优化机制。相比之下，**MuonEq** 通过轻量级的预正交化均衡修改输入到正交化的矩阵。

Muon风格方法的理论。 近期理论将Muon解释为在谱范数或核范数几何下的最速下降方法 [5, 6], 建立了随机和非凸收敛保证 [13, 10, 31, 51], 并分析了权重衰减、临界批大小和不精确正交化等实际要素 [11, 15, 16]。这部分文献与**MuonEq** 直接相关因为我们保留了与Muon相同的有限步正交化原语而仅改变了其输入的几何结构。我们的分析进一步突出了输入谱在有限步Newton-Schulz中的作用并阐明了对角均衡如何通过更强的谱校正来权衡预处理偏差。

结构化预处理、分解统计和白化启发式方法。 另一条研究线关注矩阵感知预处理包括 Shampoo [21], K-FAC [52, 53],SOAP [20], Sketchy [54], 和ASGO [55]。Adafactor [22] 从系统角度看特别相关因为它展示了如何通过行和列分解的第二矩统计将优化器状态简化为 $\mathcal{O}(m+n)$用于矩阵值参数。近期的Muon相关白化或预处理的极化方法包括FISMO和 Mousse [24, 25], 追求类似的高级目标：在施加谱约束前改善几何结构。关键区别在于这些方法依赖更丰富的矩阵或Kronecker结构预处理器而**MuonEq** 将干预限制在对角行/列均衡上。从这个意义上说，**MuonEq** 更接近低成本均衡而非完全白化。

正交化算法。 此外，也存在关于更快、更精确的极/正交化例程的补充数值文献，包括改进的Newton–Schulz变体和矩阵符号方法，如CANS和Polar Express [56, 57]。这些文献很重要，因为Muon风格的优化器在近似正交化的条件下运行，其中只能承受少量迭代；**MuonEq** 是专门为此设置设计的。

B 定理3.1的证明

证明。 令 $\alpha > 0$。对于 $k = 0$，所需形式直接来自

$$\mathbf{X}_0 = \alpha^{-1}\mathbf{G} = \mathbf{U}\,\mathrm{diag}\Big(\frac{\sigma_1}{\alpha}, \ldots, \frac{\sigma_r}{\alpha}\Big)\mathbf{V}^\top.$$

假设对于某个 $k \geq 0$,

$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)}, \ldots, s_r^{(k)}\big)\mathbf{V}^\top.$$

Then

$$\mathbf{X}_k\mathbf{X}_k^\top = \mathbf{U}\,\mathrm{diag}\big((s_1^{(k)})^2, \ldots, (s_r^{(k)})^2\big)\mathbf{U}^\top.$$

将此表达式代入迭代并使用 $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_r$ 得到

$$\mathbf{X}_{k+1} = \mathbf{U}\,\mathrm{diag}\big(\phi(s_1^{(k)}), \ldots, \phi(s_r^{(k)})\big)\mathbf{V}^\top.$$

By induction,

$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)},\dots,s_r^{(k)}\big)\mathbf{V}^\top, \qquad s_i^{(k+1)} = \phi\left(s_i^{(k)}\right).$$

We also have

$$\mathrm{Orth}(\mathbf{G}) = \mathbf{U}\mathbf{V}^\top = \mathbf{U}\mathbf{I}_r\mathbf{V}^\top, \quad \mathbf{X}_k - \mathrm{Orth}(\mathbf{G}) = \mathbf{U}\Big(\mathrm{diag}(s_1^{(k)},\dots,s_r^{(k)}) - \mathbf{I}_r\Big)\mathbf{V}^\top.$$

Left and right multiplication by matrices with orthonormal columns preserves the Frobenius norm, so

$$\big\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\big\|_F^2 = \sum_{i=1}^{r}(s_i^{(k)} - 1)^2.$$

This proves the exact error representation.

We now establish the lower bound. Recall that

$$q(t) := a + bt + ct^2.$$

By assumption,

$$0 < q(t) \le a, \qquad t \in [0,1].$$

Fix $i \in [r]$ and $k \ge 0$. If $a^k\frac{\sigma_i}{\alpha} \ge 1$, then

$$\big|1 - s_i^{(k)}\big| \ge 0 = \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

Now suppose that $a^k\sigma_i/\alpha < 1$. We claim that, for every $0 \le j \le k$,

$$0 < s_i^{(j)} \le a^j\frac{\sigma_i}{\alpha} < 1.$$

The statement is immediate for $j = 0$. Suppose it holds for some $j < k$. Then $s_i^{(j)} \in (0,1)$, and hence $(s_i^{(j)})^2 \in (0,1)$. Therefore,

$$0 < q\big((s_i^{(j)})^2\big) \le a.$$

It follows that

$$0 < s_i^{(j+1)} = s_i^{(j)}q\big((s_i^{(j)})^2\big) \le as_i^{(j)} \le a^{j+1}\frac{\sigma_i}{\alpha}.$$

Because $j + 1 \le k$,

$$a^{j+1}\frac{\sigma_i}{\alpha} \le a^k\frac{\sigma_i}{\alpha} < 1.$$

This proves the claim. In particular,

$$0 < s_i^{(k)} \le a^k\frac{\sigma_i}{\alpha} < 1,$$

and thus

$$|1 - s_i^{(k)}| = 1 - s_i^{(k)} \ge 1 - a^k\frac{\sigma_i}{\alpha} = \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

Combining the two cases gives, for every i,

$$|1 - s_i^{(k)}| \ge \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

Finally, set $\alpha = \|\mathbf{G}\|_F$. Squaring both sides, summing over i, and using the exact error representation together with

$$\frac{\sigma_i}{\|\mathbf{G}\|_F} = \frac{\sigma_i/\sigma_1}{\|\mathbf{G}\|_F/\sigma_1} = \frac{1}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}},$$

we obtain

$$\frac{1}{\sqrt{r}}\big\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\big\|_F \ge \frac{1}{\sqrt{r}}\left(\sum_{i=1}^{r}\left(1 - \frac{a^k}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}}\right)_+^2\right)^{1/2}.$$

Moreover,

$$\tau_i := \log_a\frac{\|\mathbf{G}\|_F}{\sigma_i} = \log_a\left(\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}\right),$$

so

$$\tau_1 = \log_a\sqrt{\mathrm{sr}(\mathbf{G})}, \qquad \tau_r = \log_a\left(\kappa(\mathbf{G})\sqrt{\mathrm{sr}(\mathbf{G})}\right), \qquad \tau_r - \tau_1 = \log_a\kappa(\mathbf{G}).$$

This completes the proof. □

通过归纳法，

$$\mathbf{X}_k = \mathbf{U}\,\mathrm{diag}\big(s_1^{(k)},\dots,s_r^{(k)}\big)\mathbf{V}^\top, \qquad s_i^{(k+1)} = \phi\left(s_i^{(k)}\right).$$

我们还有

$$\mathrm{Orth}(\mathbf{G}) = \mathbf{U}\mathbf{V}^\top = \mathbf{U}\mathbf{I}_r\mathbf{V}^\top, \quad \mathbf{X}_k - \mathrm{Orth}(\mathbf{G}) = \mathbf{U}\Big(\mathrm{diag}(s_1^{(k)},\dots,s_r^{(k)}) - \mathbf{I}_r\Big)\mathbf{V}^\top.$$

Left 并且右乘列正交的矩阵会保持Frobenius范数，s o

$$\big\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\big\|_F^2 = \sum_{i=1}^{r}(s_i^{(k)} - 1)^2.$$

这就证明了精确的误差表示。

我们现在建立下界。回想 t

$$q(t) := a + bt + ct^2.$$

根据假设，

$$0 < q(t) \le a, \qquad t \in [0,1].$$

固定 $i \in [r]$ 和 $k \ge 0$。如果 $a^k\frac{\sigma_i}{\alpha} \ge 1$，那么

$$\big|1 - s_i^{(k)}\big| \ge 0 = \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

现在假设 $a^k\sigma_i/\alpha < 1$。我们断言，对于每一个 $0 \le j \le k$,

$$0 < s_i^{(j)} \le a^j\frac{\sigma_i}{\alpha} < 1.$$

该命题对于 $j = 0$ 是显然的。假设它对于某个 $j < k$ 成立。那么 $s_i^{(j)} \in (0,1)$，因此 $(s_i^{(j)})^2 \in (0,1)$。因此，

$$0 < q\big((s_i^{(j)})^2\big) \le a.$$

由此可得

$$0 < s_i^{(j+1)} = s_i^{(j)}q\big((s_i^{(j)})^2\big) \le as_i^{(j)} \le a^{j+1}\frac{\sigma_i}{\alpha}.$$

因为 $j + 1 \le k$,

$$a^{j+1}\frac{\sigma_i}{\alpha} \le a^k\frac{\sigma_i}{\alpha} < 1.$$

这就证明了该断言。特别是，

$$0 < s_i^{(k)} \le a^k\frac{\sigma_i}{\alpha} < 1,$$

因此

$$|1 - s_i^{(k)}| = 1 - s_i^{(k)} \ge 1 - a^k\frac{\sigma_i}{\alpha} = \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

将两种情况结合起来，对于每一个 i,

$$|1 - s_i^{(k)}| \ge \left(1 - a^k\frac{\sigma_i}{\alpha}\right)_+.$$

最后，设置 $\alpha = \|\mathbf{G}\|_F$。对两边平方，对 i 求和，并使用精确的误差表示，

$$\frac{\sigma_i}{\|\mathbf{G}\|_F} = \frac{\sigma_i/\sigma_1}{\|\mathbf{G}\|_F/\sigma_1} = \frac{1}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}},$$

我们得到

$$\frac{1}{\sqrt{r}}\big\|\mathbf{X}_k - \mathrm{Orth}(\mathbf{G})\big\|_F \ge \frac{1}{\sqrt{r}}\left(\sum_{i=1}^{r}\left(1 - \frac{a^k}{\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}}\right)_+^2\right)^{1/2}.$$

此外，

$$\tau_i := \log_a\frac{\|\mathbf{G}\|_F}{\sigma_i} = \log_a\left(\kappa_i\sqrt{\mathrm{sr}(\mathbf{G})}\right),$$

so

$$\tau_1 = \log_a\sqrt{\mathrm{sr}(\mathbf{G})}, \qquad \tau_r = \log_a\left(\kappa(\mathbf{G})\sqrt{\mathrm{sr}(\mathbf{G})}\right), \qquad \tau_r - \tau_1 = \log_a\kappa(\mathbf{G}).$$

这完成了证明。 □

C Lemmas for Proposition 3.2

C.1 Lemma C.1

Lemma C.1. *Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric positive definite, and let $\mathbf{E} \in \mathbb{R}^{n\times n}$ be symmetric. Define $g(\mathbf{X}) = \mathbf{X}^{1/2}, f(\mathbf{X}) = \mathbf{X}^{-1/2}$. Then g and f are Fréchet differentiable at $\mathbf{A}$. We use $L_h(\mathbf{A};\mathbf{E})$ to denote the Fréchet derivative of a matrix function h at $\mathbf{A}$ applied to $\mathbf{E}$, i.e., $h(\mathbf{A}+\mathbf{E}) = h(\mathbf{A}) + L_h(\mathbf{A};\mathbf{E}) + o(\|\mathbf{E}\|_2)$. If $\mathbf{S} = \mathbf{A}^{1/2}, \mathbf{L} = L_g(\mathbf{A};\mathbf{E})$, then $\mathbf{L}$ is the unique symmetric solution of the Sylvester equation*

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

Moreover,

$$L_f(\mathbf{A};\mathbf{E}) = -\mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2}.$$

Consequently, for $\|\mathbf{E}\|_2$ sufficiently small,

$$(\mathbf{A}+\mathbf{E})^{-1/2} = \mathbf{A}^{-1/2} - \mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2} + \mathcal{O}(\|\mathbf{E}\|_2^2).$$

Proof. For completeness, differentiate the identity $g(\mathbf{X})^2 = \mathbf{X}$ at $\mathbf{A}$ in the direction $\mathbf{E}$. Writing $\mathbf{L} = L_g(\mathbf{A};\mathbf{E})$ and $\mathbf{S} = \mathbf{A}^{1/2}$ gives

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

Since $\mathbf{S} \succ 0$, the Sylvester operator $\mathbf{X} \mapsto \mathbf{SX} + \mathbf{XS}$ is invertible, so the solution $\mathbf{L}$ is unique. Next, since $f = \mathrm{inv} \circ g$, the chain rule and the Fréchet derivative of the matrix inverse,

$$L_{\mathrm{inv}}(\mathbf{Z};\mathbf{W}) = -\mathbf{Z}^{-1}\mathbf{W}\mathbf{Z}^{-1},$$

yield

$$L_f(\mathbf{A};\mathbf{E}) = -\mathbf{S}^{-1}\mathbf{L}\mathbf{S}^{-1} = -\mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2}.$$

The stated first-order expansion follows from the Fréchet differentiability of f at $\mathbf{A}$. □

C.2 Lemma C.2

Lemma C.2. *Let $\mathbf{A} \succ 0$ and set $\mathbf{S} = \mathbf{A}^{1/2} \succ 0$. Consider the Sylvester equation*

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

Its unique solution is given by [58]

$$\mathbf{L} = \int_0^\infty e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}dt.$$

Moreover, in the operator norm,

$$\|\mathbf{L}\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{S})}\|\mathbf{E}\|_2 = \frac{1}{2\sqrt{\lambda_{\min}(\mathbf{A})}}\|\mathbf{E}\|_2.$$

The same bound also holds for the Frobenius norm, and more generally for any unitarily invariant norm.

Proof. Differentiate the integrand:

$$\frac{d}{dt}\big(e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}\big) = -\mathbf{S}e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}} - e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}\mathbf{S}.$$

Integrating from 0 to ∞ yields

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E},$$

which proves the integral representation.

For the norm bound, using submultiplicativity and the fact that

$$\|e^{-t\mathbf{S}}\|_2 = e^{-t\lambda_{\min}(\mathbf{S})},$$

we have

$$\|\mathbf{L}\|_2 \le \int_0^\infty \|e^{-t\mathbf{S}}\|_2^2\|\mathbf{E}\|_2\,dt = \|\mathbf{E}\|_2\int_0^\infty e^{-2t\lambda_{\min}(\mathbf{S})}dt = \frac{1}{2\lambda_{\min}(\mathbf{S})}\|\mathbf{E}\|_2.$$

Since $\lambda_{\min}(\mathbf{S}) = \sqrt{\lambda_{\min}(\mathbf{A})}$, the stated bound follows. □

C 引理用于命题3.2

C.1 引理C.1

引理*C.1*。设$A \in \mathbb{R}^{n\times n}$为对称正定矩阵，$E \in \mathbb{R}^{n\times n}$为对称矩阵。定义 $g(X) = X^{1/2}$, $f(X) = X^{-1/2}$。则 g和 f在A处弗雷歇可微。我们用$L_h(A; E)$表示矩阵函数 h 在A处应用于E的弗雷歇导数，即 $h(A+E) = h(A) + L_h(A; E) + o(\|E\|_2)$。若$S = A^{1/2}$，$L = L_g(A; E)$，则$L$是西尔维斯特方程的唯一对称解

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

此外，

$$L_f(\mathbf{A};\mathbf{E}) = -\mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2}.$$

因此，对于足够小的 $\|\mathbf{E}\|_2$，

$$(\mathbf{A}+\mathbf{E})^{-1/2} = \mathbf{A}^{-1/2} - \mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2} + \mathcal{O}(\|\mathbf{E}\|_2^2).$$

证明。 为了完整性，在方向 $\mathbf{E}$ 中对单位矩阵 $g(\mathbf{X})^2 = \mathbf{X}$ 在 $\mathbf{A}$ 处求导。写出$\mathbf{L} = L_g(\mathbf{A};\mathbf{E})$ 和 $\mathbf{S} = \mathbf{A}^{1/2}$ 得到

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

由于 $\mathbf{S} \succ 0$，西尔维斯特算子 $\mathbf{X} \mapsto \mathbf{SX} + \mathbf{XS}$ 是可逆的，因此解 $\mathbf{L}$ 是唯一的。接下来，由于 $f = \mathrm{inv} \circ g$，根据链式法则和矩阵逆的弗雷歇导数，

$$L_{\mathrm{inv}}(\mathbf{Z};\mathbf{W}) = -\mathbf{Z}^{-1}\mathbf{W}\mathbf{Z}^{-1},$$

输出

$$L_f(\mathbf{A};\mathbf{E}) = -\mathbf{S}^{-1}\mathbf{L}\mathbf{S}^{-1} = -\mathbf{A}^{-1/2}\mathbf{L}\mathbf{A}^{-1/2}.$$

所陈述的一阶展开式源于 f 在<样式 id='2'>A</样式>处的弗雷歇可微性。 □

C.2引理C.2

<样式 id='1'>引理*C.2.* </样式>设<样式 id='3'>*A* </样式>$\succ 0$，并定义<样式 id='6'>*S* </样式>$=$ <样式 id='9'>*A*</样式>$^{1/2} \succ 0$。考虑西尔维斯特方程

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E}.$$

其唯一解由 *[58]*给出

$$\mathbf{L} = \int_0^\infty e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}dt.$$

此外，在算子范数下，

$$\|\mathbf{L}\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{S})}\|\mathbf{E}\|_2 = \frac{1}{2\sqrt{\lambda_{\min}(\mathbf{A})}}\|\mathbf{E}\|_2.$$

相同的界限也适用于*Frobenius*范数，并且更一般地适用于任何酉不变范数。

证明。 对被积函数求导:

$$\frac{d}{dt}\big(e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}\big) = -\mathbf{S}e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}} - e^{-t\mathbf{S}}\mathbf{E}e^{-t\mathbf{S}}\mathbf{S}.$$

从0积分到 ∞得到

$$\mathbf{SL} + \mathbf{LS} = \mathbf{E},$$

这证明了积分表示。

对于范数界限，利用次乘性和事实,

$$\|e^{-t\mathbf{S}}\|_2 = e^{-t\lambda_{\min}(\mathbf{S})},$$

我们有

$$\|\mathbf{L}\|_2 \le \int_0^\infty \|e^{-t\mathbf{S}}\|_2^2\|\mathbf{E}\|_2\,dt = \|\mathbf{E}\|_2\int_0^\infty e^{-2t\lambda_{\min}(\mathbf{S})}dt = \frac{1}{2\lambda_{\min}(\mathbf{S})}\|\mathbf{E}\|_2.$$

由于 $\lambda_{\min}(\mathbf{S}) = \sqrt{\lambda_{\min}(\mathbf{A})}$，所声明的界限成立。 □

D Proofs of Proposition 3.2

Proof. For the column/right statement, we have

$$\mathbf{M}^\top\mathbf{M} = \mathbf{D}_c(\mathbf{I}_n + \mathbf{C}_c)\mathbf{D}_c = \mathbf{D}_c^2 + \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c.$$

Apply Lemma C.1 with $\mathbf{A} = \mathbf{D}_c^2, \mathbf{E} = \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c$. Then

$$(\mathbf{M}^\top\mathbf{M})^{-1/2} = \mathbf{D}_c^{-1} - \mathbf{D}_c^{-1}\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2),$$

where $\mathbf{L}_c$ solves $\mathbf{D}_c\mathbf{L}_c + \mathbf{L}_c\mathbf{D}_c = \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c$.

Multiplying on the left by $\mathbf{M}$ yields

$$\begin{aligned}\mathrm{Orth}_r(\mathbf{M}) &= \mathbf{M}\mathbf{D}_c^{-1} - \mathbf{M}\mathbf{D}_c^{-1}\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2)\\ &= \mathbf{N}_c - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).\end{aligned}$$

Moreover, by Lemma C.2, we have

$$\|\mathbf{L}_c\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{D}_c)}\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2 \le \frac{\|\mathbf{D}_c\|_2^2}{2\lambda_{\min}(\mathbf{D}_c)}\|\mathbf{C}_c\|_2.$$

Hence

$$\|\mathbf{L}_c\mathbf{D}_c^{-1}\|_2 \le \|\mathbf{L}_c\|_2\|\mathbf{D}_c^{-1}\|_2 \le \frac{\kappa(\mathbf{D}_c)^2}{2}\|\mathbf{C}_c\|_2.$$

Therefore the first-order correction is $\mathcal{O}(\|\mathbf{C}_c\|_2)$, and

$$\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c + \mathcal{O}(\|\mathbf{C}_c\|_2).$$

For the row/left statement, we have

$$\mathbf{M}\mathbf{M}^\top = \mathbf{D}_r(\mathbf{I}_m + \mathbf{C}_r)\mathbf{D}_r = \mathbf{D}_r^2 + \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r.$$

Applying Lemma C.1 with $\mathbf{A} = \mathbf{D}_r^2, \mathbf{E} = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$ gives

$$(\mathbf{M}\mathbf{M}^\top)^{-1/2} = \mathbf{D}_r^{-1} - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{D}_r^{-1} + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2),$$

where $\mathbf{L}_r$ solves $\mathbf{D}_r\mathbf{L}_r + \mathbf{L}_r\mathbf{D}_r = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$.

Multiplying on the right by $\mathbf{M}$ yields

$$\begin{aligned}\mathrm{Orth}_\ell(\mathbf{M}) &= \mathbf{D}_r^{-1}\mathbf{M} - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{D}_r^{-1}\mathbf{M} + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2)\\ &= \mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).\end{aligned}$$

Again by Lemma C.2, we have

$$\|\mathbf{L}_r\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{D}_r)}\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2 \le \frac{\|\mathbf{D}_r\|_2^2}{2\lambda_{\min}(\mathbf{D}_r)}\|\mathbf{C}_r\|_2,$$

so

$$\|\mathbf{D}_r^{-1}\mathbf{L}_r\|_2 \le \|\mathbf{D}_r^{-1}\|_2\|\mathbf{L}_r\|_2 \le \frac{\kappa(\mathbf{D}_r)^2}{2}\|\mathbf{C}_r\|_2.$$

Therefore

$$\mathrm{Orth}_\ell(\mathbf{M}) = \mathbf{N}_r + \mathcal{O}(\|\mathbf{C}_r\|_2).$$

□

E Normalization and whitening

This appendix collects the normalization–whitening estimates that are omitted from the main text. We use the notation of Proposition 3.2. All $\mathcal{O}(\cdot)$ terms are taken in operator norm; the implied constants may depend on the fixed matrices in Proposition 3.2, but not on the perturbation terms.

命题3.2的证明

证明。 对于列/右侧陈述，我们有

$$\mathbf{M}^\top\mathbf{M} = \mathbf{D}_c(\mathbf{I}_n + \mathbf{C}_c)\mathbf{D}_c = \mathbf{D}_c^2 + \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c.$$

应用引理C.1和$\mathbf{A} = \mathbf{D}_c^2, \mathbf{E} = \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c$. 然后

$$(\mathbf{M}^\top\mathbf{M})^{-1/2} = \mathbf{D}_c^{-1} - \mathbf{D}_c^{-1}\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2),$$

其中$\mathbf{L}_c$ 满足$\mathbf{D}_c\mathbf{L}_c + \mathbf{L}_c\mathbf{D}_c = \mathbf{D}_c\mathbf{C}_c\mathbf{D}_c$.

在左侧乘以$\mathbf{M}$ 得到

$$\begin{aligned}\mathrm{Orth}_r(\mathbf{M}) &= \mathbf{M}\mathbf{D}_c^{-1} - \mathbf{M}\mathbf{D}_c^{-1}\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2)\\ &= \mathbf{N}_c - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).\end{aligned}$$

此外，根据引理C.2，我们有

$$\|\mathbf{L}_c\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{D}_c)}\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2 \le \frac{\|\mathbf{D}_c\|_2^2}{2\lambda_{\min}(\mathbf{D}_c)}\|\mathbf{C}_c\|_2.$$

因此

$$\|\mathbf{L}_c\mathbf{D}_c^{-1}\|_2 \le \|\mathbf{L}_c\|_2\|\mathbf{D}_c^{-1}\|_2 \le \frac{\kappa(\mathbf{D}_c)^2}{2}\|\mathbf{C}_c\|_2.$$

因此一阶修正为 $\mathcal{O}(\|\mathbf{C}_c\|_2)$，并且

$$\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c + \mathcal{O}(\|\mathbf{C}_c\|_2).$$

对于行/左边的陈述，我们有

$$\mathbf{M}\mathbf{M}^\top = \mathbf{D}_r(\mathbf{I}_m + \mathbf{C}_r)\mathbf{D}_r = \mathbf{D}_r^2 + \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r.$$

应用引理C.1，结合 $\mathbf{A} = \mathbf{D}_r^2$，$\mathbf{E} = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$ ，得到

$$(\mathbf{M}\mathbf{M}^\top)^{-1/2} = \mathbf{D}_r^{-1} - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{D}_r^{-1} + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2),$$

其中 $\mathbf{L}_r$ 满足 $\mathbf{D}_r\mathbf{L}_r + \mathbf{L}_r\mathbf{D}_r = \mathbf{D}_r\mathbf{C}_r\mathbf{D}_r$.

在右边乘以 $\mathbf{M}$ ，得到

$$\begin{aligned}\mathrm{Orth}_\ell(\mathbf{M}) &= \mathbf{D}_r^{-1}\mathbf{M} - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{D}_r^{-1}\mathbf{M} + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2)\\ &= \mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).\end{aligned}$$

再次根据引理C.2，我们有

$$\|\mathbf{L}_r\|_2 \le \frac{1}{2\lambda_{\min}(\mathbf{D}_r)}\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2 \le \frac{\|\mathbf{D}_r\|_2^2}{2\lambda_{\min}(\mathbf{D}_r)}\|\mathbf{C}_r\|_2,$$

so

$$\|\mathbf{D}_r^{-1}\mathbf{L}_r\|_2 \le \|\mathbf{D}_r^{-1}\|_2\|\mathbf{L}_r\|_2 \le \frac{\kappa(\mathbf{D}_r)^2}{2}\|\mathbf{C}_r\|_2.$$

因此

$$\mathrm{Orth}_\ell(\mathbf{M}) = \mathbf{N}_r + \mathcal{O}(\|\mathbf{C}_r\|_2).$$

□

E 归一化与白化

本附录收集了正文中省略的归一化—白化估计。我们使用命题3.2的符号。所有 $\mathcal{O}(\cdot)$项均取算子范数；隐含常数可能依赖于命题3.2中的固定矩阵，但不依赖于扰动项。

Proposition E.1. *The following estimates hold.*

In the column/right setting of Proposition 3.2, if $\|\mathbf{C}_c\|_2$ *is sufficiently small, then* $\mathrm{Orth}_r(\mathbf{N}_c) = \mathbf{N}_c(\mathbf{I}_n + \mathbf{C}_c)^{-1/2} = \mathbf{N}_c - \frac{1}{2}\mathbf{N}_c\mathbf{C}_c + \mathcal{O}(\|\mathbf{C}_c\|_2^2)$*, and therefore* $\|\mathrm{Orth}_r(\mathbf{N}_c) - \mathbf{N}_c\|_2 \le \frac{1}{2}\|\mathbf{N}_c\|_2\|\mathbf{C}_c\|_2 + \mathcal{O}(\|\mathbf{C}_c\|_2^2)$*. Moreover,*

$$\|\mathrm{Orth}_r(\mathbf{M}) - \mathbf{M}\|_2 \le \|\mathbf{N}_c\|_2\|\mathbf{D}_c - \mathbf{I}_n\|_2 + \frac{\kappa(\mathbf{D}_c)^2}{2}\|\mathbf{N}_c\|_2\|\mathbf{C}_c\|_2 + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).$$

In the row/left setting of Proposition 3.2, if $\|\mathbf{C}_r\|_2$ *is sufficiently small, then* $\mathrm{Orth}_\ell(\mathbf{N}_r) = (\mathbf{I}_m + \mathbf{C}_r)^{-1/2}\mathbf{N}_r = \mathbf{N}_r - \frac{1}{2}\mathbf{C}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{C}_r\|_2^2)$*, and therefore* $\|\mathrm{Orth}_\ell(\mathbf{N}_r) - \mathbf{N}_r\|_2 \le \frac{1}{2}\|\mathbf{N}_r\|_2\|\mathbf{C}_r\|_2 + \mathcal{O}(\|\mathbf{C}_r\|_2^2)$*. Moreover,*

$$\|\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M}\|_2 \le \|\mathbf{D}_r - \mathbf{I}_m\|_2\|\mathbf{N}_r\|_2 + \frac{\kappa(\mathbf{D}_r)^2}{2}\|\mathbf{N}_r\|_2\|\mathbf{C}_r\|_2 + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).$$

Proof. Consider first the column/right case. Applying Lemma C.1 with $\mathbf{A} = \mathbf{I}_n$ and $\mathbf{E} = \mathbf{C}_c$ gives the Sylvester equation $\mathbf{L} + \mathbf{L} = \mathbf{C}_c$. Thus $\mathbf{L} = \frac{1}{2}\mathbf{C}_c$, and $(\mathbf{I}_n + \mathbf{C}_c)^{-1/2} = \mathbf{I}_n - \frac{1}{2}\mathbf{C}_c + \mathcal{O}(\|\mathbf{C}_c\|_2^2)$. Multiplying on the left by $\mathbf{N}_c$ gives the expansion of $\mathrm{Orth}_r(\mathbf{N}_c)$, and the norm bound follows from submultiplicativity.

To compare with $\mathbf{M}$, Proposition 3.2 gives $\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2)$. Since $\mathbf{M} = \mathbf{N}_c\mathbf{D}_c$,

$$\mathrm{Orth}_r(\mathbf{M}) - \mathbf{M} = \mathbf{N}_c(\mathbf{I}_n - \mathbf{D}_c) - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).$$

Lemma C.2 yields $\|\mathbf{L}_c\mathbf{D}_c^{-1}\|_2 \le \frac{1}{2}\kappa(\mathbf{D}_c)^2\|\mathbf{C}_c\|_2$, which gives the stated estimate.

The row/left case follows by the same argument. Applying Lemma C.1 with $\mathbf{A} = \mathbf{I}_m$ and $\mathbf{E} = \mathbf{C}_r$ gives $(\mathbf{I}_m + \mathbf{C}_r)^{-1/2} = \mathbf{I}_m - \frac{1}{2}\mathbf{C}_r + \mathcal{O}(\|\mathbf{C}_r\|_2^2)$, and hence $\mathrm{Orth}_\ell(\mathbf{N}_r) = (\mathbf{I}_m + \mathbf{C}_r)^{-1/2}\mathbf{N}_r = \mathbf{N}_r - \frac{1}{2}\mathbf{C}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{C}_r\|_2^2)$. The norm bound again follows from submultiplicativity. Proposition 3.2 also gives $\mathrm{Orth}_\ell(\mathbf{M}) = \mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2)$, so

$$\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M} = (\mathbf{I}_m - \mathbf{D}_r)\mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).$$

The bound $\|\mathbf{D}_r^{-1}\mathbf{L}_r\|_2 \le \frac{1}{2}\kappa(\mathbf{D}_r)^2\|\mathbf{C}_r\|_2$ completes the proof. □

Remark E.1. *Proposition E.1 shows that one-sided normalization removes the zeroth-order marginal scale terms* $\|\mathbf{N}_c\|_2\|\mathbf{D}_c - \mathbf{I}_n\|_2$ *and* $\|\mathbf{D}_r - \mathbf{I}_m\|_2\|\mathbf{N}_r\|_2$*. Whitening then acts only on the first-order Gram error. Since* $\mathrm{diag}(\mathbf{N}_c^\top\mathbf{N}_c) = \mathbf{I}_n$ *and* $\mathrm{diag}(\mathbf{N}_r\mathbf{N}_r^\top) = \mathbf{I}_m$*, the leading correction is purely off-diagonal.*

Corollary E.2. *Assume that every row and every column of* $\mathbf{M}$ *is nonzero, and set* $\hat{\mathbf{M}} := \mathbf{D}_r^{-1/2}\mathbf{M}\mathbf{D}_c^{-1/2}$.

If $\mathbf{M}$ *has full column rank and* $\hat{\mathbf{C}}_c := \hat{\mathbf{M}}^\top\hat{\mathbf{M}} - \mathbf{I}_n$*, then for* $\|\hat{\mathbf{C}}_c\|_2$ *sufficiently small,* $\mathrm{Orth}_r(\hat{\mathbf{M}}) = \hat{\mathbf{M}}(\mathbf{I}_n + \hat{\mathbf{C}}_c)^{-1/2} = \hat{\mathbf{M}} - \frac{1}{2}\hat{\mathbf{M}}\hat{\mathbf{C}}_c + \mathcal{O}(\|\hat{\mathbf{C}}_c\|_2^2)$*, and therefore* $\|\mathrm{Orth}_r(\hat{\mathbf{M}}) - \hat{\mathbf{M}}\|_2 \le \frac{1}{2}\|\hat{\mathbf{M}}\|_2\|\hat{\mathbf{C}}_c\|_2 + \mathcal{O}(\|\hat{\mathbf{C}}_c\|_2^2)$.

If $\mathbf{M}$ *has full row rank and* $\hat{\mathbf{C}}_r := \hat{\mathbf{M}}\hat{\mathbf{M}}^\top - \mathbf{I}_m$*, then for* $\|\hat{\mathbf{C}}_r\|_2$ *sufficiently small,* $\mathrm{Orth}_\ell(\hat{\mathbf{M}}) = (\mathbf{I}_m + \hat{\mathbf{C}}_r)^{-1/2}\hat{\mathbf{M}} = \hat{\mathbf{M}} - \frac{1}{2}\hat{\mathbf{C}}_r\hat{\mathbf{M}} + \mathcal{O}(\|\hat{\mathbf{C}}_r\|_2^2)$*, and therefore* $\|\mathrm{Orth}_\ell(\hat{\mathbf{M}}) - \hat{\mathbf{M}}\|_2 \le \frac{1}{2}\|\hat{\mathbf{M}}\|_2\|\hat{\mathbf{C}}_r\|_2 + \mathcal{O}(\|\hat{\mathbf{C}}_r\|_2^2)$.

Proof. The same expansion applies to the two-sided normalized matrix $\hat{\mathbf{M}}$.

For the column-rank case, Lemma C.1 with $\mathbf{A} = \mathbf{I}_n$ and $\mathbf{E} = \hat{\mathbf{C}}_c$ gives $(\mathbf{I}_n + \hat{\mathbf{C}}_c)^{-1/2} = \mathbf{I}_n - \frac{1}{2}\hat{\mathbf{C}}_c + \mathcal{O}(\|\hat{\mathbf{C}}_c\|_2^2)$. This yields the expansion of $\mathrm{Orth}_r(\hat{\mathbf{M}})$ and the corresponding norm bound.

For the row-rank case, Lemma C.1 with $\mathbf{A} = \mathbf{I}_m$ and $\mathbf{E} = \hat{\mathbf{C}}_r$ gives $(\mathbf{I}_m + \hat{\mathbf{C}}_r)^{-1/2} = \mathbf{I}_m - \frac{1}{2}\hat{\mathbf{C}}_r + \mathcal{O}(\|\hat{\mathbf{C}}_r\|_2^2)$. This yields the expansion of $\mathrm{Orth}_\ell(\hat{\mathbf{M}})$ and the corresponding norm bound. □

命题E.1。 以下估计成立。

在命题*3.2*的列/正交设置中，如果 $\|C_c\|_2$ 足够小，则$Orth_r(N_c) = N_c(I_n + C_c)^{-1/2} = N_c - \frac{1}{2}N_cC_c + \mathcal{O}(\|C_c\|_2^2)$，因此$^-\|Orth_r(N_c) - N_c\|_2 \le \frac{1}{2}\|N_c\|_2\|C_c\|_2 + \mathcal{O}(\|C_c\|_2^2)$。此外，$^-$

$$\|\mathrm{Orth}_r(\mathbf{M}) - \mathbf{M}\|_2 \le \|\mathbf{N}_c\|_2\|\mathbf{D}_c - \mathbf{I}_n\|_2 + \frac{\kappa(\mathbf{D}_c)^2}{2}\|\mathbf{N}_c\|_2\|\mathbf{C}_c\|_2 + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).$$

在命题*3.2*的行/左设置中，如果 $\|C_r\|_2$ 足够小，则 $Orth_\ell(N_r) = (I_m + C_r)^{-1/2}N_r = N_r - \frac{1}{2}C_rN_r + \mathcal{O}(\|C_r\|_2^2)$，因此 $\|Orth_\ell(N_r) - N_r\|_2 \le \frac{1}{2}\|N_r\|_2\|C_r\|_2 + \mathcal{O}(\|C_r\|_2^2)$。此外，$^-$

$$\|\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M}\|_2 \le \|\mathbf{D}_r - \mathbf{I}_m\|_2\|\mathbf{N}_r\|_2 + \frac{\kappa(\mathbf{D}_r)^2}{2}\|\mathbf{N}_r\|_2\|\mathbf{C}_r\|_2 + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).$$

证明。 首先考虑列/右情况。应用引理C.1，使用 $\mathbf{A} = \mathbf{I}_n$ 和 $\mathbf{E} = \mathbf{C}_c$，得到西尔维斯特方程 $\mathbf{L} + \mathbf{L} = \mathbf{C}_c$。因此 $\mathbf{L} = \frac{1}{2}\mathbf{C}_c$，并且 $(\mathbf{I}_n + \mathbf{C}_c)^{-1/2} = \mathbf{I}_n - \frac{1}{2}\mathbf{C}_c + \mathcal{O}(\|\mathbf{C}_c\|_2^2)$。在左侧乘以 $\mathbf{N}_c$ 得到 $\mathrm{Orth}_r(\mathbf{N}_c)$ 的展开式，范数界则由次乘性得出。

与$\mathbf{M}$进行比较，命题3.2给出$\mathrm{Orth}_r(\mathbf{M}) = \mathbf{N}_c - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2)$. 由于$\mathbf{M} = \mathbf{N}_c\mathbf{D}_c$,

$$\mathrm{Orth}_r(\mathbf{M}) - \mathbf{M} = \mathbf{N}_c(\mathbf{I}_n - \mathbf{D}_c) - \mathbf{N}_c\mathbf{L}_c\mathbf{D}_c^{-1} + \mathcal{O}(\|\mathbf{D}_c\mathbf{C}_c\mathbf{D}_c\|_2^2).$$

引理C.2给出 $\|\mathbf{L}_c\mathbf{D}_c^{-1}\|_2 \le \frac{1}{2}\kappa(\mathbf{D}_c)^2\|\mathbf{C}_c\|_2$, 这就得到了所要求的估计。

行/左边的情形通过同样的论证得出。应用引理C.1，其中$\mathrm{A} = \mathrm{I}_m$，$\mathrm{E} = \mathrm{C}_r$，给出$(\mathrm{I}_m + \mathrm{C}_r)^{-1/2} = \mathrm{I}_m - \frac{1}{2}\mathrm{C}_r + \mathcal{O}(\|\mathrm{C}_r\|_2^2)$，因此$\mathrm{Orth}_\ell(\mathrm{N}_r) = (\mathrm{I}_m + \mathrm{C}_r)^{-1/2}\mathrm{N}_r = \mathrm{N}_r - \frac{1}{2}\mathrm{C}_r\mathrm{N}_r + \mathcal{O}(\|\mathrm{C}_r\|_2^2)$。范数界再次来自次乘法性。命题3.2还给出$\mathrm{Orth}_\ell(\mathrm{M}) = \mathrm{N}_r - \mathrm{D}_r^{-1}\mathrm{L}_r\mathrm{N}_r + \mathcal{O}(\|\mathrm{D}_r\mathrm{C}_r\mathrm{D}_r\|_2^2)$，所以

$$\mathrm{Orth}_\ell(\mathbf{M}) - \mathbf{M} = (\mathbf{I}_m - \mathbf{D}_r)\mathbf{N}_r - \mathbf{D}_r^{-1}\mathbf{L}_r\mathbf{N}_r + \mathcal{O}(\|\mathbf{D}_r\mathbf{C}_r\mathbf{D}_r\|_2^2).$$

界 $\|\mathbf{D}_r^{-1}\mathbf{L}_r\|_2 \le \frac{1}{2}\kappa(\mathbf{D}_r)^2\|\mathbf{C}_r\|_2$ 完成了证明。 □

备注 E.1。 命题 *E.1* 表明，单边归一化消除了零阶边缘尺度项 $\|\mathbf{N}_c\|_2\|\mathbf{D}_c - \mathbf{I}_n\|_2$ 和 $\|\mathbf{D}_r - \mathbf{I}_m\|_2\|\mathbf{N}_r\|_2$。白化仅作用于一阶格拉姆误差。由于 $\mathrm{diag}(\mathbf{N}_c^\top\mathbf{N}_c) = \mathbf{I}_n$ 和 $\mathrm{diag}(\mathbf{N}_r\mathbf{N}_r^\top) = \mathbf{I}_m$，主校正项纯粹是离对角线项。

ˆ推论 E.2。 假设 $\mathbf{M}$ 的每一行和每一列都不为零，并设 $\mathbf{M} := \mathbf{D}_r^{-1/2}\mathbf{M}\mathbf{D}_c^{-1/2}$。

如果 M 满列秩且 $\hat{C}_c := \hat{M}^\top\hat{M} - I_n$，那么对于 $\|\hat{C}_c\|_2$ 足够小，$Orth_r(\hat{M}) = \hat{M}(I_n + \hat{C}_c)^{-1/2} = \hat{M} - \frac{1}{2}\hat{M}\hat{C}_c + \mathcal{O}(\|\hat{C}_c\|_2^2)$，因此 $\|Orth_r(\hat{M}) - \hat{M}\|_2 \le \frac{1}{2}\|\hat{M}\|_2\|\hat{C}_c\|_2 + \mathcal{O}(\|\hat{C}_c\|_2^2)$。

如果M满行秩且$\hat{C}_r := \hat{M}\hat{M}^\top - I_m$，那么对于 $\|\hat{C}_r\|_2$ 足够小，$Orth_\ell(\hat{M}) = (I_m + \hat{C}_r)^{-1/2}\hat{M} = \hat{M} - \frac{1}{2}\hat{C}_r\hat{M} + \mathcal{O}(\|\hat{C}_r\|_2^2)$，因此 $\|Orth_\ell(\hat{M}) - \hat{M}\|_2 \le \frac{1}{2}\|\hat{M}\|_2\|\hat{C}_r\|_2 + \mathcal{O}(\|\hat{C}_r\|_2^2)$。

ˆ证明。 同样的展开式也适用于双边归一化矩阵$\mathbf{M}$。

对于列秩情况，引理C.1与$\mathbf{A} = \mathbf{I}_n$ 和$\mathbf{E} = \hat{\mathbf{C}}_c$ 一起给出$(\mathbf{I}_n + \hat{\mathbf{C}}_c)^{-1/2} = \mathbf{I}_n - \frac{1}{2}\hat{\mathbf{C}}_c + \mathcal{O}(\|\hat{\mathbf{C}}_c\|_2^2)$。这给出了$\mathrm{Orth}_r(\hat{\mathbf{M}})$的展开式和相应的范数界。

对于行秩情况，引理C.1在<样式 id='1'>A</样式> = <样式 id='4'>I</样式>$_m$ 和<样式 id='7'>E</样式> = ˆ<样式 id='10'>C</样式>$_r$下给出（<样式 id='13'>I</样式>$_m$ + ˆ<样式 id='16'>C</样式>$_r$）$^{-1/2}$ = <样式 id='20'>I</样式>$_m$ − $\frac{1}{2}$ˆ<样式 id='23'>C</样式>$_r$ +$\mathcal{O}$（∥ˆ<样式 id='28'>C</样式>$_r$∥$_2^2$）。这给出了Orth$_\ell$（ˆ<样式 id='32'>M</样式>）的展开式以及相应的范数界。 □

Remark E.2. *After two-sided normalization, whitening is centered at the identity Gram matrix of* $\hat{\mathbf{M}}$. *It no longer corrects marginal row or column scales; it only removes the residual mismatch left after diagonal equilibration.*

F Lemmas for Proposition 3.4

F.1 Lemma F.1

Lemma F.1. *Let the momentum iterate satisfy* $\mathbf{M}_t = \beta\mathbf{M}_{t-1} + (1-\beta)\mathbf{G}_t, \beta \in [0,1)$, *and assume there exists a constant* $G_\infty > 0$ *such that* $\|\mathbf{G}_t\|_\infty \le G_\infty, \|\mathbf{M}_0\|_\infty \le G_\infty$. *For* $\varepsilon > 0$, *define* $\mathbf{D}_{r,t} := \operatorname{diag}\big(\operatorname{rowsum}(\mathbf{M}_t \odot \mathbf{M}_t) + \varepsilon\big), \mathbf{D}_{c,t} := \operatorname{diag}\big(\operatorname{colsum}(\mathbf{M}_t \odot \mathbf{M}_t) + \varepsilon\big)$, *and* $\mathbf{P}_t := \mathbf{D}_{r,t}^{-1/2}, \mathbf{Q}_t := \mathbf{D}_{c,t}^{-1/2}$. *Then, for all* t,

$$(nG_\infty^2 + \varepsilon)^{-1/2}\mathbf{I}_m \preceq \mathbf{P}_t \preceq \varepsilon^{-1/2}\mathbf{I}_m, \quad (mG_\infty^2 + \varepsilon)^{-1/2}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq \varepsilon^{-1/2}\mathbf{I}_n.$$

Moreover,

$$\sqrt{\varepsilon}\mathbf{I}_m \preceq \mathbf{P}_t^{-1} \preceq \sqrt{nG_\infty^2 + \varepsilon}\mathbf{I}_m, \quad \sqrt{\varepsilon}\mathbf{I}_n \preceq \mathbf{Q}_t^{-1} \preceq \sqrt{mG_\infty^2 + \varepsilon}\mathbf{I}_n.$$

Proof. We first show that the momentum iterate $\mathbf{M}_t$ is uniformly bounded entrywise.

Since

$$\mathbf{M}_t = \beta\mathbf{M}_{t-1} + (1-\beta)\mathbf{G}_t,$$

we have

$$\|\mathbf{M}_t\|_\infty \le \beta\|\mathbf{M}_{t-1}\|_\infty + (1-\beta)\|\mathbf{G}_t\|_\infty \le \beta\|\mathbf{M}_{t-1}\|_\infty + (1-\beta)G_\infty.$$

Using $\|\mathbf{M}_0\|_\infty = 0 \le G_\infty$, an induction on t yields

$$\|\mathbf{M}_t\|_\infty \le G_\infty, \qquad \forall t.$$

Therefore, every entry of $\mathbf{M}_t$ satisfies

$$|(\mathbf{M}_t)_{ij}| \le G_\infty, \qquad \forall i, j, t.$$

Now fix any row i. Since that row has n entries,

$$\sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 \le nG_\infty^2.$$

Hence each diagonal entry of $\mathbf{D}_{r,t}$ satisfies

$$\varepsilon \le (\mathbf{D}_{r,t})_{ii} = \sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 + \varepsilon \le nG_\infty^2 + \varepsilon.$$

Similarly, for any column j,

$$\sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 \le mG_\infty^2,$$

so each diagonal entry of $\mathbf{D}_{c,t}$ satisfies

$$\varepsilon \le (\mathbf{D}_{c,t})_{jj} = \sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 + \varepsilon \le mG_\infty^2 + \varepsilon.$$

Since $\mathbf{P}_t = \mathbf{D}_{r,t}^{-1/2}$ and $\mathbf{Q}_t = \mathbf{D}_{c,t}^{-1/2}$ are diagonal matrices, their diagonal entries are

$$(\mathbf{P}_t)_{ii} = \left(\sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 + \varepsilon\right)^{-1/2}, \quad (\mathbf{Q}_t)_{jj} = \left(\sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 + \varepsilon\right)^{-1/2}.$$

Using the bounds above, we obtain

$$(nG_\infty^2 + \varepsilon)^{-1/2} \le (\mathbf{P}_t)_{ii} \le \varepsilon^{-1/2}, \quad (mG_\infty^2 + \varepsilon)^{-1/2} \le (\mathbf{Q}_t)_{jj} \le \varepsilon^{-1/2}.$$

<样式 id='1'>^</样式><样式 id='3'>注解E.2.</样式>在双边归一化后，白化以<样式 id='5'>M</样式>的单位格拉姆矩阵为中心。它不再修正边缘行或列尺度；它仅移除对角均衡后留下的残差不匹配。

F 对命题 3.4 的引理

F.1 引理 F.1

引理 *F.1.* 设动量迭代满足 $M_t = \beta M_{t-1} + (1-\beta)G_t, \beta \in [0,1)$, 并假设存在常数 $G_\infty > 0$ 使得 $\|G_t\|_\infty \le G_\infty, \|M_0\|_\infty \le G_\infty$. 对于 $\varepsilon > 0$, 定义 $D_{r,t} := diag(rowsum(M_t \odot M_t) + \varepsilon)$, $D_{c,t} := diag(colsum(M_t \odot M_t) + \varepsilon)$, 和 $P_t := D_{r\,t}^{-1/2}$, , $Q_t := D_{c\,t}^{-1/2}$, . 那么 , 对于所有 t,

$$(nG_\infty^2 + \varepsilon)^{-1/2}\mathbf{I}_m \preceq \mathbf{P}_t \preceq \varepsilon^{-1/2}\mathbf{I}_m, \quad (mG_\infty^2 + \varepsilon)^{-1/2}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq \varepsilon^{-1/2}\mathbf{I}_n.$$

此外,

$$\sqrt{\varepsilon}\mathbf{I}_m \preceq \mathbf{P}_t^{-1} \preceq \sqrt{nG_\infty^2 + \varepsilon}\mathbf{I}_m, \quad \sqrt{\varepsilon}\mathbf{I}_n \preceq \mathbf{Q}_t^{-1} \preceq \sqrt{mG_\infty^2 + \varepsilon}\mathbf{I}_n.$$

证明。 我们首先证明动量迭代 $\mathbf{M}_t$ 逐项一致有界。

由于

$$\mathbf{M}_t = \beta\mathbf{M}_{t-1} + (1-\beta)\mathbf{G}_t,$$

我们已有

$$\|\mathbf{M}_t\|_\infty \le \beta\|\mathbf{M}_{t-1}\|_\infty + (1-\beta)\|\mathbf{G}_t\|_\infty \le \beta\|\mathbf{M}_{t-1}\|_\infty + (1-\beta)G_\infty.$$

使用 $\|\mathbf{M}_0\|_\infty = 0 \le G_\infty$，对 t 进行归纳得到

$$\|\mathbf{M}_t\|_\infty \le G_\infty, \qquad \forall t.$$

因此，$\mathbf{M}_t$ 的每个元素都满足

$$|(\mathbf{M}_t)_{ij}| \le G_\infty, \qquad \forall i, j, t.$$

现在固定任意一行 i。由于该行有 n 个元素，

$$\sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 \le nG_\infty^2.$$

因此 $\mathbf{D}_{r,t}$ 的每个对角线元素都满足

$$\varepsilon \le (\mathbf{D}_{r,t})_{ii} = \sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 + \varepsilon \le nG_\infty^2 + \varepsilon.$$

类似地，对于任意一列 j,

$$\sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 \le mG_\infty^2,$$

因此，$\mathbf{D}_{c,t}$ 的每个对角线元素都满足

$$\varepsilon \le (\mathbf{D}_{c,t})_{jj} = \sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 + \varepsilon \le mG_\infty^2 + \varepsilon.$$

由于 $\mathbf{P}_t = \mathbf{D}_{r\,t}^{-1/2}$,和 $\mathbf{Q}_t = \mathbf{D}_{c\,t}^{-1/2}$,都是对角矩阵，它们的对角线元素是

$$(\mathbf{P}_t)_{ii} = \left(\sum_{j=1}^{n} (\mathbf{M}_t)_{ij}^2 + \varepsilon\right)^{-1/2}, \quad (\mathbf{Q}_t)_{jj} = \left(\sum_{i=1}^{m} (\mathbf{M}_t)_{ij}^2 + \varepsilon\right)^{-1/2}.$$

根据上述界限，我们得到

$$(nG_\infty^2 + \varepsilon)^{-1/2} \le (\mathbf{P}_t)_{ii} \le \varepsilon^{-1/2}, \quad (mG_\infty^2 + \varepsilon)^{-1/2} \le (\mathbf{Q}_t)_{jj} \le \varepsilon^{-1/2}.$$

Since positive diagonal matrices are ordered entrywise in the Loewner sense, it follows that

$$(nG_\infty^2+\varepsilon)^{-1/2}\mathbf{I}_m \preceq \mathbf{P}_t \preceq \varepsilon^{-1/2}\mathbf{I}_m, \quad (mG_\infty^2+\varepsilon)^{-1/2}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq \varepsilon^{-1/2}\mathbf{I}_n.$$

Because $\mathbf{P}_t$ and $\mathbf{Q}_t$ are symmetric positive definite diagonal matrices, their singular values are exactly their diagonal entries. Therefore,

$$(nG_\infty^2+\varepsilon)^{-1/2} \le \sigma_{\min}(\mathbf{P}_t) \le \|\mathbf{P}_t\|_2 \le \varepsilon^{-1/2}, \quad (mG_\infty^2+\varepsilon)^{-1/2} \le \sigma_{\min}(\mathbf{Q}_t) \le \|\mathbf{Q}_t\|_2 \le \varepsilon^{-1/2}.$$

Finally, inverting the above matrix inequalities gives

$$\sqrt{\varepsilon}\mathbf{I}_m \preceq \mathbf{P}_t^{-1} \preceq \sqrt{nG_\infty^2+\varepsilon}\mathbf{I}_m, \quad \sqrt{\varepsilon}\mathbf{I}_n \preceq \mathbf{Q}_t^{-1} \preceq \sqrt{mG_\infty^2+\varepsilon}\mathbf{I}_n.$$

This completes the proof. □

F.2 Lemma F.2

Lemma F.2. *Suppose that $\{E_t, A_t\}$ are nonnegative sequences. Assume $E_{t+1} \le (1-\alpha_{t+1})E_t + A_{t+1}$ where $\alpha_t = t^{-p}$, $p \in (0,1]$. Then $\alpha_t E_t \le 2(E_t - E_{t+1} + A_{t+1})$.*

Proof. See [10], Lemma A.3. □

F.3 Lemma F.3

Lemma F.3. *For Algorithm 1, the accumulated error between the momentum term and the true gradient is bounded for $t \ge 1$:*

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le& \beta_{t+1}\mathbb{E}\left[\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2\right] \\ &+ \frac{\beta_{t+1}^2}{1-\beta_{t+1}} L^2\eta_t^2\|a\mathbf{O}_t + \lambda_t\mathbf{X}_t\|_F^2 + (1-\beta_{t+1})^2\sigma^2.\end{aligned}$$

Proof. First, we have

$$\begin{aligned}&\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &= \|\beta_{t+1}\mathbf{M}_t + (1-\beta_{t+1})\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &= \|\beta_{t+1}(\mathbf{M}_t - \nabla f(\mathbf{X}_t)) + (1-\beta_{t+1})(\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})) \\ &\quad + \beta_{t+1}(\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1}))\|_F^2 \\ &= \beta_{t+1}^2\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2 + \beta_{t+1}^2\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &\quad + (1-\beta_{t+1})^2\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &\quad + 2\beta_{t+1}^2\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle_F \\ &\quad + 2\beta_{t+1}(1-\beta_{t+1})\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\rangle_F \\ &\quad + 2\beta_{t+1}(1-\beta_{t+1})\langle \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1}), \nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\rangle_F.\end{aligned}$$

According to Assumption 3.3. Taking the expectation of its squared norm, and using the unbiasedness and independence of the stochastic gradient, we obtain:

$$\begin{aligned}\mathbb{E}[\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2] =& \beta_{t+1}^2\mathbb{E}[\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2] \\ &+ (1-\beta_{t+1})^2\mathbb{E}[\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2] \\ &+ \beta_{t+1}^2\mathbb{E}[\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2] \\ &+ 2\beta_{t+1}^2\mathbb{E}[\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle].\end{aligned}$$

Applying Young's inequality with a parameter ($ab \le \frac{\epsilon}{2}a^2 + \frac{1}{2\epsilon}b^2$), we have

$$\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle_F \le \frac{\epsilon}{2}\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2 + \frac{1}{2\epsilon}\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2.$$

由于正对角矩阵在Loewner意义下按分量排序，因此可知

$$(nG_\infty^2+\varepsilon)^{-1/2}\mathbf{I}_m \preceq \mathbf{P}_t \preceq \varepsilon^{-1/2}\mathbf{I}_m, \quad (mG_\infty^2+\varepsilon)^{-1/2}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq \varepsilon^{-1/2}\mathbf{I}_n.$$

因为 $\mathbf{P}_t$ 和 $\mathbf{Q}_t$ 是对称正定对角矩阵，它们的奇异值正好等于它们的对角线元素。因此，

$$(nG_\infty^2+\varepsilon)^{-1/2} \le \sigma_{\min}(\mathbf{P}_t) \le \|\mathbf{P}_t\|_2 \le \varepsilon^{-1/2}, \quad (mG_\infty^2+\varepsilon)^{-1/2} \le \sigma_{\min}(\mathbf{Q}_t) \le \|\mathbf{Q}_t\|_2 \le \varepsilon^{-1/2}.$$

最后，通过上述矩阵不等式求逆得到

$$\sqrt{\varepsilon}\mathbf{I}_m \preceq \mathbf{P}_t^{-1} \preceq \sqrt{nG_\infty^2+\varepsilon}\mathbf{I}_m, \quad \sqrt{\varepsilon}\mathbf{I}_n \preceq \mathbf{Q}_t^{-1} \preceq \sqrt{mG_\infty^2+\varepsilon}\mathbf{I}_n.$$

这就完成了证明。 □

引理F.2

引理F.2. 假设 $\{E_t, A_t\}$ 是非负序列。设 $E_{t+1} \le (1-\alpha_{t+1})E_t + A_{t+1}$，其中 $\alpha_t = t^{-p}$, $p \in (0,1]$。则 $\alpha_t E_t \le 2(E_t - E_{t+1} + A_{t+1})$。

证明。 参见 [10],引理A.3。 □

F.3 引理 F.3

Lemma F.3. 对于算法*1*，动量项与真实梯度之间的累积误差在内是有界的： *e*
gradient 是有界的，对于 $t \ge 1$*:*

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le& \beta_{t+1}\mathbb{E}\left[\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2\right] \\ &+ \frac{\beta_{t+1}^2}{1-\beta_{t+1}} L^2\eta_t^2\|a\mathbf{O}_t + \lambda_t\mathbf{X}_t\|_F^2 + (1-\beta_{t+1})^2\sigma^2.\end{aligned}$$

证明。 首先，我们有

$$\begin{aligned}&\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &= \|\beta_{t+1}\mathbf{M}_t + (1-\beta_{t+1})\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &= \|\beta_{t+1}(\mathbf{M}_t - \nabla f(\mathbf{X}_t)) + (1-\beta_{t+1})(\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})) \\ &\quad + \beta_{t+1}(\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1}))\|_F^2 \\ &= \beta_{t+1}^2\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2 + \beta_{t+1}^2\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &\quad + (1-\beta_{t+1})^2\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2 \\ &\quad + 2\beta_{t+1}^2\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle_F \\ &\quad + 2\beta_{t+1}(1-\beta_{t+1})\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\rangle_F \\ &\quad + 2\beta_{t+1}(1-\beta_{t+1})\langle \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1}), \nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\rangle_F.\end{aligned}$$

根据假设3.3。取其平方范数的期望，并利用随机梯度的无偏性和独立性，我们得到：

$$\begin{aligned}\mathbb{E}[\|\mathbf{M}_{t+1} - \nabla f(\mathbf{X}_{t+1})\|_F^2] =& \beta_{t+1}^2\mathbb{E}[\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2] \\ &+ (1-\beta_{t+1})^2\mathbb{E}[\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1}) - \nabla f(\mathbf{X}_{t+1})\|_F^2] \\ &+ \beta_{t+1}^2\mathbb{E}[\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2] \\ &+ 2\beta_{t+1}^2\mathbb{E}[\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle].\end{aligned}$$

使用参数 ($ab \le \frac{\epsilon}{2}a^2 + \frac{1}{2\epsilon}b^2$) 应用杨氏不等式，我们得到 —

$$\langle \mathbf{M}_t - \nabla f(\mathbf{X}_t), \nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\rangle_F \le \frac{\epsilon}{2}\|\mathbf{M}_t - \nabla f(\mathbf{X}_t)\|_F^2 + \frac{1}{2\epsilon}\|\nabla f(\mathbf{X}_t) - \nabla f(\mathbf{X}_{t+1})\|_F^2.$$

Thus, we have:

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}^2(1+\epsilon)\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\beta_{t+1}^2\left(1+\frac{1}{\epsilon}\right)\mathbb{E}\left[\|\nabla f(\mathbf{X}_t)-\nabla f(\mathbf{X}_{t+1})\|_F^2\right]\\ &+(1-\beta_{t+1})^2\mathbb{E}\left[\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1})-\nabla f(\mathbf{X}_{t+1})\|_F^2\right].\end{aligned}$$

According to Assumption 3.2,

$$\begin{aligned}\|\nabla f(\mathbf{X}_t)-\nabla f(\mathbf{X}_{t+1})\|_F^2 &\le L^2\|\mathbf{X}_t-\mathbf{X}_{t+1}\|_F^2\\ &= L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2\end{aligned}$$

Therefore:

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}^2(1+\epsilon)\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\beta_{t+1}^2\left(1+\frac{1}{\epsilon}\right)L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2+(1-\beta_{t+1})^2\sigma^2.\end{aligned}$$

Then, by letting $\epsilon := \frac{1-\beta_{t+1}}{\beta_{t+1}}, t \ge 1$, we have

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\frac{\beta_{t+1}^2}{1-\beta_{t+1}}L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2+(1-\beta_{t+1})^2\sigma^2.\end{aligned}\tag{6}$$

□

G Proofs of Proposition 3.4

Proof. By Assumption 3.2 and the descent lemma,

$$\begin{aligned}f(\mathbf{X}_{t+1}) &\le f(\mathbf{X}_t)+\langle\nabla f(\mathbf{X}_t),\mathbf{X}_{t+1}-\mathbf{X}_t\rangle+\frac{L}{2}\|\mathbf{X}_{t+1}-\mathbf{X}_t\|_F^2\\ &= f(\mathbf{X}_t)-a\eta_t\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle+\frac{La^2\eta_t^2}{2}\|\mathbf{O}_t\|_F^2\\ &\le f(\mathbf{X}_t)-a\eta_t\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle+\frac{La^2n}{2}\eta_t^2,\end{aligned}$$

where the last step uses $\|\mathbf{O}_t\|_F^2 \le n$.

Let $\mathbf{S}_t := \nabla f(\mathbf{X}_t)-\mathbf{M}_t, r_\varepsilon := \sqrt{\varepsilon}, R_\varepsilon := \sqrt{nG_\infty^2+\varepsilon}, C_\varepsilon := \sqrt{mG_\infty^2+\varepsilon}$.

Define the midpoint-rescaled matrices

$$\widetilde{\mathbf{P}}_t := \frac{R_\varepsilon+r_\varepsilon}{2}\mathbf{P}_t, \widetilde{\mathbf{Q}}_t := \frac{C_\varepsilon+r_\varepsilon}{2}\mathbf{Q}_t, \widetilde{\mathbf{A}}_t := \widetilde{\mathbf{P}}_t^{-1}, \widetilde{\mathbf{B}}_t := \widetilde{\mathbf{Q}}_t^{-1}.$$

Since the rescaling is by positive scalars,

$$\operatorname{polar}(\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t) = \operatorname{polar}(\mathbf{P}_t\mathbf{M}_t\mathbf{Q}_t) = \mathbf{O}_t.$$

Hence

$$\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle = \left\langle\widetilde{\mathbf{A}}_t\nabla f(\mathbf{X}_t)\widetilde{\mathbf{B}}_t,\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle.$$

By Lemma F.1, we have

$$R_\varepsilon^{-1}\mathbf{I}_m \preceq \mathbf{P}_t \preceq r_\varepsilon^{-1}\mathbf{I}_m, \qquad C_\varepsilon^{-1}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq r_\varepsilon^{-1}\mathbf{I}_n.$$

Since $R_\varepsilon = r_\varepsilon\rho_r, C_\varepsilon = r_\varepsilon\rho_c$, we obtain

$$\frac{\rho_r+1}{2\rho_r}\mathbf{I}_m \preceq \widetilde{\mathbf{P}}_t \preceq \frac{\rho_r+1}{2}\mathbf{I}_m, \qquad \frac{\rho_c+1}{2\rho_c}\mathbf{I}_n \preceq \widetilde{\mathbf{Q}}_t \preceq \frac{\rho_c+1}{2}\mathbf{I}_n.$$

因此，我们有：

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}^2(1+\epsilon)\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\beta_{t+1}^2\left(1+\frac{1}{\epsilon}\right)\mathbb{E}\left[\|\nabla f(\mathbf{X}_t)-\nabla f(\mathbf{X}_{t+1})\|_F^2\right]\\ &+(1-\beta_{t+1})^2\mathbb{E}\left[\|\nabla f(\mathbf{X}_{t+1};\xi_{t+1})-\nabla f(\mathbf{X}_{t+1})\|_F^2\right].\end{aligned}$$

根据假设 3.2，

$$\begin{aligned}\|\nabla f(\mathbf{X}_t)-\nabla f(\mathbf{X}_{t+1})\|_F^2 &\le L^2\|\mathbf{X}_t-\mathbf{X}_{t+1}\|_F^2\\ &= L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2\end{aligned}$$

因此：

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}^2(1+\epsilon)\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\beta_{t+1}^2\left(1+\frac{1}{\epsilon}\right)L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2+(1-\beta_{t+1})^2\sigma^2.\end{aligned}$$

然后，通过让 $\epsilon := \frac{1-\beta_{t+1}}{\beta_{t+1}}, t \ge 1$，我们得到

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}-\nabla f(\mathbf{X}_{t+1})\|_F^2\right] \le{}& \beta_{t+1}\mathbb{E}\left[\|\mathbf{M}_t-\nabla f(\mathbf{X}_t)\|_F^2\right]\\ &+\frac{\beta_{t+1}^2}{1-\beta_{t+1}}L^2\eta_t^2\|a\mathbf{O}_t+\lambda_t\mathbf{X}_t\|_F^2+(1-\beta_{t+1})^2\sigma^2.\end{aligned}\tag{6}$$

□

G 证明命题 3.4

证明。 根据假设 3.2 和下降引理，

$$\begin{aligned}f(\mathbf{X}_{t+1}) &\le f(\mathbf{X}_t)+\langle\nabla f(\mathbf{X}_t),\mathbf{X}_{t+1}-\mathbf{X}_t\rangle+\frac{L}{2}\|\mathbf{X}_{t+1}-\mathbf{X}_t\|_F^2\\ &= f(\mathbf{X}_t)-a\eta_t\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle+\frac{La^2\eta_t^2}{2}\|\mathbf{O}_t\|_F^2\\ &\le f(\mathbf{X}_t)-a\eta_t\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle+\frac{La^2n}{2}\eta_t^2,\end{aligned}$$

其中最后一步使用了 $\|\mathbf{O}_t\|_F^2 \le n$。

Let $\mathbf{S}_t := \nabla f(\mathbf{X}_t)-\mathbf{M}_t, r_\varepsilon := \sqrt{\varepsilon}, R_\varepsilon := \sqrt{nG_\infty^2+\varepsilon}, C_\varepsilon := \sqrt{mG_\infty^2+\varepsilon}$.

定义中点缩放矩阵

$$\widetilde{\mathbf{P}}_t := \frac{R_\varepsilon+r_\varepsilon}{2}\mathbf{P}_t, \widetilde{\mathbf{Q}}_t := \frac{C_\varepsilon+r_\varepsilon}{2}\mathbf{Q}_t, \widetilde{\mathbf{A}}_t := \widetilde{\mathbf{P}}_t^{-1}, \widetilde{\mathbf{B}}_t := \widetilde{\mathbf{Q}}_t^{-1}.$$

由于缩放是通过正标量进行的，

$$\operatorname{polar}(\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t) = \operatorname{polar}(\mathbf{P}_t\mathbf{M}_t\mathbf{Q}_t) = \mathbf{O}_t.$$

因此

$$\langle\nabla f(\mathbf{X}_t),\mathbf{O}_t\rangle = \left\langle\widetilde{\mathbf{A}}_t\nabla f(\mathbf{X}_t)\widetilde{\mathbf{B}}_t,\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle.$$

根据引理F.1，我们有

$$R_\varepsilon^{-1}\mathbf{I}_m \preceq \mathbf{P}_t \preceq r_\varepsilon^{-1}\mathbf{I}_m, \qquad C_\varepsilon^{-1}\mathbf{I}_n \preceq \mathbf{Q}_t \preceq r_\varepsilon^{-1}\mathbf{I}_n.$$

由于 $R_\varepsilon = r_\varepsilon\rho_r$， $C_\varepsilon = r_\varepsilon\rho_c$， 我们得到

$$\frac{\rho_r+1}{2\rho_r}\mathbf{I}_m \preceq \widetilde{\mathbf{P}}_t \preceq \frac{\rho_r+1}{2}\mathbf{I}_m, \qquad \frac{\rho_c+1}{2\rho_c}\mathbf{I}_n \preceq \widetilde{\mathbf{Q}}_t \preceq \frac{\rho_c+1}{2}\mathbf{I}_n.$$

Therefore, we have

$$\|\widetilde{\mathbf{A}}_t\|_2 \le \frac{2\rho_r}{\rho_r+1}, \quad \|\widetilde{\mathbf{B}}_t\|_2 \le \frac{2\rho_c}{\rho_c+1}, \quad \|\widetilde{\mathbf{A}}_t - \mathbf{I}_m\|_2 \le \frac{\rho_r-1}{\rho_r+1}, \quad \|\widetilde{\mathbf{B}}_t - \mathbf{I}_n\|_2 \le \frac{\rho_c-1}{\rho_c+1}.$$

Using $\nabla f(\mathbf{X}_t) = \mathbf{M}_t + \mathbf{S}_t$, we split

$$\begin{aligned}\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle &= \left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle + \left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle \\ &= \|\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t\|_* + \left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle + \left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle.\end{aligned}$$

(i) First, by the singular-value inequality

$$\sigma_i(\mathbf{AXB}) \ge \sigma_{\min}(\mathbf{A})\sigma_i(\mathbf{X})\sigma_{\min}(\mathbf{B}),$$

we have

$$\|\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t\|_* \ge \frac{(\rho_r+1)(\rho_c+1)}{4\rho_r\rho_c}\|\mathbf{M}_t\|_*.$$

(ii) Second, using Hölder's inequality, $\|\mathbf{O}_t\|_2 = 1$, and the bounds above,

$$\begin{aligned}\left|\left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle\right| &\le \|\widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t\|_*\|\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \|\widetilde{\mathbf{A}}_t\|_2\|\widetilde{\mathbf{B}}_t\|_2\|\mathbf{S}_t\|_*\|\widetilde{\mathbf{P}}_t\|_2\|\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \rho_r\rho_c\|\mathbf{S}_t\|_* \\ &\le \rho_r\rho_c\sqrt{n}\|\mathbf{S}_t\|_F.\end{aligned}$$

(iii) Third, we use the symmetric decomposition

$$\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t = (\widetilde{\mathbf{A}}_t - \mathbf{I}_m)\mathbf{M}_t + \mathbf{M}_t(\widetilde{\mathbf{B}}_t - \mathbf{I}_n) + (\widetilde{\mathbf{A}}_t - \mathbf{I}_m)\mathbf{M}_t(\widetilde{\mathbf{B}}_t - \mathbf{I}_n).$$

Hence

$$\|\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t\|_* \le \left(\frac{\rho_r-1}{\rho_r+1} + \frac{\rho_c-1}{\rho_c+1} + \frac{(\rho_r-1)(\rho_c-1)}{(\rho_r+1)(\rho_c+1)}\right)\|\mathbf{M}_t\|_*.$$

Therefore, we have

$$\begin{aligned}\left|\left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle\right| &\le \|\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t\|_*\|\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \frac{3\rho_r\rho_c - \rho_r - \rho_c - 1}{4}\|\mathbf{M}_t\|_*.\end{aligned}$$

Combining the above bounds, we obtain

$$\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle \ge \chi_\varepsilon\|\mathbf{M}_t\|_* - \rho_r\rho_c\sqrt{n}\|\mathbf{S}_t\|_F.$$

Under the setting $\varepsilon \ge \frac{4}{5}G_\infty^2\max\{m,n\}$, we have $\rho_r = \sqrt{1+\frac{nG_\infty^2}{\varepsilon}} \le \sqrt{1+\frac{5n}{4\max\{m,n\}}} \le \frac{3}{2}$, $\rho_c = \sqrt{1+\frac{mG_\infty^2}{\varepsilon}} \le \sqrt{1+\frac{5m}{4\max\{m,n\}}} \le \frac{3}{2}$.

Let $\psi(x,y) := \frac{(x+1)(y+1)}{4xy} - \frac{3xy-x-y-1}{4}, x, y \ge 1$. A direct calculation gives $\partial_x\psi(x,y) = \frac{1-x^{-2}-x^{-2}y^{-1}-3y}{4} < 0, \partial_y\psi(x,y) = \frac{1-y^{-2}-x^{-1}y^{-2}-3x}{4} < 0$, for all $x, y \ge 1$. Hence ψ is decreasing in each argument on $[1,\infty)^2$. Therefore, we have $\chi_\varepsilon = \psi(\rho_r,\rho_c) \ge \psi\left(\frac{3}{2},\frac{3}{2}\right) = \frac{1}{144} > 0$. Since $\chi_\varepsilon > 0$, $\|\mathbf{M}_t\|_* \ge \|\mathbf{M}_t\|_F$, and

$$\|\mathbf{M}_t\|_F \ge \|\nabla f(\mathbf{X}_t)\|_F - \|\mathbf{S}_t\|_F,$$

it follows that

$$\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle \ge \chi_\varepsilon\|\nabla f(\mathbf{X}_t)\|_F - \left(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\right)\|\mathbf{S}_t\|_F.$$

Substituting the above inequality into the descent bound yields

$$\begin{aligned}f(\mathbf{X}_{t+1}) \le& f(\mathbf{X}_t) - a\chi_\varepsilon\eta_t\|\nabla f(\mathbf{X}_t)\|_F \\ &+ a\left(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\right)\eta_t\|\mathbf{S}_t\|_F + \frac{La^2n}{2}\eta_t^2.\end{aligned}$$

因此，我们有

$$\|\widetilde{\mathbf{A}}_t\|_2 \le \frac{2\rho_r}{\rho_r+1}, \quad \|\widetilde{\mathbf{B}}_t\|_2 \le \frac{2\rho_c}{\rho_c+1}, \quad \|\widetilde{\mathbf{A}}_t - \mathbf{I}_m\|_2 \le \frac{\rho_r-1}{\rho_r+1}, \quad \|\widetilde{\mathbf{B}}_t - \mathbf{I}_n\|_2 \le \frac{\rho_c-1}{\rho_c+1}.$$

使用 $\nabla f(\mathbf{X}_t) = \mathbf{M}_t + \mathbf{S}_t$，我们将

$$\begin{aligned}\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle &= \left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle + \left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle \\ &= \|\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t\|_* + \left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle + \left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle.\end{aligned}$$

*(i)*首先，根据奇异值不等式

$$\sigma_i(\mathbf{AXB}) \ge \sigma_{\min}(\mathbf{A})\sigma_i(\mathbf{X})\sigma_{\min}(\mathbf{B}),$$

我们有

$$\|\widetilde{\mathbf{P}}_t\mathbf{M}_t\widetilde{\mathbf{Q}}_t\|_* \ge \frac{(\rho_r+1)(\rho_c+1)}{4\rho_r\rho_c}\|\mathbf{M}_t\|_*.$$

(ii) 其次，利用 Hölder 不等式、$\|\mathbf{O}_t\|_2 = 1$，以及上述界限，

$$\begin{aligned}\left|\left\langle \widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle\right| &\le \|\widetilde{\mathbf{A}}_t\mathbf{S}_t\widetilde{\mathbf{B}}_t\|_*\|\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \|\widetilde{\mathbf{A}}_t\|_2\|\widetilde{\mathbf{B}}_t\|_2\|\mathbf{S}_t\|_*\|\widetilde{\mathbf{P}}_t\|_2\|\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \rho_r\rho_c\|\mathbf{S}_t\|_* \\ &\le \rho_r\rho_c\sqrt{n}\|\mathbf{S}_t\|_F.\end{aligned}$$

*(iii)*第三，我们使用对称分解

$$\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t = (\widetilde{\mathbf{A}}_t - \mathbf{I}_m)\mathbf{M}_t + \mathbf{M}_t(\widetilde{\mathbf{B}}_t - \mathbf{I}_n) + (\widetilde{\mathbf{A}}_t - \mathbf{I}_m)\mathbf{M}_t(\widetilde{\mathbf{B}}_t - \mathbf{I}_n).$$

因此

$$\|\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t\|_* \le \left(\frac{\rho_r-1}{\rho_r+1} + \frac{\rho_c-1}{\rho_c+1} + \frac{(\rho_r-1)(\rho_c-1)}{(\rho_r+1)(\rho_c+1)}\right)\|\mathbf{M}_t\|_*.$$

因此，我们得到

$$\begin{aligned}\left|\left\langle \widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t, \widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\right\rangle\right| &\le \|\widetilde{\mathbf{A}}_t\mathbf{M}_t\widetilde{\mathbf{B}}_t - \mathbf{M}_t\|_*\|\widetilde{\mathbf{P}}_t\mathbf{O}_t\widetilde{\mathbf{Q}}_t\|_2 \\ &\le \frac{3\rho_r\rho_c - \rho_r - \rho_c - 1}{4}\|\mathbf{M}_t\|_*.\end{aligned}$$

结合上述界限，我们得到

$$\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle \ge \chi_\varepsilon\|\mathbf{M}_t\|_* - \rho_r\rho_c\sqrt{n}\|\mathbf{S}_t\|_F.$$

在 $\varepsilon \ge \frac{4}{5}G_\infty^2\max\{m,n\}$ 设置下，我们有 $\rho_r = \sqrt{1+\frac{nG_\infty^2}{\varepsilon}} \le \sqrt{1+\frac{5n}{4\max\{m,n\}}} \le \frac{3}{2}$, $\rho_c = \sqrt{1+\frac{mG_\infty^2}{\varepsilon}} \le \sqrt{1+\frac{5m}{4\max\{m,n\}}} \le \frac{3}{2}$.

令 $\psi(x,y) := \frac{(x+1)(y+1)}{4xy} - \frac{3xy-x-y-1}{4}, x, y \ge 1$. 直接计算得到 $\partial_x\psi(x,y) = \frac{1-x^{-2}-x^{-2}y^{-1}-3y}{4} < 0, \partial_y\psi(x,y) = \frac{1-y^{-2}-x^{-1}y^{-2}-3x}{4} < 0$，对所有 $x, y \ge 1$。因此 ψ 在每个参数上关于 $[1,\infty)^2$ 是递减的。因此，我们得到 $\chi_\varepsilon = \psi(\rho_r,\rho_c) \ge \psi\left(\frac{3}{2},\frac{3}{2}\right) = \frac{1}{144} > 0$。由于 $\chi_\varepsilon > 0$, $\|\mathbf{M}_t\|_* \ge \|\mathbf{M}_t\|_F$，和

$$\|\mathbf{M}_t\|_F \ge \|\nabla f(\mathbf{X}_t)\|_F - \|\mathbf{S}_t\|_F,$$

因此

$$\langle \nabla f(\mathbf{X}_t), \mathbf{O}_t\rangle \ge \chi_\varepsilon\|\nabla f(\mathbf{X}_t)\|_F - \left(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\right)\|\mathbf{S}_t\|_F.$$

将上述不等式代入下降界得到

$$\begin{aligned}f(\mathbf{X}_{t+1}) \le& f(\mathbf{X}_t) - a\chi_\varepsilon\eta_t\|\nabla f(\mathbf{X}_t)\|_F \\ &+ a\left(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\right)\eta_t\|\mathbf{S}_t\|_F + \frac{La^2n}{2}\eta_t^2.\end{aligned}$$

Apply Young's inequality with parameter h_{t+1}/L:

$$a\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)\eta_t\|\mathbf{S}_t\|_F\le\frac{ah_{t+1}}{2L}\|\mathbf{S}_t\|_F^2+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\frac{\eta_t^2}{h_{t+1}}.$$

Hence

$$\begin{aligned}f(\mathbf{X}_{t+1})\le&f(\mathbf{X}_t)-a\chi_\varepsilon\eta_t\|\nabla f(\mathbf{X}_t)\|_F+\frac{ah_{t+1}}{2L}\|\mathbf{S}_t\|_F^2\\&+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\frac{\eta_t^2}{h_{t+1}}+\frac{La^2n}{2}\eta_t^2.\end{aligned}$$

Taking expectations and summing the above inequality over $t=1,\dots,T$, then using $f(\mathbf{X}_{T+1})\ge f^\star$, we have

$$\begin{aligned}a\chi_\varepsilon\sum_{t=1}^T\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F\le&f(\mathbf{X}_1)-f^\star+\frac{a}{2L}\sum_{t=1}^T h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2\\&+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\sum_{t=1}^T\frac{\eta_t^2}{h_{t+1}}+\frac{La^2n}{2}\sum_{t=1}^T\eta_t^2.\end{aligned}$$

By Lemma F.3 with $\lambda=0$,

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2\le(1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2+\frac{(1-h_{t+1})^2}{h_{t+1}}L^2\eta_t^2a^2n+h_{t+1}^2\sigma^2.$$

Since $(1-h_{t+1})^2\le1$ and

$$\frac{\eta_t^2}{h_{t+1}}=\frac{t^{-3/2}}{(t+1)^{-1/2}}=\frac{\sqrt{t+1}}{t^{3/2}}\le\frac{2\sqrt{2}}{t+1}=2\sqrt{2}h_{t+1}^2,$$

we have

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2\le(1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2+h_{t+1}^2(2\sqrt{2}L^2a^2n+\sigma^2).$$

According to Lemma F.2, by letting $E_t=\mathbb{E}\|\mathbf{S}_t\|_F^2, A_{t+1}=h_{t+1}^2(2\sqrt{2}L^2a^2n+\sigma^2)$, we have

$$h_t\mathbb{E}\|\mathbf{S}_t\|_F^2\le2\left(\mathbb{E}\|\mathbf{S}_t\|_F^2-\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2+A_{t+1}\right).$$

Moreover, since $\beta_1=0$,

$$\begin{aligned}\mathbb{E}\|\mathbf{S}_1\|_F^2&=\mathbb{E}\|\nabla f(\mathbf{X}_1)-\mathbf{M}_1\|_F^2\\&=\mathbb{E}\|\nabla f(\mathbf{X}_1)-\nabla f(\mathbf{X}_1;\xi_1)\|_F^2\le\sigma^2.\end{aligned}$$

It follows that

$$\begin{aligned}\sum_{t=1}^T h_t\mathbb{E}\|\mathbf{S}_t\|_F^2&\le2\sum_{t=1}^T\left(\mathbb{E}\|\mathbf{S}_t\|_F^2-\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2+A_{t+1}\right)\\&\le2\mathbb{E}\|\mathbf{S}_1\|_F^2+2(2\sqrt{2}L^2a^2n+\sigma^2)\sum_{t=1}^T\frac{1}{t+1}\\&\le2\sigma^2+2(2\sqrt{2}L^2a^2n+\sigma^2)(1+\ln T).\end{aligned}$$

Hence, since $h_{t+1}\le h_t$,

$$\sum_{t=1}^T h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2\le2\sigma^2+2(2\sqrt{2}L^2a^2n+\sigma^2)(1+\ln T).$$

Also,

$$\sum_{t=1}^T\frac{\eta_t^2}{h_{t+1}}=\sum_{t=1}^T\frac{t^{-3/2}}{(t+1)^{-1/2}}\le2(1+\ln T),\qquad\sum_{t=1}^T\eta_t^2=\sum_{t=1}^T t^{-3/2}\le3.$$

Substituting these bounds into the previous inequality, we get

$$a\chi_\varepsilon\sum_{t=1}^T\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F\le f(\mathbf{X}_1)-f^\star+C_1(1+\ln T)+C_2,$$

应用Young不等式，参数为 h_{t+1}/L:

$$a\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)\eta_t\|\mathbf{S}_t\|_F\le\frac{ah_{t+1}}{2L}\|\mathbf{S}_t\|_F^2+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\frac{\eta_t^2}{h_{t+1}}.$$

因此

$$\begin{aligned}f(\mathbf{X}_{t+1})\le&f(\mathbf{X}_t)-a\chi_\varepsilon\eta_t\|\nabla f(\mathbf{X}_t)\|_F+\frac{ah_{t+1}}{2L}\|\mathbf{S}_t\|_F^2\\&+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\frac{\eta_t^2}{h_{t+1}}+\frac{La^2n}{2}\eta_t^2.\end{aligned}$$

取期望并对 $t=1,\ \dots,\ T$ 上的上述不等式求和，然后使用 $f(\mathbf{X}_{T+1})\ge f^\star$，我们得到

$$\begin{aligned}a\chi_\varepsilon\sum_{t=1}^T\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F\le&f(\mathbf{X}_1)-f^\star+\frac{a}{2L}\sum_{t=1}^T h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2\\&+\frac{aL}{2}\big(\chi_\varepsilon+\rho_r\rho_c\sqrt{n}\big)^2\sum_{t=1}^T\frac{\eta_t^2}{h_{t+1}}+\frac{La^2n}{2}\sum_{t=1}^T\eta_t^2.\end{aligned}$$

根据引理F.3 和 $\lambda=0$,

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2\le(1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2+\frac{(1-h_{t+1})^2}{h_{t+1}}L^2\eta_t^2a^2n+h_{t+1}^2\sigma^2.$$

由于 $(1-h_{t+1})^2\le1$ 并且

$$\frac{\eta_t^2}{h_{t+1}}=\frac{t^{-3/2}}{(t+1)^{-1/2}}=\frac{\sqrt{t+1}}{t^{3/2}}\le\frac{2\sqrt{2}}{t+1}=2\sqrt{2}h_{t+1}^2,$$

我们得到

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2\le(1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2+h_{t+1}^2(2\sqrt{2}L^2a^2n+\sigma^2).$$

根据引理F.2，通过令 $E_t=\mathbb{E}\|\mathbf{S}_t\|_F^2, A_{t+1}=h_{t+1}^2(2\sqrt{2}L^2a^2n+\sigma^2)$，我们得到

$$h_t\mathbb{E}\|\mathbf{S}_t\|_F^2\le2\left(\mathbb{E}\|\mathbf{S}_t\|_F^2-\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2+A_{t+1}\right).$$

此外，由于 $\beta_1=0$,

$$\begin{aligned}\mathbb{E}\|\mathbf{S}_1\|_F^2&=\mathbb{E}\|\nabla f(\mathbf{X}_1)-\mathbf{M}_1\|_F^2\\&=\mathbb{E}\|\nabla f(\mathbf{X}_1)-\nabla f(\mathbf{X}_1;\xi_1)\|_F^2\le\sigma^2.\end{aligned}$$

因此

$$\begin{aligned}\sum_{t=1}^T h_t\mathbb{E}\|\mathbf{S}_t\|_F^2&\le2\sum_{t=1}^T\left(\mathbb{E}\|\mathbf{S}_t\|_F^2-\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2+A_{t+1}\right)\\&\le2\mathbb{E}\|\mathbf{S}_1\|_F^2+2(2\sqrt{2}L^2a^2n+\sigma^2)\sum_{t=1}^T\frac{1}{t+1}\\&\le2\sigma^2+2(2\sqrt{2}L^2a^2n+\sigma^2)(1+\ln T).\end{aligned}$$

因此，由于 $h_{t+1}\le h_t$,

$$\sum_{t=1}^T h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2\le2\sigma^2+2(2\sqrt{2}L^2a^2n+\sigma^2)(1+\ln T).$$

此外,

$$\sum_{t=1}^T\frac{\eta_t^2}{h_{t+1}}=\sum_{t=1}^T\frac{t^{-3/2}}{(t+1)^{-1/2}}\le2(1+\ln T),\qquad\sum_{t=1}^T\eta_t^2=\sum_{t=1}^T t^{-3/2}\le3.$$

将这些边界代入之前的不等式，我们得到

$$a\chi_\varepsilon\sum_{t=1}^T\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F\le f(\mathbf{X}_1)-f^\star+C_1(1+\ln T)+C_2,$$

where

$$C_1 = \frac{a}{L}\big(2\sqrt{2}L^2a^2n + \sigma^2\big) + aL\big(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\big)^2, \quad C_2 = \frac{a\sigma^2}{L} + \frac{3La^2n}{2}.$$

Finally, since $\eta_t = t^{-3/4} \geq T^{-3/4}$ for all $1 \leq t \leq T$,

$$T^{-3/4}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \leq \sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F.$$

Therefore, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \leq \frac{f(\mathbf{X}_1) - f^\star + C_1(1+\ln T) + C_2}{a\chi_\varepsilon \cdot T^{1/4}}.$$

This completes the proof. □

H Lemmas for Theorem 3.5 and Corollary 3.6

H.1 Lemma H.1

Lemma H.1. *Let*

$$\mathbf{D}_{r,t} := \mathrm{diag}\big(\mathrm{rowsum}(\mathbf{M}_t \odot \mathbf{M}_t)\big), \quad \mathbf{P}_{r,t} := (\mathbf{D}_{r,t}^{1/2})^\dagger, \quad \mathbf{R}_t := \mathbf{P}_{r,t}\mathbf{M}_t, \quad \mathbf{O}_t := \mathrm{Orth}(\mathbf{R}_t),$$

where † denotes the Moore–Penrose pseudoinverse, so zero rows of $\mathbf{M}_t$ remain zero after row normalization, and $\mathrm{Orth}(\mathbf{0}) := \mathbf{0}$. Then, for all t,

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\sqrt{m}}\|\mathbf{M}_t\|_F, \qquad \|\mathbf{O}_t\|_F^2 = \mathrm{rank}(\mathbf{R}_t) \leq n.$$

Proof. (i) If $\mathbf{M}_t = \mathbf{0}$, then $\mathbf{R}_t = \mathbf{O}_t = \mathbf{0}$ and the claim is immediate.

(ii) Assume $\mathbf{M}_t \neq \mathbf{0}$. Let

$$d_{i,t} := \|(\mathbf{M}_t)_{i,:}\|_2, \qquad \mathbf{D}_t := \mathrm{diag}(d_{1,t}, \ldots, d_{m,t}),$$

so that $\mathbf{D}_t = \mathbf{D}_{r,t}^{1/2}$ and $\mathbf{M}_t = \mathbf{D}_t\mathbf{R}_t$. Let $\mathbf{H}_t := (\mathbf{R}_t\mathbf{R}_t^\top)^{1/2}$. If $\mathbf{R}_t = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top$ is a compact SVD, then $\mathbf{O}_t = \mathbf{U}_t\mathbf{V}_t^\top$ and therefore $\mathbf{O}_t\mathbf{R}_t^\top = \mathbf{H}_t$. Hence

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle = \mathrm{tr}(\mathbf{O}_t^\top\mathbf{D}_t\mathbf{R}_t) = \mathrm{tr}(\mathbf{D}_t\mathbf{O}_t\mathbf{R}_t^\top) = \mathrm{tr}(\mathbf{D}_t\mathbf{H}_t) = \sum_{i=1}^{m} d_{i,t}(\mathbf{H}_t)_{ii}.$$

If $d_{i,t} > 0$, then the ith row of $\mathbf{R}_t$ has unit Euclidean norm, so

$$1 = \mathbf{e}_i^\top\mathbf{R}_t\mathbf{R}_t^\top\mathbf{e}_i = \mathbf{e}_i^\top\mathbf{H}_t^2\mathbf{e}_i \leq \|\mathbf{H}_t\|_2\mathbf{e}_i^\top\mathbf{H}_t\mathbf{e}_i = \|\mathbf{R}_t\|_2(\mathbf{H}_t)_{ii}.$$

Thus $(\mathbf{H}_t)_{ii} \geq \|\mathbf{R}_t\|_2^{-1}$ whenever $d_{i,t} > 0$. If $d_{i,t} = 0$, the corresponding term vanishes. Therefore,

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\|\mathbf{R}_t\|_2}\sum_{i=1}^{m} d_{i,t}.$$

Now each nonzero row of $\mathbf{R}_t$ has norm 1, so $\|\mathbf{R}_t\|_F^2 \leq m$, hence $\|\mathbf{R}_t\|_2 \leq \sqrt{m}$. Also, $\sum_{i=1}^{m} d_{i,t} \geq \|\mathbf{M}_t\|_F$. It follows that

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\sqrt{m}}\|\mathbf{M}_t\|_F.$$

Finally, if $r_t := \mathrm{rank}(\mathbf{R}_t)$, then the nonzero singular values of $\mathbf{O}_t$ are all equal to 1, so $\|\mathbf{O}_t\|_F^2 = r_t \leq n$. □

其中

$$C_1 = \frac{a}{L}\big(2\sqrt{2}L^2a^2n + \sigma^2\big) + aL\big(\chi_\varepsilon + \rho_r\rho_c\sqrt{n}\big)^2, \quad C_2 = \frac{a\sigma^2}{L} + \frac{3La^2n}{2}.$$

最后，由于 $\eta_t = t^{-3/4} \geq T^{-3/4}$ 对所有 $1 \leq t \leq T$ 成立，

$$T^{-3/4}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \leq \sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F.$$

因此，我们有

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \leq \frac{f(\mathbf{X}_1) - f^\star + C_1(1+\ln T) + C_2}{a\chi_\varepsilon \cdot T^{1/4}}.$$

这完成了证明。 □

定理3.5和推论3.6的H引理

H.1 引理H.1

引理H.1. 令

$$\mathbf{D}_{r,t} := \mathrm{diag}\big(\mathrm{rowsum}(\mathbf{M}_t \odot \mathbf{M}_t)\big), \quad \mathbf{P}_{r,t} := (\mathbf{D}_{r,t}^{1/2})^\dagger, \quad \mathbf{R}_t := \mathbf{P}_{r,t}\mathbf{M}_t, \quad \mathbf{O}_t := \mathrm{Orth}(\mathbf{R}_t),$$

其中 † 表示 *Moore-Penrose* 伪逆，因此经过行归一化后，$\mathbf{M}_t$ 的零行仍然为零，并且 $\mathrm{Orth}(\mathbf{0}) := \mathbf{0}$。然后，对于所有 t，

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\sqrt{m}}\|\mathbf{M}_t\|_F, \qquad \|\mathbf{O}_t\|_F^2 = \mathrm{rank}(\mathbf{R}_t) \leq n.$$

证明。*(i)*如果 $\mathbf{M}_t = \mathbf{0}$，那么 $\mathbf{R}_t = \mathbf{O}_t = \mathbf{0}$，因此该命题立即成立。

*(ii)*假设 $\mathbf{M}_t \neq \mathbf{0}$。令

$$d_{i,t} := \|(\mathbf{M}_t)_{i,:}\|_2, \qquad \mathbf{D}_t := \mathrm{diag}(d_{1,t}, \ldots, d_{m,t}),$$

因此 $\mathbf{D}_t = \mathbf{D}_{r\,t}^{1/2}$，并且 $\mathbf{M}_t = \mathbf{D}_t\mathbf{R}_t$。令 $\mathbf{H}_t := (\mathbf{R}_t\mathbf{R}_t^\top)^{1/2}$。如果 $\mathbf{R}_t = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top$ 是一个紧致奇异值分解，那么 $\mathbf{O}_t = \mathbf{U}_t\mathbf{V}_t^\top$，因此 $\mathbf{O}_t\mathbf{R}_t^\top = \mathbf{H}_t$。因此

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle = \mathrm{tr}(\mathbf{O}_t^\top\mathbf{D}_t\mathbf{R}_t) = \mathrm{tr}(\mathbf{D}_t\mathbf{O}_t\mathbf{R}_t^\top) = \mathrm{tr}(\mathbf{D}_t\mathbf{H}_t) = \sum_{i=1}^{m} d_{i,t}(\mathbf{H}_t)_{ii}.$$

如果 $d_{i,t} > 0$，那么 $\mathbf{R}_t$ 的第 i 行具有单位欧几里得范数，因此

$$1 = \mathbf{e}_i^\top\mathbf{R}_t\mathbf{R}_t^\top\mathbf{e}_i = \mathbf{e}_i^\top\mathbf{H}_t^2\mathbf{e}_i \leq \|\mathbf{H}_t\|_2\mathbf{e}_i^\top\mathbf{H}_t\mathbf{e}_i = \|\mathbf{R}_t\|_2(\mathbf{H}_t)_{ii}.$$

Thus $(\mathbf{H}_t)_{ii} \geq \|\mathbf{R}_t\|_2^{-1}$，只要 $d_{i,t} > 0$。如果 $d_{i,t} = 0$，则相应的项消失。因此 e,

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\|\mathbf{R}_t\|_2}\sum_{i=1}^{m} d_{i,t}.$$

现在，$\mathbf{R}$ t 1, $\|\ \ _t\|_F^2 \leq m$ $\|\ \ _t\|_2 \leq \sqrt{m}$ $\sum_{i=1}^{m} d_{i\,t} \geq$ 的每一行非零行都有范数，因此 $\mathbf{R}$，所以 $\mathbf{R}$。此外，$\mathbf{M}$。由此可知

$$\langle \mathbf{M}_t, \mathbf{O}_t\rangle \geq \frac{1}{\sqrt{m}}\|\mathbf{M}_t\|_F.$$

最后，如果 $r_t := \mathrm{rank}(\mathbf{R}_t)$，那么 $\mathbf{O}_t$ 的非零奇异值都等于 1，所以 $\|\mathbf{O}_t\|_F^2 = r_t \leq n$。 □

H.2 Lemma H.2

Lemma H.2. *Let $\{\hat{\mathbf{M}}_t\}_{t=1}^T \subset \mathbb{R}^{m\times n}$ be any finite algorithmic trajectory, and set $\mathrm{Orth}(\mathbf{0}) := \mathbf{0}$. Consider the five-step pre-scaled Newton–Schulz map with the degree-two Taylor coefficients analyzed in [15, Definition 2]. For any $\mathbf{A} \in \mathbb{R}^{m\times n}$, define*

$$\alpha(\mathbf{A}) := \max\{1, \|\mathbf{A}\|_F\}, \qquad \mathbf{Y}^{(0)}(\mathbf{A}) := \frac{\mathbf{A}}{\alpha(\mathbf{A})},$$

and, for $k = 0, 1, 2, 3, 4$,

$$\mathbf{Y}^{(k+1)}(\mathbf{A}) = p_{\mathrm{ns}}\big(\mathbf{Y}^{(k)}(\mathbf{A})\mathbf{Y}^{(k)}(\mathbf{A})^\top\big)\mathbf{Y}^{(k)}(\mathbf{A}),$$

where

$$p_{\mathrm{ns}}(\mathbf{Z}) := \frac{15}{8}\mathbf{I}_m - \frac{5}{4}\mathbf{Z} + \frac{3}{8}\mathbf{Z}^2.$$

We define

$$\mathrm{NS5}(\mathbf{A}) := \mathbf{Y}^{(5)}(\mathbf{A}).$$

For each t, let Π_t be the orthogonal projector onto $\mathrm{range}(\hat{\mathbf{M}}_t)$, and define

$$\delta_{t,0} := \big\|\Pi_t - \mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)^\top\big\|_2, \qquad \delta_{0,T} := \max_{1\le t\le T} \delta_{t,0}.$$

Then $\delta_{0,T} < 1$. Moreover, the trajectory-wise NS5 polar-approximation error

$$\varepsilon_{\mathrm{ns},T} := \max_{1\le t\le T} \big\|\mathrm{NS5}(\hat{\mathbf{M}}_t) - \mathrm{Orth}(\hat{\mathbf{M}}_t)\big\|_2 < 1.$$

Proof. If $\hat{\mathbf{M}}_t = \mathbf{0}$, then $\Pi_t = \mathbf{0}$ and $\delta_{t,0} = 0$. Otherwise, the pre-scaling by $\alpha(\hat{\mathbf{M}}_t) = \max\{1, \|\hat{\mathbf{M}}_t\|_F\}$ gives

$$0 < \lambda_{\min}^{+}\big(\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)^\top\big) \le 1$$

on $\mathrm{range}(\hat{\mathbf{M}}_t)$, and hence $\delta_{t,0} < 1$. Since the horizon T is finite, $\delta_{0,T} = \max_{1\le t\le T} \delta_{t,0} < 1$.

The polynomial above is the degree-two Taylor Newton–Schulz polynomial in [15, Definition 2]. Applying [15, Theorem 2] with $\kappa = 2$ and $q = 5$ yields

$$\varepsilon_{\mathrm{ns},T} \le 1 - \sqrt{1 - \delta_{0,T}^{(\kappa+1)^q}} = 1 - \sqrt{1 - \delta_{0,T}^{3^5}}.$$

Finally, since $1 - \sqrt{1-x} \le x$ for $x \in [0, 1]$ and $\delta_{0,T} < 1$, we obtain

$$\varepsilon_{\mathrm{ns},T} \le \delta_{0,T}^{3^5} < 1.$$

□

H.3 Lemma H.3

Lemma H.3. *Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. If $\mathbf{A} = \mathbf{0}$, define $\mathrm{Orth}(\mathbf{A}) = \mathbf{0}$ and $\mathrm{NS5}(\mathbf{A}) = \mathbf{0}$. Otherwise, let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the compact SVD, and set $\mathbf{Q} := \mathrm{Orth}(\mathbf{A}) = \mathbf{U}\mathbf{V}^\top$ and $\mathbf{O} := \mathrm{NS5}(\mathbf{A})$. Then there exists a diagonal matrix $\widetilde{\mathbf{\Sigma}}$ such that $\mathbf{O} = \mathbf{U}\widetilde{\mathbf{\Sigma}}\mathbf{V}^\top$. Moreover, for every row-normalized input $\mathbf{A} = \hat{\mathbf{M}}_t$ along the algorithmic trajectory,*

$$\|\mathrm{NS5}(\mathbf{A}) - \mathrm{Orth}(\mathbf{A})\|_2 \le \varepsilon_{\mathrm{ns}}.$$

Consequently, every diagonal entry $\widetilde{\sigma}_i$ of $\widetilde{\mathbf{\Sigma}}$ satisfies

$$1 - \varepsilon_{\mathrm{ns}} \le \widetilde{\sigma}_i \le 1 + \varepsilon_{\mathrm{ns}}, \qquad i \in [\mathrm{rank}(\mathbf{A})].$$

In particular, whenever $\mathbf{A} = \hat{\mathbf{M}}_t \neq \mathbf{0}$, we have $\mathrm{Orth}(\mathrm{NS5}(\mathbf{A})) = \mathrm{Orth}(\mathbf{A})$.

H.2引理H.2

引理 H.2。设 $\{\hat{\mathbf{M}}_t\}_{t=1}^T \subset \mathbb{R}^{m\times n}$ 为任意有限算法轨迹，并定义 $\mathrm{Orth}(\mathbf{0}) := \mathbf{0}$。考虑在 *[15, 定义 2]* 中分析的五步预缩放牛顿-舒尔茨映射。对于任意 $\mathbf{A} \in \mathbb{R}^{m\times n}$，定义

$$\alpha(\mathbf{A}) := \max\{1, \|\mathbf{A}\|_F\}, \qquad \mathbf{Y}^{(0)}(\mathbf{A}) := \frac{\mathbf{A}}{\alpha(\mathbf{A})},$$

以及，对于 $k = 0$，1、2、3、4，

$$\mathbf{Y}^{(k+1)}(\mathbf{A}) = p_{\mathrm{ns}}\big(\mathbf{Y}^{(k)}(\mathbf{A})\mathbf{Y}^{(k)}(\mathbf{A})^\top\big)\mathbf{Y}^{(k)}(\mathbf{A}),$$

其中

$$p_{\mathrm{ns}}(\mathbf{Z}) := \frac{15}{8}\mathbf{I}_m - \frac{5}{4}\mathbf{Z} + \frac{3}{8}\mathbf{Z}^2.$$

我们定义

$$\mathrm{NS5}(\mathbf{A}) := \mathbf{Y}^{(5)}(\mathbf{A}).$$

对于每个 t，令 Π_t 是到<code>range(ˆM_t)</code>的正交投影器，并定义

$$\delta_{t,0} := \big\|\Pi_t - \mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)^\top\big\|_2, \qquad \delta_{0,T} := \max_{1\le t\le T} \delta_{t,0}.$$

那么$\delta_{0,\ T} < 1$。此外，轨迹方向的*NS5*极近似误差

$$\varepsilon_{\mathrm{ns},T} := \max_{1\le t\le T} \big\|\mathrm{NS5}(\hat{\mathbf{M}}_t) - \mathrm{Orth}(\hat{\mathbf{M}}_t)\big\|_2 < 1.$$

证明。如果 $\hat{\mathbf{M}}_t = \mathbf{0}$，那么 $\Pi_t = \mathbf{0}$ 和 $\delta_{t,\ 0} = 0$。否则，通过 $\alpha(\hat{\mathbf{M}}_t) = \max\{1, \|\hat{\mathbf{M}}_t\|_F\}$ 给出

$$0 < \lambda_{\min}^{+}\big(\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)\mathbf{Y}^{(0)}(\hat{\mathbf{M}}_t)^\top\big) \le 1$$

在范围($\hat{\mathbf{M}}_t$)上，并且因此 $\delta_{t,0} < 1$。由于地平线 T是有限的,$\delta_{0,T} = \max_{1\le t\le T} \delta_{t,0} < 1$。

上述多项式是 [15, 定义 2]中的二阶泰勒牛顿-舒尔茨多项式。应用 [15, 定理 2]，并使用 $\kappa = 2$ 和 $q = 5$，得到

$$\varepsilon_{\mathrm{ns},T} \le 1 - \sqrt{1 - \delta_{0,T}^{(\kappa+1)^q}} = 1 - \sqrt{1 - \delta_{0,T}^{3^5}}.$$

最后，由于 $1 - \sqrt{1-x} \le x$ 对 $x \in [0, 1]$ 和 $\delta_{0,T} < 1$，我们得到

$$\varepsilon_{\mathrm{ns},T} \le \delta_{0,T}^{3^5} < 1.$$

□

H.3引理H.3

H.3 引理 *H.3*。设 $\mathbf{A} \in \mathbb{R}^{m\times n}$。若 $\mathbf{A} = \mathbf{0}$，定义 $\mathrm{Orth(A)} = \mathbf{0}$ 和 $\mathrm{NS5(A)} = \mathbf{0}$。否则，设 $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ 为紧凑*SVD*，并令 $\mathbf{Q} := \mathrm{Orth(A)} = \mathbf{U}\mathbf{V}^\top$ 和 $\mathbf{O} := \mathrm{NS5(A)}$。那么存在一个对角矩阵 $\widetilde{\mathbf{\Sigma}}$，使得 $\mathbf{O} = \mathbf{U}\widetilde{\mathbf{\Sigma}}\mathbf{V}^\top$. 。此外，对于算法轨迹上的任意行归一化输入 $\hat{\mathbf{A}} = \mathbf{M}_t$，

$$\|\mathrm{NS5}(\mathbf{A}) - \mathrm{Orth}(\mathbf{A})\|_2 \le \varepsilon_{\mathrm{ns}}.$$

˜因此，每个对角矩阵 $\widetilde{\sigma}_i$ 的 $\mathbf{\Sigma}$ 都满足

$$1 - \varepsilon_{\mathrm{ns}} \le \widetilde{\sigma}_i \le 1 + \varepsilon_{\mathrm{ns}}, \qquad i \in [\mathrm{rank}(\mathbf{A})].$$

特别是，每当 $\mathbf{A} = \hat{\mathbf{M}}_t \neq \mathbf{0}$ 时，我们有 $\mathrm{Orth}(\mathrm{NS5}(\mathbf{A})) = \mathrm{Orth}(\mathbf{A})$。

Proof. If $\mathbf{A} = \mathbf{0}$, the conclusion is immediate by definition. Assume $\mathbf{A} \neq \mathbf{0}$.

The internal scaling multiplies $\mathbf{A}$ by a positive scalar and hence does not change its left or right singular vectors. Thus

$$\mathbf{Y}^{(0)}(\mathbf{A}) = \mathbf{U}\mathbf{\Sigma}^{(0)}\mathbf{V}^\top$$

for some diagonal matrix $\mathbf{\Sigma}^{(0)}$. Assume inductively that

$$\mathbf{Y}^{(k)}(\mathbf{A}) = \mathbf{U}\mathbf{\Sigma}^{(k)}\mathbf{V}^\top$$

with diagonal $\mathbf{\Sigma}^{(k)}$. Then

$$\mathbf{Y}^{(k)}(\mathbf{A})\mathbf{Y}^{(k)}(\mathbf{A})^\top = \mathbf{U}\big(\mathbf{\Sigma}^{(k)}\big)^2\mathbf{U}^\top,$$

and therefore

$$\mathbf{Y}^{(k+1)}(\mathbf{A}) = \mathbf{U}p_{\mathrm{ns}}\big((\mathbf{\Sigma}^{(k)})^2\big)\mathbf{\Sigma}^{(k)}\mathbf{V}^\top.$$

Hence every iterate has the same left and right singular vectors as $\mathbf{A}$, and in particular

$$\mathrm{NS5}(\mathbf{A}) = \mathbf{U}\widetilde{\mathbf{\Sigma}}\mathbf{V}^\top$$

for some diagonal matrix $\widetilde{\mathbf{\Sigma}}$.

Now, if $\mathbf{A} = \hat{\mathbf{M}}_t$ is a row-normalized input on the algorithmic trajectory, then by the definition of $\varepsilon_{\mathrm{ns}}$,

$$\|\mathrm{NS5}(\mathbf{A}) - \mathrm{Orth}(\mathbf{A})\|_2 = \|\mathbf{U}(\widetilde{\mathbf{\Sigma}} - \mathbf{I})\mathbf{V}^\top\|_2 = \|\widetilde{\mathbf{\Sigma}} - \mathbf{I}\|_2 \le \varepsilon_{\mathrm{ns}}.$$

Hence

$$\max_i |\widetilde{\sigma}_i - 1| \le \varepsilon_{\mathrm{ns}},$$

which is equivalent to

$$1 - \varepsilon_{\mathrm{ns}} \le \widetilde{\sigma}_i \le 1 + \varepsilon_{\mathrm{ns}} \qquad \text{for all } i.$$

Since $\varepsilon_{\mathrm{ns}} < 1$, these diagonal entries are strictly positive. Therefore

$$\mathrm{Orth}(\mathrm{NS5}(\mathbf{A})) = \mathrm{Orth}(\mathbf{A})$$

whenever $\mathbf{A} = \hat{\mathbf{M}}_t \neq \mathbf{0}$. □

H.4 Lemma H.4

Lemma H.4. *Under Lemma H.3, for all t,*

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle \ge \frac{1 - \varepsilon_{\mathrm{ns}}}{\sqrt{m}}\|\mathbf{M}_t\|_F, \qquad \|\mathbf{O}_t\|_F^2 \le (1 + \varepsilon_{\mathrm{ns}})^2 n.$$

Proof. If $\mathbf{M}_t = \mathbf{0}$, then $\hat{\mathbf{M}}_t = \mathbf{O}_t = \mathbf{0}$ and the claim is immediate. Assume $\mathbf{M}_t \neq \mathbf{0}$. We have

$$d_{i,t} := \|(\mathbf{M}_t)_{i,:}\|_2, \qquad \mathbf{D}_t := \mathbf{D}_{r,t}^{1/2} = \mathrm{diag}(d_{1,t}, \dots, d_{m,t}),$$

so that

$$\mathbf{M}_t = \mathbf{D}_t\hat{\mathbf{M}}_t.$$

Let

$$\hat{\mathbf{M}}_t = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top, \qquad \mathbf{H}_t := (\hat{\mathbf{M}}_t\hat{\mathbf{M}}_t^\top)^{1/2} = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top.$$

By Lemma H.3,

$$\mathbf{O}_t = \mathbf{U}_t\widetilde{\mathbf{\Sigma}}_t\mathbf{V}_t^\top$$

with diagonal entries of $\widetilde{\mathbf{\Sigma}}_t$ lying in $[1 - \varepsilon_{\mathrm{ns}}, 1 + \varepsilon_{\mathrm{ns}}]$. Therefore,

$$\mathbf{O}_t\hat{\mathbf{M}}_t^\top = \mathbf{U}_t\widetilde{\mathbf{\Sigma}}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top \succeq (1 - \varepsilon_{\mathrm{ns}})\mathbf{U}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top = (1 - \varepsilon_{\mathrm{ns}})\mathbf{H}_t.$$

Hence

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle = \mathrm{tr}(\mathbf{O}_t^\top\mathbf{D}_t\hat{\mathbf{M}}_t) = \mathrm{tr}(\mathbf{D}_t\mathbf{O}_t\hat{\mathbf{M}}_t^\top) \ge (1 - \varepsilon_{\mathrm{ns}})\,\mathrm{tr}(\mathbf{D}_t\mathbf{H}_t) = (1 - \varepsilon_{\mathrm{ns}})\sum_{i=1}^m d_{i,t}(\mathbf{H}_t)_{ii}.$$

证明。 如果 $\mathbf{A} = \mathbf{0}$，则根据定义结论立即成立。假设 $\mathbf{A} \neq \mathbf{0}$。

内部缩放乘以 $\mathbf{A}$ 的正标量，因此不会改变其左或右奇异向量。因此

$$\mathbf{Y}^{(0)}(\mathbf{A}) = \mathbf{U}\mathbf{\Sigma}^{(0)}\mathbf{V}^\top$$

对于某个对角矩阵 $\mathbf{\Sigma}^{(0)}$。假设归纳地，

$$\mathbf{Y}^{(k)}(\mathbf{A}) = \mathbf{U}\mathbf{\Sigma}^{(k)}\mathbf{V}^\top$$

对角线为 $\mathbf{\Sigma}^{(k)}$。那么

$$\mathbf{Y}^{(k)}(\mathbf{A})\mathbf{Y}^{(k)}(\mathbf{A})^\top = \mathbf{U}\big(\mathbf{\Sigma}^{(k)}\big)^2\mathbf{U}^\top,$$

因此

$$\mathbf{Y}^{(k+1)}(\mathbf{A}) = \mathbf{U}p_{\mathrm{ns}}\big((\mathbf{\Sigma}^{(k)})^2\big)\mathbf{\Sigma}^{(k)}\mathbf{V}^\top.$$

因此每个迭代都具有与 $\mathbf{A}$ 相同的左和右奇异向量，特别是

$$\mathrm{NS5}(\mathbf{A}) = \mathbf{U}\widetilde{\mathbf{\Sigma}}\mathbf{V}^\top$$

˜对于某个对角矩阵 $\mathbf{\Sigma}$。

ˆ现在，如果 $\mathbf{A} = \mathbf{M}_t$ 是算法轨迹上的行归一化输入，那么根据$\varepsilon_{\mathrm{ns}}$的定义，

$$\|\mathrm{NS5}(\mathbf{A}) - \mathrm{Orth}(\mathbf{A})\|_2 = \|\mathbf{U}(\widetilde{\mathbf{\Sigma}} - \mathbf{I})\mathbf{V}^\top\|_2 = \|\widetilde{\mathbf{\Sigma}} - \mathbf{I}\|_2 \le \varepsilon_{\mathrm{ns}}.$$

因此

$$\max_i |\widetilde{\sigma}_i - 1| \le \varepsilon_{\mathrm{ns}},$$

这等价于

$$1 - \varepsilon_{\mathrm{ns}} \le \widetilde{\sigma}_i \le 1 + \varepsilon_{\mathrm{ns}} \qquad \text{for all } i.$$

由于 $\varepsilon_{\mathrm{ns}} < 1$，这些对角线元素都是严格正的。因此

$$\mathrm{Orth}(\mathrm{NS5}(\mathbf{A})) = \mathrm{Orth}(\mathbf{A})$$

ˆ只要$\mathbf{A} = \mathbf{M}_t \neq \mathbf{0}$。 □

H.4 定理 H.4

引理 H.4。 在引理 *H.3* 下，对于所有 t,

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle \ge \frac{1 - \varepsilon_{\mathrm{ns}}}{\sqrt{m}}\|\mathbf{M}_t\|_F, \qquad \|\mathbf{O}_t\|_F^2 \le (1 + \varepsilon_{\mathrm{ns}})^2 n.$$

ˆ证明。 如果 $\mathbf{M}_t = \mathbf{0}$，那么$\mathbf{M}_t = \mathbf{O}_t = \mathbf{0}$ 并且结论立即成立。假设 $\mathbf{M}_t \neq \mathbf{0}$。我们有

$$d_{i,t} := \|(\mathbf{M}_t)_{i,:}\|_2, \qquad \mathbf{D}_t := \mathbf{D}_{r,t}^{1/2} = \mathrm{diag}(d_{1,t}, \dots, d_{m,t}),$$

所以

$$\mathbf{M}_t = \mathbf{D}_t\hat{\mathbf{M}}_t.$$

Let

$$\hat{\mathbf{M}}_t = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top, \qquad \mathbf{H}_t := (\hat{\mathbf{M}}_t\hat{\mathbf{M}}_t^\top)^{1/2} = \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top.$$

根据引理 H.3,

$$\mathbf{O}_t = \mathbf{U}_t\widetilde{\mathbf{\Sigma}}_t\mathbf{V}_t^\top$$

˜其对角线元素为 $\mathbf{\Sigma}_t$，且属于 $[1 - \varepsilon_{\mathrm{ns}}, 1 + \varepsilon_{\mathrm{ns}}]$。因此,

$$\mathbf{O}_t\hat{\mathbf{M}}_t^\top = \mathbf{U}_t\widetilde{\mathbf{\Sigma}}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top \succeq (1 - \varepsilon_{\mathrm{ns}})\mathbf{U}_t\mathbf{\Sigma}_t\mathbf{U}_t^\top = (1 - \varepsilon_{\mathrm{ns}})\mathbf{H}_t.$$

因此

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle = \mathrm{tr}(\mathbf{O}_t^\top\mathbf{D}_t\hat{\mathbf{M}}_t) = \mathrm{tr}(\mathbf{D}_t\mathbf{O}_t\hat{\mathbf{M}}_t^\top) \ge (1 - \varepsilon_{\mathrm{ns}})\,\mathrm{tr}(\mathbf{D}_t\mathbf{H}_t) = (1 - \varepsilon_{\mathrm{ns}})\sum_{i=1}^m d_{i,t}(\mathbf{H}_t)_{ii}.$$

If $d_{i,t} > 0$, then the i-th row of $\hat{\mathbf{M}}_t$ has unit Euclidean norm, so

$$1 = \mathbf{e}_i^\top \hat{\mathbf{M}}_t \hat{\mathbf{M}}_t^\top \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{H}_t^2 \mathbf{e}_i \le \|\mathbf{H}_t\|_2 \mathbf{e}_i^\top \mathbf{H}_t \mathbf{e}_i = \|\hat{\mathbf{M}}_t\|_2 (\mathbf{H}_t)_{ii}.$$

Thus $(\mathbf{H}_t)_{ii} \ge \|\hat{\mathbf{M}}_t\|_2^{-1}$ whenever $d_{i,t} > 0$. If $d_{i,t} = 0$, the corresponding term vanishes. Consequently,

$$\operatorname{tr}(\mathbf{D}_t \mathbf{H}_t) \ge \|\hat{\mathbf{M}}_t\|_2^{-1} \sum_{i=1}^{m} d_{i,t}.$$

Now each nonzero row of $\hat{\mathbf{M}}_t$ has norm 1, so $\|\hat{\mathbf{M}}_t\|_F^2 \le m$, hence $\|\hat{\mathbf{M}}_t\|_2 \le \sqrt{m}$. Also,

$$\sum_{i=1}^{m} d_{i,t} \ge \Big(\sum_{i=1}^{m} d_{i,t}^2 \Big)^{1/2} = \|\mathbf{M}_t\|_F.$$

Combining the above bounds gives

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle \ge \frac{1 - \varepsilon_{\mathrm{ns}}}{\sqrt{m}} \|\mathbf{M}_t\|_F.$$

For the norm bound, let $r_t := \operatorname{rank}(\hat{\mathbf{M}}_t)$. Again by Lemma H.3, every nonzero singular value of $\mathbf{O}_t$ is at most $1 + \varepsilon_{\mathrm{ns}}$. Therefore

$$\|\mathbf{O}_t\|_F^2 = \sum_{i=1}^{r_t} \tilde{\sigma}_{i,t}^2 \le (1 + \varepsilon_{\mathrm{ns}})^2 r_t \le (1 + \varepsilon_{\mathrm{ns}})^2 n.$$

This completes the proof. □

H.5 Lemma H.5

Lemma H.5. *Consider the row-normalized update with decoupled weight decay*

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t (a \mathbf{O}_t + \lambda_t \mathbf{X}_t),$$

where $\mathbf{O}_t = \mathrm{NS5}(\hat{\mathbf{M}}_t)$ *and Lemma H.4 holds. Let* $\gamma := 1 - \varepsilon_{\mathrm{ns}}, 0 \le \rho < \frac{a\gamma}{\sqrt{m}}$*, and assume*

$$0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n}} t^{-1/4}. \tag{7}$$

Then, along every realization,

$$\lambda_t \|\mathbf{X}_t\|_F \le \rho, \qquad \|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F \le (a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} + \rho) \cdot \eta_t.$$

Proof. Let $\lambda_0 := \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}}$. Since $\rho < a\gamma_{\mathrm{ns}}/\sqrt{m} \le a/\sqrt{m}$, we have

$$\lambda_0 \le \frac{1}{4(1 + \varepsilon_{\mathrm{ns}})\sqrt{mn}} \le 1.$$

Hence

$$0 \le \eta_t \lambda_t \le \lambda_0 t^{-1} \le 1.$$

Using Lemma H.4,

$$\|\mathbf{O}_t\|_F \le (1 + \varepsilon_{\mathrm{ns}})\sqrt{n}.$$

Therefore,

$$\begin{aligned} \|\mathbf{X}_{t+1}\|_F &= \|(1 - \eta_t \lambda_t)\mathbf{X}_t - a\eta_t \mathbf{O}_t\|_F \\ &\le (1 - \eta_t \lambda_t)\|\mathbf{X}_t\|_F + a\eta_t \|\mathbf{O}_t\|_F \\ &\le \|\mathbf{X}_t\|_F + a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n}\eta_t. \end{aligned}$$

By induction and $\sum_{s=1}^{t-1} s^{-3/4} \le 4t^{1/4}$,

$$\|\mathbf{X}_t\|_F \le \|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} t^{1/4}.$$

Thus

$$\lambda_t \|\mathbf{X}_t\|_F \le \lambda_0 t^{-1/4} \Big(\|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} t^{1/4} \Big) \le \rho.$$

Finally,

$$\begin{aligned} \|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F &= \eta_t \|a\mathbf{O}_t + \lambda_t \mathbf{X}_t\|_F \\ &\le \eta_t \Big(a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} + \rho \Big) \end{aligned}$$

This completes the proof. □

ˆ 如果 $d_{i,t} > 0$，那么$\mathbf{M}_t$ 的第 i-行具有单位欧几里得范数，所以

$$1 = \mathbf{e}_i^\top \hat{\mathbf{M}}_t \hat{\mathbf{M}}_t^\top \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{H}_t^2 \mathbf{e}_i \le \|\mathbf{H}_t\|_2 \mathbf{e}_i^\top \mathbf{H}_t \mathbf{e}_i = \|\hat{\mathbf{M}}_t\|_2 (\mathbf{H}_t)_{ii}.$$

因此（$\mathbf{H}_t$)$_{ii} \ge \| \hat{\mathbf{M}}_t\|_2^{-1}$ whenever $d_{i,t} > 0$。如果 $d_{i,t} = 0$，则相应的项消失。因此，

$$\operatorname{tr}(\mathbf{D}_t \mathbf{H}_t) \ge \|\hat{\mathbf{M}}_t\|_2^{-1} \sum_{i=1}^{m} d_{i,t}.$$

现在 $\hat{\mathbf{M}}_t$ 的每一行非零行的范数为1，所以 $\| \hat{\mathbf{M}}_t\|_F^2 \le m$，因此 $\| \hat{\mathbf{M}}_t\|_2 \le \sqrt{m}$。此外，

$$\sum_{i=1}^{m} d_{i,t} \ge \Big(\sum_{i=1}^{m} d_{i,t}^2 \Big)^{1/2} = \|\mathbf{M}_t\|_F.$$

结合上述界限得到

$$\langle \mathbf{M}_t, \mathbf{O}_t \rangle \ge \frac{1 - \varepsilon_{\mathrm{ns}}}{\sqrt{m}} \|\mathbf{M}_t\|_F.$$

对于范数界限，设 $r_t :=$ 秩($\hat{\mathbf{M}}_t$)。再次根据引理 H.3，$\mathbf{O}_t$ 的每个非零奇异值至多为 $1 + \varepsilon_{\mathrm{ns}}$。因此

$$\|\mathbf{O}_t\|_F^2 = \sum_{i=1}^{r_t} \tilde{\sigma}_{i,t}^2 \le (1 + \varepsilon_{\mathrm{ns}})^2 r_t \le (1 + \varepsilon_{\mathrm{ns}})^2 n.$$

这完成了证明。 □

H.5 引理 H.5

引理 H.5。 考虑行归一化的更新，其中权重衰减是解耦的

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t (a \mathbf{O}_t + \lambda_t \mathbf{X}_t),$$

其中 $\mathbf{O}_t = \mathrm{NS5}(\hat{\mathbf{M}}_t)$，并且引理 *H.4* 成立。令 $\gamma := 1 - \varepsilon_{\mathrm{ns}}, 0 \le \rho < \frac{a\gamma}{\sqrt{m}}$，并假设

$$0 \le \lambda_t \le \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n}} t^{-1/4}. \tag{7}$$

那么，对于每一个实现，

$$\lambda_t \|\mathbf{X}_t\|_F \le \rho, \qquad \|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F \le (a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} + \rho) \cdot \eta_t.$$

证明。 令 $\lambda_0 := \frac{\rho}{\|\mathbf{X}_1\|_F + 4a(1+\varepsilon_{\mathrm{ns}})\sqrt{n}}$。因为 $\rho < a\gamma_{\mathrm{ns}}/\sqrt{m} \le a/\sqrt{m}$，所以我们有

$$\lambda_0 \le \frac{1}{4(1 + \varepsilon_{\mathrm{ns}})\sqrt{mn}} \le 1.$$

因此

$$0 \le \eta_t \lambda_t \le \lambda_0 t^{-1} \le 1.$$

利用引理 H.4,

$$\|\mathbf{O}_t\|_F \le (1 + \varepsilon_{\mathrm{ns}})\sqrt{n}.$$

因此，

$$\begin{aligned} \|\mathbf{X}_{t+1}\|_F &= \|(1 - \eta_t \lambda_t)\mathbf{X}_t - a\eta_t \mathbf{O}_t\|_F \\ &\le (1 - \eta_t \lambda_t)\|\mathbf{X}_t\|_F + a\eta_t \|\mathbf{O}_t\|_F \\ &\le \|\mathbf{X}_t\|_F + a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n}\eta_t. \end{aligned}$$

通过归纳和 $\sum_{s=1}^{t-1} s^{-3/4} \le 4t^{1/4}$,

$$\|\mathbf{X}_t\|_F \le \|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} t^{1/4}.$$

Thus

$$\lambda_t \|\mathbf{X}_t\|_F \le \lambda_0 t^{-1/4} \Big(\|\mathbf{X}_1\|_F + 4a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} t^{1/4} \Big) \le \rho.$$

最后，

$$\begin{aligned} \|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F &= \eta_t \|a\mathbf{O}_t + \lambda_t \mathbf{X}_t\|_F \\ &\le \eta_t \Big(a(1 + \varepsilon_{\mathrm{ns}})\sqrt{n} + \rho \Big) \end{aligned}$$

这完成了证明。 □

I Proofs of Theorem 3.5 and Corollary 3.6

Proof. Let $\mathbf{S}_t := \nabla f(\mathbf{X}_t) - \mathbf{M}_t, \gamma := 1 - \varepsilon_{\text{ns}}$, and define $c^{\text{ns}}_{m,n} := \frac{\gamma}{\sqrt{m}} + (1+\varepsilon_{\text{ns}})\sqrt{n}$.

By (7), we have $\frac{a\gamma}{\sqrt{m}} - \rho > 0$. By Assumption 3.2 and the descent lemma,

$$\begin{aligned} f(\mathbf{X}_{t+1}) &\le f(\mathbf{X}_t) + \langle \nabla f(\mathbf{X}_t), \mathbf{X}_{t+1} - \mathbf{X}_t \rangle + \frac{L}{2}\|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F^2 \\ &= f(\mathbf{X}_t) - a\eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle - \lambda_t \eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{X}_t \rangle + \frac{L\eta_t^2}{2}\|a\mathbf{O}_t + \lambda_t \mathbf{X}_t\|_F^2 \\ &\le f(\mathbf{X}_t) - a\eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle + \rho\eta_t \|\nabla f(\mathbf{X}_t)\|_F + \frac{L}{2}\big(a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho\big)^2 \eta_t^2, \end{aligned} \tag{8}$$

where the last step uses Lemma H.5 and Cauchy's inequality.

Again by Lemma H.4, we have

$$\begin{aligned} \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle &= \langle \mathbf{M}_t, \mathbf{O}_t \rangle + \langle \mathbf{S}_t, \mathbf{O}_t \rangle \\ &\ge \frac{\gamma}{\sqrt{m}} \|\mathbf{M}_t\|_F - \|\mathbf{S}_t\|_F \|\mathbf{O}_t\|_F \\ &\ge \frac{\gamma}{\sqrt{m}} \|\mathbf{M}_t\|_F - (1+\varepsilon_{\text{ns}})\sqrt{n} \|\mathbf{S}_t\|_F \\ &\ge \frac{\gamma}{\sqrt{m}} \|\nabla f(\mathbf{X}_t)\|_F - \Big(\frac{\gamma}{\sqrt{m}} + (1+\varepsilon_{\text{ns}})\sqrt{n}\Big) \|\mathbf{S}_t\|_F. \end{aligned} \tag{9}$$

Substituting (9) into (8), we have

$$f(\mathbf{X}_{t+1}) \le f(\mathbf{X}_t) - \left(\frac{a\gamma}{\sqrt{m}} - \rho\right) \eta_t \|\nabla f(\mathbf{X}_t)\|_F + a c^{\text{ns}}_{m,n} \eta_t \|\mathbf{S}_t\|_F + \frac{L}{2}\big(a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho\big)^2 \eta_t^2. \tag{10}$$

Apply Young's inequality with parameter h_{t+1}/L:

$$a c^{\text{ns}}_{m,n} \eta_t \|\mathbf{S}_t\|_F \le \frac{a h_{t+1}}{2L} \|\mathbf{S}_t\|_F^2 + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \frac{\eta_t^2}{h_{t+1}}.$$

Let $d^{\text{ns}}_\rho := a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho$ and $\kappa^{\text{ns}}_\rho := \frac{a\gamma}{\sqrt{m}} - \rho$. Hence

$$\begin{aligned} f(\mathbf{X}_{t+1}) \le & f(\mathbf{X}_t) - \kappa^{\text{ns}}_\rho \eta_t \|\nabla f(\mathbf{X}_t)\|_F + \frac{a h_{t+1}}{2L} \|\mathbf{S}_t\|_F^2 \\ & + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \frac{\eta_t^2}{h_{t+1}} + \frac{L}{2}\big(d^{\text{ns}}_\rho\big)^2 \eta_t^2. \end{aligned}$$

Taking expectations and summing from $t = 1$ to T, then using $f(\mathbf{X}_{T+1}) \ge f^\star$, we obtain

$$\begin{aligned} \kappa^{\text{ns}}_\rho \sum_{t=1}^T \eta_t \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le & f(\mathbf{X}_1) - f^\star + \frac{a}{2L} \sum_{t=1}^T h_{t+1} \mathbb{E}\|\mathbf{S}_t\|_F^2 \\ & + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \sum_{t=1}^T \frac{\eta_t^2}{h_{t+1}} + \frac{L}{2}\big(d^{\text{ns}}_\rho\big)^2 \sum_{t=1}^T \eta_t^2. \end{aligned} \tag{11}$$

By Lemma F.3, using $\|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F \le d^{\text{ns}}_\rho \eta_t$ from Lemma H.5, we have

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 \le (1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2 + \frac{(1-h_{t+1})^2}{h_{t+1}} L^2 \big(d^{\text{ns}}_\rho\big)^2 \eta_t^2 + h_{t+1}^2 \sigma^2.$$

Since $(1-h_{t+1})^2 \le 1$ and

$$\frac{\eta_t^2}{h_{t+1}} = \frac{t^{-3/2}}{(t+1)^{-1/2}} = \frac{\sqrt{t+1}}{t^{3/2}} \le \frac{2\sqrt{2}}{t+1} = 2\sqrt{2} h_{t+1}^2,$$

we have

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 \le (1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2 + h_{t+1}^2 \Big(2\sqrt{2} L^2 \big(d^{\text{ns}}_\rho\big)^2 + \sigma^2\Big).$$

I 定理3.5和推论3.6的证明

证明。 设 $\mathbf{S}_t := \nabla f(\mathbf{X}_t) - \mathbf{M}_t, \gamma := 1 - \varepsilon_{\text{ns}}$, 并定义 $c^{\text{ns}}_{m,n} := \frac{\gamma}{\sqrt{m}} + (1+\varepsilon_{\text{ns}})\sqrt{n}$.

由(7)知，我们有 $\frac{a\gamma}{\sqrt{m}} - \rho > 0$。根据假设3.2和下降引理,

$$\begin{aligned} f(\mathbf{X}_{t+1}) &\le f(\mathbf{X}_t) + \langle \nabla f(\mathbf{X}_t), \mathbf{X}_{t+1} - \mathbf{X}_t \rangle + \frac{L}{2}\|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F^2 \\ &= f(\mathbf{X}_t) - a\eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle - \lambda_t \eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{X}_t \rangle + \frac{L\eta_t^2}{2}\|a\mathbf{O}_t + \lambda_t \mathbf{X}_t\|_F^2 \\ &\le f(\mathbf{X}_t) - a\eta_t \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle + \rho\eta_t \|\nabla f(\mathbf{X}_t)\|_F + \frac{L}{2}\big(a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho\big)^2 \eta_t^2, \end{aligned} \tag{8}$$

其中最后一步使用了引理H.5和柯西不等式。

再次根据引理H.4，我们有

$$\begin{aligned} \langle \nabla f(\mathbf{X}_t), \mathbf{O}_t \rangle &= \langle \mathbf{M}_t, \mathbf{O}_t \rangle + \langle \mathbf{S}_t, \mathbf{O}_t \rangle \\ &\ge \frac{\gamma}{\sqrt{m}} \|\mathbf{M}_t\|_F - \|\mathbf{S}_t\|_F \|\mathbf{O}_t\|_F \\ &\ge \frac{\gamma}{\sqrt{m}} \|\mathbf{M}_t\|_F - (1+\varepsilon_{\text{ns}})\sqrt{n} \|\mathbf{S}_t\|_F \\ &\ge \frac{\gamma}{\sqrt{m}} \|\nabla f(\mathbf{X}_t)\|_F - \Big(\frac{\gamma}{\sqrt{m}} + (1+\varepsilon_{\text{ns}})\sqrt{n}\Big) \|\mathbf{S}_t\|_F. \end{aligned} \tag{9}$$

将(9)代入(8)，我们有

$$f(\mathbf{X}_{t+1}) \le f(\mathbf{X}_t) - \left(\frac{a\gamma}{\sqrt{m}} - \rho\right) \eta_t \|\nabla f(\mathbf{X}_t)\|_F + a c^{\text{ns}}_{m,n} \eta_t \|\mathbf{S}_t\|_F + \frac{L}{2}\big(a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho\big)^2 \eta_t^2. \tag{10}$$

应用参数为 h_{t+1}/L的杨氏不等式:

$$a c^{\text{ns}}_{m,n} \eta_t \|\mathbf{S}_t\|_F \le \frac{a h_{t+1}}{2L} \|\mathbf{S}_t\|_F^2 + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \frac{\eta_t^2}{h_{t+1}}.$$

令 $d^{\text{ns}}_\rho := a(1+\varepsilon_{\text{ns}})\sqrt{n} + \rho$，且 $\kappa^{\text{ns}}_\rho := \frac{a\gamma}{\sqrt{m}} - \rho$.，因此

$$\begin{aligned} f(\mathbf{X}_{t+1}) \le & f(\mathbf{X}_t) - \kappa^{\text{ns}}_\rho \eta_t \|\nabla f(\mathbf{X}_t)\|_F + \frac{a h_{t+1}}{2L} \|\mathbf{S}_t\|_F^2 \\ & + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \frac{\eta_t^2}{h_{t+1}} + \frac{L}{2}\big(d^{\text{ns}}_\rho\big)^2 \eta_t^2. \end{aligned}$$

取期望并从 $t = 1$到 T求和，然后使用 $f(\mathbf{X}_{T+1}) \ge f^\star$，我们得到

$$\begin{aligned} \kappa^{\text{ns}}_\rho \sum_{t=1}^T \eta_t \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le & f(\mathbf{X}_1) - f^\star + \frac{a}{2L} \sum_{t=1}^T h_{t+1} \mathbb{E}\|\mathbf{S}_t\|_F^2 \\ & + \frac{aL}{2}\big(c^{\text{ns}}_{m,n}\big)^2 \sum_{t=1}^T \frac{\eta_t^2}{h_{t+1}} + \frac{L}{2}\big(d^{\text{ns}}_\rho\big)^2 \sum_{t=1}^T \eta_t^2. \end{aligned} \tag{11}$$

根据引理 F.3，使用 $\|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F \le d^{\text{ns}}_\rho \eta_t$ 和引理 H.5 中的 $\|\mathbf{X}_t\|_F \le d^{\text{ns}}_\rho \eta_t$，我们有

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 \le (1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2 + \frac{(1-h_{t+1})^2}{h_{t+1}} L^2 \big(d^{\text{ns}}_\rho\big)^2 \eta_t^2 + h_{t+1}^2 \sigma^2.$$

由于 $(1-h_{t+1})^2 \le 1$，且

$$\frac{\eta_t^2}{h_{t+1}} = \frac{t^{-3/2}}{(t+1)^{-1/2}} = \frac{\sqrt{t+1}}{t^{3/2}} \le \frac{2\sqrt{2}}{t+1} = 2\sqrt{2} h_{t+1}^2,$$

我们已有

$$\mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 \le (1-h_{t+1})\mathbb{E}\|\mathbf{S}_t\|_F^2 + h_{t+1}^2 \Big(2\sqrt{2} L^2 \big(d^{\text{ns}}_\rho\big)^2 + \sigma^2\Big).$$

Applying Lemma F.2 with

$$E_t = \mathbb{E}\|\mathbf{S}_t\|_F^2, \qquad A_{t+1} = h_{t+1}^2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big),$$

we get

$$h_t\mathbb{E}\|\mathbf{S}_t\|_F^2 \le 2\Big(\mathbb{E}\|\mathbf{S}_t\|_F^2 - \mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 + A_{t+1}\Big).$$

Moreover, since $\beta_1 = 0$,

$$\mathbb{E}\|\mathbf{S}_1\|_F^2 = \mathbb{E}\|\nabla f(\mathbf{X}_1) - \mathbf{M}_1\|_F^2 = \mathbb{E}\|\nabla f(\mathbf{X}_1) - \nabla f(\mathbf{X}_1;\xi_1)\|_F^2 \le \sigma^2.$$

Thus,

$$\begin{aligned}\sum_{t=1}^{T} h_t\mathbb{E}\|\mathbf{S}_t\|_F^2 &\le 2\sum_{t=1}^{T}\Big(\mathbb{E}\|\mathbf{S}_t\|_F^2 - \mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 + A_{t+1}\Big)\\ &\le 2\mathbb{E}\|\mathbf{S}_1\|_F^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)\sum_{t=1}^{T}\frac{1}{t+1}\\ &\le 2\sigma^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)(1+\ln T).\end{aligned}$$

Hence, since $h_{t+1} \le h_t$, we have

$$\sum_{t=1}^{T} h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2 \le 2\sigma^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)(1+\ln T).$$

Also,

$$\sum_{t=1}^{T}\frac{\eta_t^2}{h_{t+1}} = \sum_{t=1}^{T}\frac{t^{-3/2}}{(t+1)^{-1/2}} \le 2(1+\ln T), \qquad \sum_{t=1}^{T}\eta_t^2 = \sum_{t=1}^{T} t^{-3/2} \le 3.$$

Substituting the above bounds into (11),

$$\kappa_\rho^{\mathrm{ns}}\sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}.$$

where

$$C_{1,\rho}^{\mathrm{ns}} := \frac{a}{L}\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big) + aL\big(c_{m,n}^{\mathrm{ns}}\big)^2, C_{2,\rho}^{\mathrm{ns}} := \frac{a\sigma^2}{L} + \frac{3L}{2}\big(d_\rho^{\mathrm{ns}}\big)^2.$$

Finally, since $\eta_t = t^{-3/4} \ge T^{-3/4}$ for all $1 \le t \le T$,

$$T^{-3/4}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F.$$

Therefore,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}}{\kappa_\rho^{\mathrm{ns}}\cdot T^{1/4}}.$$

Setting $\varepsilon_{\mathrm{ns}}$ to zero yields the corresponding exact-polar result for Theorem 3.5 with decoupled weight decay. Setting additionally $\rho = 0$ recovers the zero weight decay proof. This completes the proof. □

J More Results

► **Learning Rate Search.** Figure 5 shows the learning-rate sweeps for LLaMA2-130M and LLaMA2-350M trained on C4 for 2.6B and 7.5B tokens, respectively.

► **Module-wise Pre-NS Momentum Analysis.** Since **MuonEq** acts directly on the matrix fed into Newton–Schulz, we analyze pre-NS momentum matrices along a 2.6B-token LLaMA2-130M/C4 training trajectory, tracking the `query`, `key`, `value`, `output`, `gate`, `down`, and `up` weights in layers 1, 5, 9, and 12 at 1%, 10%, 50%, and 100% of training. Figure 6 reports the singular-value entropy and stable rank of these matrices; empirically, `RowColNorm` consistently improves both metrics

应用引理F.2,

$$E_t = \mathbb{E}\|\mathbf{S}_t\|_F^2, \qquad A_{t+1} = h_{t+1}^2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big),$$

我们得到

$$h_t\mathbb{E}\|\mathbf{S}_t\|_F^2 \le 2\Big(\mathbb{E}\|\mathbf{S}_t\|_F^2 - \mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 + A_{t+1}\Big).$$

此外，由于 $\beta_1 = 0$,

$$\mathbb{E}\|\mathbf{S}_1\|_F^2 = \mathbb{E}\|\nabla f(\mathbf{X}_1) - \mathbf{M}_1\|_F^2 = \mathbb{E}\|\nabla f(\mathbf{X}_1) - \nabla f(\mathbf{X}_1;\xi_1)\|_F^2 \le \sigma^2.$$

因此，

$$\begin{aligned}\sum_{t=1}^{T} h_t\mathbb{E}\|\mathbf{S}_t\|_F^2 &\le 2\sum_{t=1}^{T}\Big(\mathbb{E}\|\mathbf{S}_t\|_F^2 - \mathbb{E}\|\mathbf{S}_{t+1}\|_F^2 + A_{t+1}\Big)\\ &\le 2\mathbb{E}\|\mathbf{S}_1\|_F^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)\sum_{t=1}^{T}\frac{1}{t+1}\\ &\le 2\sigma^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)(1+\ln T).\end{aligned}$$

因此，由于 $h_{t+1} \le h_t$，我们有

$$\sum_{t=1}^{T} h_{t+1}\mathbb{E}\|\mathbf{S}_t\|_F^2 \le 2\sigma^2 + 2\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big)(1+\ln T).$$

另外，

$$\sum_{t=1}^{T}\frac{\eta_t^2}{h_{t+1}} = \sum_{t=1}^{T}\frac{t^{-3/2}}{(t+1)^{-1/2}} \le 2(1+\ln T), \qquad \sum_{t=1}^{T}\eta_t^2 = \sum_{t=1}^{T} t^{-3/2} \le 3.$$

将上述界限代入(11),

$$\kappa_\rho^{\mathrm{ns}}\sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}.$$

在

$$C_{1,\rho}^{\mathrm{ns}} := \frac{a}{L}\Big(2\sqrt{2}L^2\big(d_\rho^{\mathrm{ns}}\big)^2 + \sigma^2\Big) + aL\big(c_{m,n}^{\mathrm{ns}}\big)^2, C_{2,\rho}^{\mathrm{ns}} := \frac{a\sigma^2}{L} + \frac{3L}{2}\big(d_\rho^{\mathrm{ns}}\big)^2.$$

最后，由于 $\eta_t = t^{-3/4} \ge T^{-3/4}$ 对所有 $1 \le t \le T$,

$$T^{-3/4}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \sum_{t=1}^{T}\eta_t\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F.$$

因此,

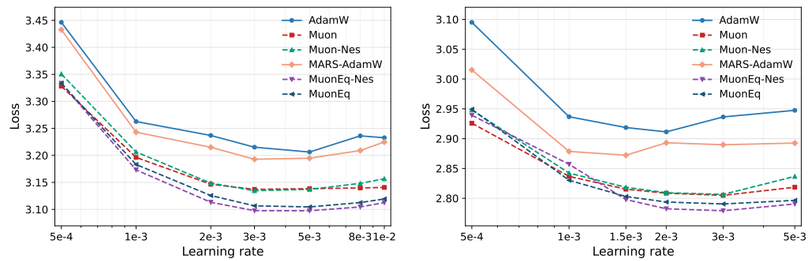
$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_F \le \frac{f(\mathbf{X}_1) - f^\star + C_{1,\rho}^{\mathrm{ns}}(1+\ln T) + C_{2,\rho}^{\mathrm{ns}}}{\kappa_\rho^{\mathrm{ns}}\cdot T^{1/4}}.$$

将 $\varepsilon_{\mathrm{ns}}$ 设置为零得到定理3.5的相应精确极化结果，权重衰减解耦。进一步设置 $\rho = 0$ 恢复零权重衰减证明。这完成了证明。 □

J 更多结果

►**学习率搜索**。 图5显示了LLaMA2-130M和LLaMA2- 350M在C4上分别训练2.6B和7.5B个token的学习率扫描结果。

►**模块级预NS动量分析**。 由于 **MuonEq** 直接作用于输入Newton–Schulz的矩阵，我们沿着2.6B-token LLaMA2-130M/C4训练轨迹分析预NS动量矩阵，追踪查询、键、值、输出、门、向下和向上权重在训练1%、10%、50%和100%时在层1、5、9和12的情况。图6报告了这些矩阵的奇异值熵和稳定秩；经验上，行列归一化始终提高这两个指标



(a) 130M (b) 350M

Figure 5: Learning-rate sweeps for LLaMA2-130M and LLaMA2-350M trained on C4 for 2.6B and 7.5B tokens, respectively.

over `direct`, `ColNorm`, and `RowNorm`. Figure 7 provides the empirical counterpart of Eq. (5): the top row measures the finite-step NS5 approximation error after preprocessing, and the bottom row the bias introduced relative to the unpreconditioned polar direction. At 1% and 10% of training, two-sided `RowColNorm` is usually lowest or near-lowest in the top panels across layers, indicating that early RC equilibration most effectively improves the geometry seen by NS5, but it also yields the largest preconditioning bias. By 50% and 100%, the top-panel differences shrink substantially while the extra bias of the two-sided map remains visible. These plots illustrate the same trade-off as Eq. (5): RC provides stronger two-sided spectral correction but also larger preprocessing bias. For the hidden-weight setting studied here, R remains the more suitable default one-sided variant.

► **Sensitivity Analysis.** Figure 8 analyzes hyperparameter sensitivity on GPT2-small trained on FineWeb for 2.6B tokens by sweeping the learning rate (x-axis) and momentum (y-axis) at $K = 4$ and $K = 5$. Across both NS budgets, especially at $K = 5$, the default R variant of **MuonEq**-Nes shows a broader low-perplexity region than Muon-Nes, indicating reduced sensitivity and stronger robustness to hyperparameter choice.

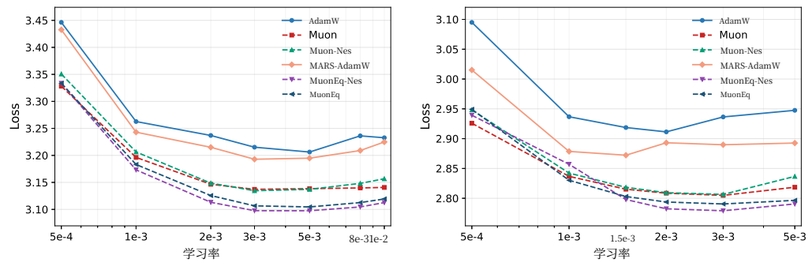
K Experimental Details

K.1 Pretraining on C4

► **Experimental setup.** We use the same 8-node Ascend 910C cluster for all LLaMA experiments; 130M uses 32 NPUs, whereas 350M and 1B use 64 NPUs. We compare AdamW, MARS-AdamW, Muon, Muon-Nes, and **MuonEq/MuonEq**-Nes on LLaMA2-130M, LLaMA2-350M and LLaMA2-1B [29] trained on C4 [30]. We keep the model architecture, dataset, and training recipe fixed across optimizers. The global batch size is 128 for LLaMA2-130M, 256 for LLaMA2-350M and 512 for LLaMA2-1B, and the maximum sequence length is 4096 for both models. Hyperparameters are selected according to validation performance.

► **Hyperparameter search.** The selected hyperparameters are summarized in Tables 4 , 5 and 6. In the main comparison, LLaMA2-130M, LLaMA2-350M, and LLaMA2-1B are trained on 10.5B, 21.0B, and 21.0B tokens, respectively. Learning-rate sweeps are conducted on shorter pilot runs of 2.6B tokens for LLaMA2-130M and 7.5B tokens for LLaMA2-350M. AdamW and MARS-AdamW use $(\beta_1, \beta_2) = (0.9, 0.95)$, $\epsilon = 10^{-8}$, and weight decay 0.1. Muon, Muon-Nes, and **MuonEq/MuonEq**-Nes use Muon momentum 0.95 and weight decay 0.1. We set $\gamma = 0.025$ for MARS-AdamW and $\gamma = 0.05$ for Muon-Nes and **MuonEq/MuonEq**-Nes. We set $\varepsilon = 10^{-8}$ for **MuonEq/MuonEq**-Nes. For all Muon-type optimizers, the number of NS iterations is set to 5 by default. For Muon, Muon-Nes, and **MuonEq/MuonEq**-Nes, we use the Muon implementation from `Moonlight`[2]. All 2D weight matrices except embeddings are optimized with the corresponding Muon-type optimizer, while 1D parameters and embeddings are optimized with AdamW.

[2] `https://github.com/MoonshotAI/Moonlight`



(a) 130M (b) 350M

图5：针对LLaMA2-130M和LLaMA2-350M在C4上分别训练2.6B和7.5B token的学习率扫描结果。

优于直接、列归一化和行归一化。图7提供了等式(5)的经验对应物：顶行测量预处理后的有限步NS5近似误差，底行测量相对于未预处理的极方向引入的偏差。在训练1%和10%时，顶层面板中双行列归一化通常在所有层中最低或接近最低，表明早期的RC均衡最有效地改善了NS5看到的几何形状，但它也产生了最大的预条件偏差。到50%和100%时，顶层面板的差异显著缩小，而双面映射的额外偏差仍然可见。这些图说明了与等式(5)相同的权衡：RC提供更强的双面谱校正，但也产生更大的预处理偏差。对于此处研究的隐藏权重设置，R仍然是更合适的默认单面变体。

► **敏感性分析**。 图8分析了在FineWeb上训练的GPT2-small（2.6B token）的超参数敏感性，通过扫描学习率（x轴）和动量（y轴）在 $K = 4$和 $K = 5$。在两种NS预算下，尤其是在 $K = 5$，默认的R变体的**MuonEq**-Nes比Muon-Nes显示出更宽广的低困惑度区域，表明其敏感性降低，对超参数选择具有更强的鲁棒性。

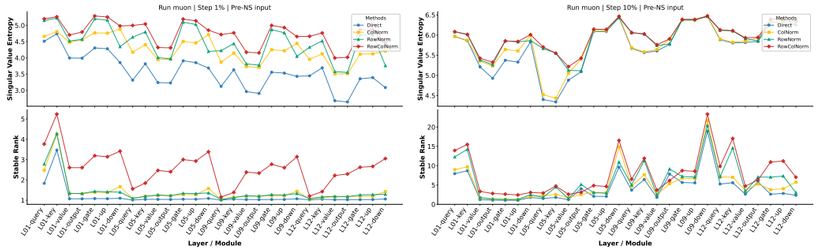
K 实验细节

K.1 在 C4 上的预训练

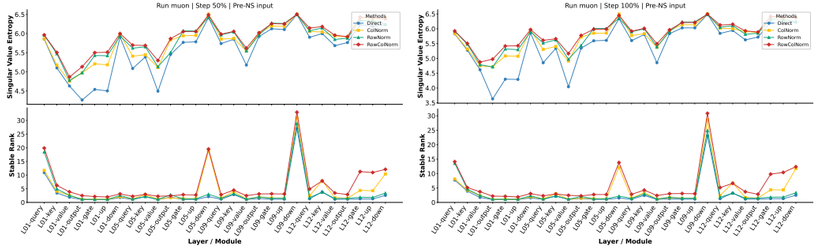
► **实验设置**。 我们为所有 LLaMA 实验使用相同的 8 节点 Ascend 910C 集群；130M 使用 32 个 NPU，而 350M 和 1B 使用 64 个 NPU。我们在 LLaMA2-130M、LLaMA2-350M 和 LLaMA2- 1 B [29] 上比较 AdamW、MARS-AdamW、Muon、Muon-Nes 以及 **MuonEq**/**MuonEq**-Nes，它们都在 C4 [30] 上进行训练。我们保持模型架构、数据集和训练配方在所有优化器之间固定。LLaMA2-130M 的全局批大小为 128，LLaMA2-350M 为 256，LLaMA2-1B 为 512，两个模型的最大序列长度均为 4096。超参数根据验证性能进行选择。

► **超参数搜索**。 选定的超参数汇总于表4、表5和表6。在主要比较中，LLaMA2-130M、LLaMA2-350M和LLaMA2-1B分别使用10.5B、21.0B和21.0B个token进行训练。对于LLaMA2-130M和LLaMA2-350M，学习率扫描在2.6B token的较短预运行中执行。AdamW和MARS-AdamW使用$(\beta_1, \beta_2) = (0.9, 0.95)$，$\epsilon = 10^{-8}$，以及权重衰减0.1。Muon、Muon-Nes和MuonEq/MuonEq-Nes使用Muon动量0.95和权重衰减0.1。我们为MARS-AdamW设置 $\gamma = 0.025$，为Muon-Nes和MuonEq/MuonEq-Nes设置 $\gamma = 0.05$ 。我们为MuonEq/MuonEq-Nes设置 $\varepsilon = 10^{-8}$ 。对于所有Muon类型优化器，默认情况下NS迭代次数设置为5。对于Muon、Muon-Nes和MuonEq/MuonEq-Nes，我们使用Moonlight[2]提供的Muon实现。除了嵌入之外，所有2D权重矩阵都使用相应的Muon类型优化器进行优化，而1D参数和嵌入则使用AdamW进行优化。

[2] https://github.com/MoonshotAI/Moonlight



(a) Step 1%　　(b) Step 10%



(c) Step 50%　　(d) Step 100%

Figure 6: Singular-value entropy and stable rank of Muon momentum matrices under different normalization schemes during LLaMA2-130M training on C4 for 2.6B tokens. `RowColNorm` consistently improves both metrics over `Direct`, `ColNorm`, and `RowNorm`.

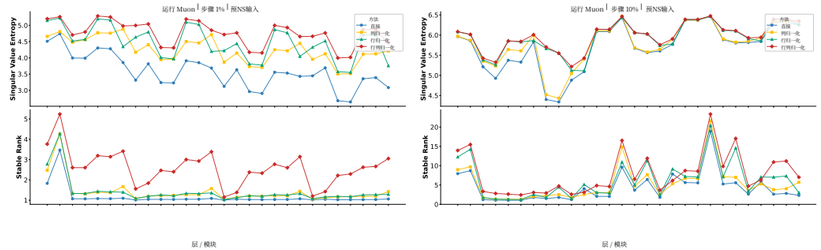
► **Clarification of γ.** The symbol γ in the hyperparameter tables denotes an algorithm-dependent correction coefficient. For MARS-AdamW, γ is the variance-reduction correction coefficient used by MARS. For Muon-Nes and MuonEq-Nes, γ denotes the correction coefficient in the following one-batch Nesterov/MVR parameterization [3, 10]. Let $\mathbf{G}_t := \nabla f(\mathbf{X}_t; \xi_t)$, and let $\mathbf{M}_t^{\mathrm{Nes}}$ denote the momentum matrix passed to the diagonal preconditioning and Newton–Schulz steps. In the constant-momentum setting used in our experiments, the Nesterov buffer can be written as

$$\mathbf{M}_t^{\mathrm{Nes}} = \beta \mathbf{M}_{t-1}^{\mathrm{Nes}} + (1-\beta)\mathbf{G}_t + \gamma\beta\big(\mathbf{G}_t - \mathbf{G}_{t-1}\big), \qquad \mathbf{G}_{t-1} := \nabla f(\mathbf{X}_{t-1}; \xi_{t-1}).$$

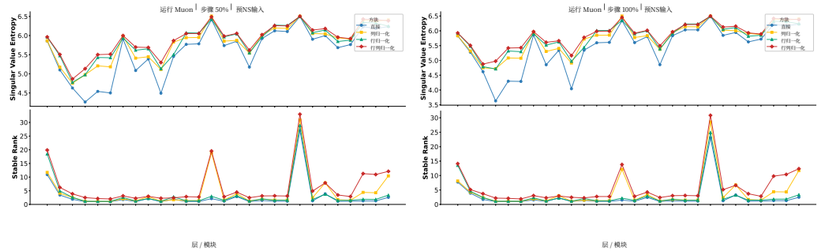
The **MuonEq**-Nes convention used in Algorithm 1 corresponds to $\gamma = 1 - \beta$. Indeed, if the EMA buffer is $\mathbf{M}_t^{\mathrm{EMA}} = \beta \mathbf{M}_{t-1}^{\mathrm{EMA}} + (1-\beta)\mathbf{G}_t$ and the Nesterov input is $\widetilde{\mathbf{M}}_t = \beta \mathbf{M}_t^{\mathrm{EMA}} + (1-\beta)\mathbf{G}_t$, then $\widetilde{\mathbf{M}}_t$ satisfies the recurrence above with $\gamma = 1-\beta$. Thus the reported Muon momentum $\beta = 0.95$ corresponds to $\gamma = 0.05$; the actual coefficient multiplying $\mathbf{G}_t - \mathbf{G}_{t-1}$ is $\gamma\beta = 0.0475$.

Table 4: Hyperparameters used for training LLaMA2-130M on C4

Hyper-parameter	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
Max Learning Rate	5e-3	3e-3	3e-3	3e-3	3e-3	3e-3
Warmup Steps	500					
Batch Size	128					
Maximum Length	4096					
Weight Decay	0.1					
(β_1, β_2)	(0.9,0.95)					
Muon-Momentum	✗	✗	0.95			
Gamma	✗	0.025	✗	0.05	✗	0.05



(a) 步骤 1%　　(b) 步骤 10%



(c) 步骤50%　　(d) 步骤100%

图6：在LLaMA2-130M训练C4上使用2.6B个token时，不同归一化方案下Muon动量矩阵的奇异值熵和稳定秩。RowColNorm始终比Direct、ColNorm和RowNorm更优地提升这两个指标。

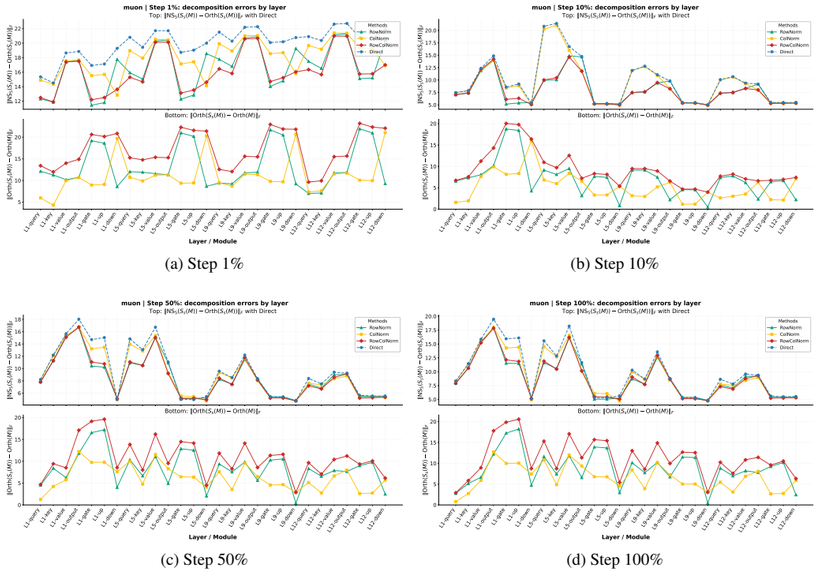
► **关于 γ的说明**。超参数表中的符号 γ 表示算法相关的校正系数。对于MARS-AdamW， γ 是MARS使用的方差减少校正系数。对于Muon-Nes和MuonEq-Nes， γ 表示以下单批Nesterov/MVR参数化 [3, 10]中的校正系数。令 $\mathbf{G}_t := \nabla f(\mathbf{X}_t;\ \xi_t)$，并令 $\mathbf{M}_t^{\mathrm{Nes}}$表示传递给对角预条件和Newton–Schulz步骤的动量矩阵。在我们的实验中使用的恒定动量设置下，Nesterov缓冲区可以写为

$$\mathbf{M}_t^{\mathrm{Nes}} = \beta \mathbf{M}_{t-1}^{\mathrm{Nes}} + (1-\beta)\mathbf{G}_t + \gamma\beta\big(\mathbf{G}_t - \mathbf{G}_{t-1}\big), \qquad \mathbf{G}_{t-1} := \nabla f(\mathbf{X}_{t-1}; \xi_{t-1}).$$

算法1中使用的MuonEq-Nes规范对应于 $\gamma = 1-\beta$。确实，如果EMA缓冲区是$\mathbf{M}_t^{\mathrm{EMA}} = \beta \mathbf{M}_{t-1}^{\mathrm{EMA}}$ +($1-\beta)\mathbf{G}_t$，并且Nesterov输入是 $\widetilde{\ }\mathbf{M}_t = \beta \mathbf{M}_t^{\mathrm{EMA}} + (1-\beta)\mathbf{G}_t$， $\widetilde{\ }$，那么$\mathbf{M}_t$满足上述递归关系式，其中 $\gamma = 1-\beta$。因此，报告的Muon动量 $\beta = 0.95$对应于 $\gamma = 0.05$；实际乘以$\mathbf{G}_t - \mathbf{G}_{t-1}$ 的系数是 $\gamma\beta = 0.0475$。

表4：在C4上训练LLaMA2-130M所使用的超参数

超参数	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
最大学习率	5e-3	3e-3	3e-3	3e-3	3e-3	3e-3
预热步数	500					
批大小	128					
最大长度	4096					
权重衰减	0.1					
$(\beta_1,\ \beta_2)$	(0.9,0.95)					
μ子动量	✗	✗	0.95			
伽马	✗	0.025	✗	0.05	✗	0.05



(a) Step 1%

(b) Step 10%

(c) Step 50%

(d) Step 100%

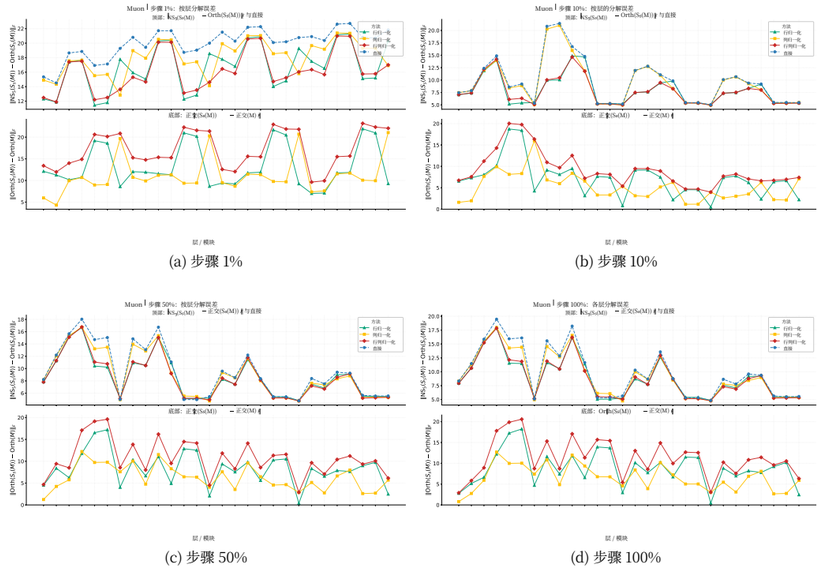
Figure 7: Layerwise Muon NS5 bias decomposition under different normalization schemes during LLaMA2-130M training on C4 for 2.6B tokens. The top panel reports $\|\mathrm{NS}_5(S(\mathbf{M})) - \mathrm{Orth}(S(\mathbf{M}))\|_F$, while the bottom panel reports $\|\mathrm{Orth}(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F$, both evaluated across layers at the same training step. These plots reveal how the choice of normalization shifts the error budget between the **finite-step approximation error** term and the **preconditioning bias** term.

Table 5: Hyperparameters used for training LLaMA2-350M on C4

Hyper-parameter	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
Max Learning Rate	1.5e-3	2e-3	1.5e-3	2e-3	2e-3	2e-3
Warmup Steps	500					
Batch Size	256					
Maximum Length	4096					
Weight Decay	0.1					
(β_1, β_2)	(0.9,0.95)					
Muon-Momentum	✗	✗	0.95			
Gamma	✗	0.025	✗	0.05	✗	0.05

Table 6: Hyperparameters used for training LLaMA2-1B on C4

Hyper-parameter	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
Max Learning Rate	{5e-4, 8e-4}					
Warmup Steps	1000					
Batch Size	512					
Maximum Length	4096					
Weight Decay	0.1					
(β_1, β_2)	(0.9,0.95)					
Muon-Momentum	✗	✗	0.95			
Gamma	✗	0.025	✗	0.05	✗	0.05



(a) 步骤 1%

(b) 步骤 10%

(c) 步骤 50%

(d) 步骤 100%

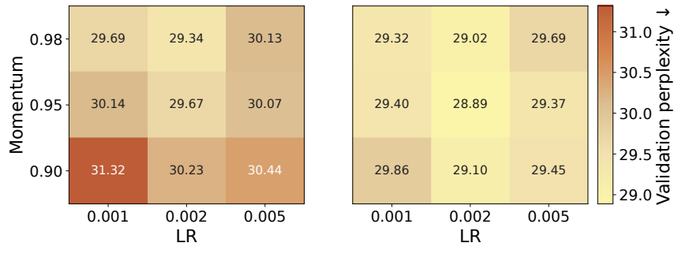
图7：在不同归一化方案下，在C4上对LLaMA2-130M进行2.6B token训练时的逐层μ子NS5偏差分解。顶部面板报告了 $\|\mathrm{NS}_5(S(\mathbf{M})) - \mathrm{Orth}(S(\mathbf{M}))\|_F$，底部面板报告了 $\|\mathrm{Orth}(S(\mathbf{M})) - \mathrm{Orth}(\mathbf{M})\|_F$，两者均在相同训练步骤的各层进行评估。这些图表揭示了归一化选择如何改变误差预算在<样式 id='19'>有限步近似误差</样式>项和<样式 id='21'>预条件偏差</样式>项之间的分配。

表5：用于在C4上训练LLaMA2-350M的超参数

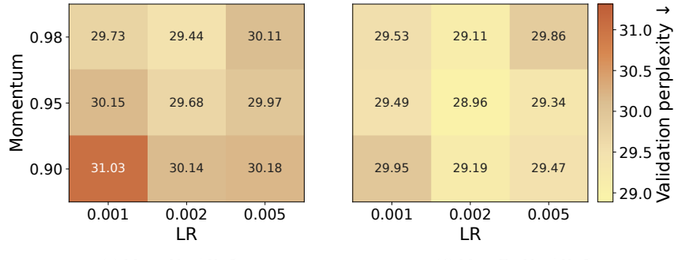
超参数	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
最大学习率	1.5e-3	2e-3	1.5e-3	2e-3	2e-3	2e-3
预热步数	500					
批大小	256					
最大长度	4096					
权重衰减	0.1					
(β_1, β_2)	(0.9,0.95)					
μ子动量	✗	✗	0.95			
伽马	✗	0.025	✗	0.05	✗	0.05

表6：用于在C4上训练LLaMA2-1B的超参数

超参数	AdamW	MARS-AdamW	Muon	Muon-Nes	MuonEq	MuonEq-Nes
最大学习率	{5e-4, 8e-4}					
预热步数	1000					
批大小	512					
最大长度	4096					
权重衰减	0.1					
(β_1, β_2)	(0.9,0.95)					
μ子动量	✗	✗	0.95			
伽马	✗	0.025	✗	0.05	✗	0.05



(a) Muon-Nes | K=4 (b) MuonEq-Nes | K=4



(c) Muon-Nes | K=5 (d) MuonEq-Nes | K=5

Figure 8: Validation perplexity heatmaps on GPT2-small/FineWeb (2.6B tokens) over learning rate and momentum at $K = 4$ and $K = 5$.

Table 7: Hyperparameters used for training ResNet-18 on CIFAR10

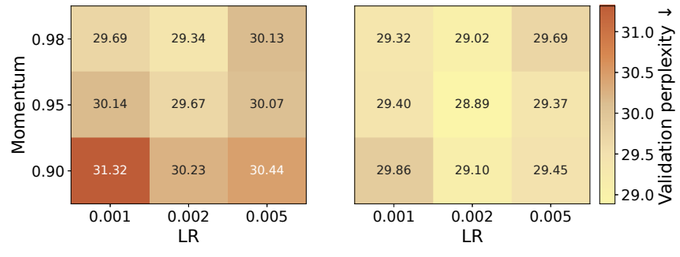
Hyper-parameter	SGD	AdamW	Muon	Muon-Nes	MuonEq-Nes (RC)	MuonEq-Nes (C)	MuonEq-Nes (R)
Max Learning Rate	5e-2	1e-3	5e-2	5e-2	5e-2	5e-2	5e-2
Warmup Steps					0		
Epochs					100		
Batch Size					128		
(β_1, β_2)		(0.9,0.999)			(0.9,0.95)		
Muon-Momentum	✗	✗			0.9		

Table 8: Hyperparameters used for training GPT2-small on FineWeb

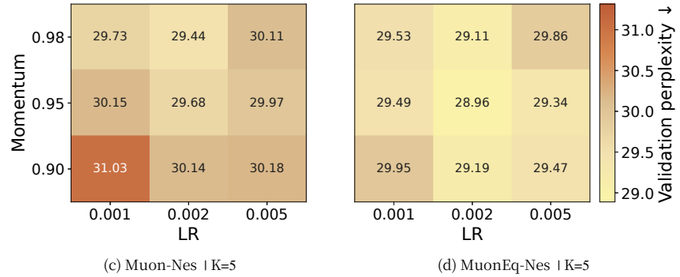
Hyper-parameter	AdamW	AdaMuon	Muon+ (colrow)	Mousse	Muon-Nes	MuonEq-Nes (RC)	MuonEq-Nes (C)	MuonEq-Nes (R)
Max Learning Rate	2e-3	5e-3	5e-3	2e-3	2e-3	2e-3	2e-3	2e-3
Warmup Steps					1000			
Total Steps					20000			
Batch Size					128			
Maximum Length					4096			
Weight Decay					0.1			
(β_1, β_2)					(0.9,0.95)			
Muon-Momentum	✗					0.95		
Gamma	✗	✗	✗	✗	✗		0.05	

K.2 Ablation Study Details

► **Experimental setup.** All ablation experiments are conducted on 4 RTX Pro6000 (96GB) GPUs, with results averaged over three random seeds. We use two workloads: CIFAR-10 [59] with ResNet-18 [60] and FineWeb [61] pretraining of GPT2-small [62] up to 10.5B tokens. Unless otherwise stated, the architecture, data pipeline, training budget, learning-rate schedule, warmup length, weight decay, momentum, Newton–Schulz iteration budget, and parameter grouping are



(a) Muon-Nes | K=4 (b) MuonEq-Nes | K=4



(c) Muon-Nes | K=5 (d) MuonEq-Nes | K=5

图8：在GPT2-small/FineWeb（2.6B个token）上，关于学习率和动量的验证困惑度热图，在 $K = 4$ 和 $K = 5$ 处。

表7：用于在CIFAR10上训练ResNet-18的超参数

超参数	SGD	AdamW	Muon	Muon-Nes	MuonEq-Nes (RC)	MuonEq-Nes (C)	MuonEq-Nes (R)
最大学习率	5e-2	1e-3	5e-2	5e-2	5e-2	5e-2	5e-2
预热步数					0		
周期					100		
批大小					128		
(β_1, β_2)		(0.9,0.999)			(0.9,0.95)		
μ子动量	✗	✗			0.9		

表8：FineWeb上训练GPT2-small所使用的超参数

超参数	AdamW	AdaMuon	Muon+ (列行)	Mousse	Muon-Nes	MuonEq-Nes (RC)	MuonEq-Nes (C)	MuonEq-Nes (R)
最大学习率	2e-3	5e-3	5e-3	2e-3	2e-3	2e-3	2e-3	2e-3
预热步数					1000			
总步数					20000			
批大小					128			
最大长度					4096			
权重衰减					0.1			
(β_1, β_2)					(0.9,0.95)			
μ子动量	✗					0.95		
伽马	✗	✗	✗	✗	✗		0.05	

K.2 消融研究详情

► **实验设置**。 所有消融实验均在 4 块 RTX Pro6000 (96GB) GPU 上进行，结果在三个随机种子上取平均。我们使用两种工作负载：CIFAR-10 [59]with ResNet-18 [60] 和 FineWeb [61] 的 GPT2-small [62] 预训练，直到 10.5B 个 token。除非另有说明，架构、数据管道、训练预算、学习率计划、预热长度、权重衰减、动量、Newton–Schulz 迭代预算和参数分组是

kept identical to the corresponding reference recipe; only the diagonal map before orthogonalization is changed. The selected hyperparameters are summarized in Tables 7 and 8. For learning-rate tuning, we use discrete search grids that depend on the workload and optimizer family. On CIFAR-10 with ResNet-18, the SGD baseline is tuned over $\{1e{-}1, 5e{-}2, 1e{-}2, 5e{-}3\}$, AdamW over $\{5e{-}3, 1e{-}3, 5e{-}4, 1e{-}4\}$, and Muon-type optimizers over $\{1e{-}1, 5e{-}2, 1e{-}2, 5e{-}3, 1e{-}3\}$. We use the Muon implementation from `cifar10-airbench`[3]. On FineWeb pretraining of GPT2-small, all methods are tuned over the same learning-rate grid $\{5e{-}4, 1e{-}3, 2e{-}3, 5e{-}3\}$. We use the Muon implementation from `Moonlight`[4]

We compare the three **MuonEq**-Nes variants RC, R, and C. RC applies two-sided row/column normalization throughout training, R applies row normalization throughout training, and C applies column normalization throughout training. We take R as the default variant. In all cases, the orthogonalization routine, optimizer states, and scalar hyperparameters are unchanged.

The purpose of this ablation is to isolate the effect of pre-orthogonalization geometry rather than to redesign the optimizer for each variant. We therefore use the same training budget and evaluation protocol across all runs. On CIFAR-10 we report final test accuracy. On FineWeb/GPT2-small we report validation perplexity at the end of the 10.5B-token budget.

This ablation is directly connected to Eq. (5). RC provides the strongest two-sided spectral correction and mainly targets the finite-step Newton–Schulz term. R and C test the two one-sided geometries under the same optimizer configuration. Since **MuonEq/MuonEq**-Nes targets hidden matrix weights, R is the more relevant default in our setting, while C is included as the column-sided companion form.

K.3 A Note on Row/Column Terminology in Prior Work

A careful comparison to prior row/column normalization papers requires fixing the matrix convention. [44] write linear weights as $\theta \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and define column-wise normalization along the output dimension. Under the standard deep-learning storage layout $\mathbf{W} = \theta^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, this corresponds to row normalization of the stored hidden-weight matrix. Their empirical observation that row-wise normalization can be unstable is also traced largely to the LM-head, where row-wise normalization produces extreme gradient values. This is not the same setting as **MuonEq/MuonEq**-Nes, which is aimed at hidden matrix weights.

By contrast, [43] use the standard convention $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, so their ColNorm is genuinely column-wise in the same orientation as ours. Their main recommendation for language models is first-layer ColNorm, hidden-layer Spectral, and last-layer Sign. We therefore interpret our row-sided default choice as directly aligned with [28], compatible with [44] after accounting for layout conventions, and not in tension with [43] once first-layer/input geometry is separated from hidden-layer geometry.

L Limitations

MuonEq is designed as a lightweight pre-orthogonalization correction for matrix-valued hidden weights. The row-normalized variant is our default in this setting, but it should not be viewed as a universally optimal choice for every parameter block or architecture. Our analysis isolates the core geometry of the pre-balanced orthogonalized update, while broader coverage of implementation variants and training regimes remains an open direction. Our convergence analysis for the RC variant is restricted to an auxiliary exact-polar regime: Proposition 3.4 assumes Nesterov momentum is disabled, no decoupled weight decay is used, and the stabilization parameter ε is sufficiently large. Therefore, this RC guarantee does not formally cover the practical RC ablations, which use finite-step Newton–Schulz iterations and the small numerical stabilization $\varepsilon = 10^{-8}$. Extending the RC theory to this practical regime is left for future work. This limitation is specific to the auxiliary RC analysis; the main guarantee for the default R variant is given by Theorem 3.5 and Corollary 3.6. Empirically, our large-scale evaluation focuses on LLaMA-style pretraining on C4, complemented by smaller-scale ablations.

[3] `https://github.com/KellerJordan/cifar10-airbench`
[4] `https://github.com/MoonshotAI/Moonlight`

保持与相应参考配方完全相同；仅在正交化前的对角矩阵有所改变。所选超参数总结在表7和表8中。对于学习率调整，我们使用依赖于工作负载和优化器系列的离散搜索网格。在CIFAR-10与ResNet-18上，SGD基线在 $\{1e{-}1, 5e{-}2, 1e{-}2, 5e{-}3\}$, AdamW在$\{5e{-}3, 1e{-}3, 5e{-}4, 1e{-}4\}$, 以及Muon类型的优化器在 $\{1e{-}1, 5e{-}2, 1e{-}2, 5e{-}3, 1e{-}3\}$。我们使用来自cifar10-airbench[3]的Muon实现。在GPT2- small的FineWeb预训练中，所有方法都在相同的学习率网格 $\{5e{-}4, 1e{-}3, 2e{-}3, 5e{-}3\}$上进行调整。我们使用来自Moonlight[4]的Muon实现

我们比较了三个 **MuonEq**-Nes 变体 RC、R 和 C。RC 在整个训练过程中应用双边行/列归一化，R 在整个训练过程中应用行归一化，C 在整个训练过程中应用列归一化。我们以 R 作为默认变体。在所有情况下，正交化程序、优化器状态和标量超参数均保持不变。

此消融实验的目的是隔离预正交化几何形状的影响，而不是为每个变体重新设计优化器。因此，我们在所有运行中使用了相同的训练预算和评估协议。在CIFAR-10上，我们报告最终测试准确率。在FineWeb/GPT2-small上，我们报告在10.5B令牌预算结束时的验证困惑度。

此消融直接与公式(5)相关。RC提供最强的双边谱校正，主要针对有限步Newton-Schulz项。R和C在相同的优化器配置下测试两种单边几何形状。由于MuonEq/MuonEq-Nestargets隐藏矩阵权重，在我们的设置中R是更相关的默认选项，而C作为列侧的伴随形式被包含。

K.3 关于 PriorWork 中行/列术语的说明

与先前的行/列归一化论文进行仔细比较需要固定矩阵约定。[44] 将线性权重写为 $\theta \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ 并沿输出维度定义按列归一化。在标准的深度学习存储布局 $\mathbf{W} = \theta^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ 下，这对应于存储的隐藏权重矩阵的行归一化。他们关于行归一化可能不稳定的经验观察也主要归因于 LM 头，在行归一化会产生极端梯度值。这不同于 **MuonEq**/**MuonEq**-Nes，后者旨在处理隐藏矩阵权重。

相比之下， [43] 使用标准约定 $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$，因此他们的 ColNorm 真正是在与我们相同的方向上按列进行的。他们为语言模型的主要建议是第一层 ColNorm、隐藏层 Spectral 和最后一层 Sign。因此，我们将我们行侧的默认选择解释为与 [28]，直接对齐，在考虑布局约定后与 [44] 兼容，并且在第一层/输入几何与隐藏层几何分离后，与 [43] 不冲突。

L 局限性

MuonEq被设计为一种轻量级的预正交化校正方法，用于矩阵值隐藏权重。行归一化变体是我们在此设置中的默认选择，但它不应被视为适用于所有参数块或架构的通用最优选择。我们的分析隔离了预平衡正交化更新的核心几何结构，而更广泛的实现变体和训练机制的研究仍是一个开放方向。我们对RC变体的收敛性分析仅限于辅助精确极化R regime：命题3.4假设禁用了Nesterov动量，未使用解耦权重衰减，且稳定化参数 ε 足够大。因此，此RC保证并未正式涵盖实际RC消融实验，后者使用有限步Newton-Schulz迭代和小的数值稳定化 $\varepsilon = 10^{-8}$。将RC理论扩展到此实际regime是未来工作的任务。此限制仅针对辅助RC分析；默认R变体的主要保证由定理3.5和推论3.6给出。在实证中，我们的大规模评估专注于在C4上的LLaMA风格预训练，辅以较小规模的消融实验。

[3] https://github.com/KellerJordan/cifar10-airbench[4]
https://github.com/MoonshotAI/Moonlight

M Broader Impacts

This work studies an application-agnostic optimizer that may improve training efficiency for matrix-valued neural network parameters without introducing new data, data collection, or deployment pipelines. Its direct societal risks are therefore mainly inherited from the downstream models and datasets on which it is used. As with other efficiency improvements, **MuonEq** could lower the cost of both beneficial and harmful model training, so appropriate downstream safeguards remain necessary. Any environmental benefit depends on whether efficiency gains are used to reduce compute for a fixed objective rather than to scale up training.

M 更广泛的影响

本研究研究了一种与具体应用无关的优化器，它可以在不引入新数据、数据收集或部署管道的情况下，提高矩阵值神经网络参数的训练效率。因此，其直接的社会风险主要来自使用它的下游模型和数据集。与其他效率改进类似，**MuonEq** 可能降低有益和有害模型训练的成本，因此适当的下游安全措施仍然是必要的。任何环境效益取决于效率提升是否用于在固定目标下减少计算量，而不是用于扩大训练规模。